Geometric Class Field Theory for Curves

Assaf Marzan

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TODO

- Check the references at Bib.
- exclude or minimize the frame-torsor equivliance. and make sure to mention it in ramification settings.
- Go over which results rely on k being algebraically closed (for example, Theorem 58), and verify that this assumption is stated and used consistently throughout the thesis.

Abstract

We calculate the ramification of the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ of a local system $\mathcal{F}$ on a curve $C \setminus \mathfrak{m}$. Assuming the ramification of $\mathcal{F}$ is bounded by $\mathfrak{m}$, we find that the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ is at most tamely ramified at the generic point of the exceptional divisor $E_{\mathfrak{m}}$ of the blowup of $C^{(\deg \mathfrak{m})}$ at $\mathfrak{m}$. As a primary application, we utilize this result to prove geometric class field theory for curves. Our approach builds upon the geometric framework for the unramified case originally established by Deligne for the rank-one Langlands correspondence, following the subsequent extensions to the ramified case developed by Takeuchi and Guignard.

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Introduction

In this thesis, we give a direct proof of a key geometric result in Deligne's approach to geometric class field theory. We work primarily over a perfect field k. Let C be a smooth, projective, geometrically connected curve over k, of genus g. Let $\mathfrak{m} = \sum_i n_i P_i$ be an effective divisor on C, called a modulus, and set $U = C \setminus \mathfrak{m}$. Geometric class field theory asserts that every rank-one local system $\mathcal{F}$ on U whose ramification is bounded by $\mathfrak{m}$ arises uniquely as the pullback of a multiplicative rank-one local system on $\mathrm{Pic}_{C,\mathfrak{m}}$ along the Abel–Jacobi morphism:

$$\Phi : U \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}\,.$$

One of the main geometric ingredients in Deligne's approach to this assertion is the analysis of the symmetric powers $\mathcal{F}^{[d]}$ on $U^{(d)}$, and in particular the proof that, for all sufficiently large d, they descend along the degree d Abel-Jacobi morphism:

$$\Phi_d : U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}\,.$$

The geometric-fibers of Φ_d (for $d \geq \deg \mathfrak{m} + 2g - 1$) over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m}-g+1}_{k^{sep}} & \text{if } \mathfrak{m} > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \text{if } \mathfrak{m} = 0 \end{cases}$$

Where g is the genus of the curve C. The unramified case ($\mathfrak{m} = 0$) is relatively simple, as the Abel-Jacobi map is proper, surjective with geometrically connected fibers, which follows from the fact that it is a fibration in projective spaces. Thus, by using the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $U^{(d)} (= C^{(d)})$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}} (= \mathrm{Pic}^d_C)$.

The ramified case ($\mathfrak{m} > 0$) is more subtle, as the Abel-Jacobi map is not proper anymore, and one needs to analyze the ramification of $\mathcal{F}^{[d]}$ "along the boundary" of $U^{(d)}$ in $C^{(d)}$.

Previous work has generalized Deligne's approach to the ramified case, most notably by Guignard [Gui19] and Takeuchi [Tak18]. Their approaches differ. To descend, Guignard proves that the restriction of $\mathcal{F}^{[d]}$ to any line in the fiber of the degree d Abel-Jacobi map is a constant étale sheaf. He achieves this by demonstrating that the restriction is at most tamely ramified and invoking the triviality of the tame fundamental group of $\mathbb{A}^1_k$. His analysis relies on local geometric class field theory. It is also worth noting that Guignard's method generalizes to relative curves over arbitrary base schemes. Takeuchi, on the other hand, constructs a compactification of $U^{(d)}$ by blowing up $C^{(d)}$ along certain well-chosen centers. This compactification, denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$, has $U^{(d)}$ as an open subscheme with a codimension 1 closed subscheme H as complement. He then shows that the Abel-Jacobi map extends to a proper morphism from $\tilde{C}^{(d)}_{\mathfrak{m}}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, which is a fibration in

projective spaces. Thus, by the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $\tilde{C}_{\mathfrak{m}}^{(d)}$ and that of $\mathrm{Pic}_{C,\mathfrak{m}}^{d}$. To conclude the descent, Takeuchi analyzes the ramification of $\mathcal{F}^{[d]}$ along the boundary $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$ of $U^{(d)}$ in $\tilde{C}_{\mathfrak{m}}^{(d)}$, showing that it is tamely ramified there, which suffices. His method relies on the theory of refined Swan conductors.

For an account of these approaches, see [Gui19] and [Tak18]. For a full approach following Deligne's method in the unramified case, and the tamely ramified case see [Ten15], and [Tót11].

In this thesis, we take a more elementary approach and calculate the ramification of $\mathcal{F}^{[d]}$ at the generic point of H directly, using explicit ramification theory rather than refined Swan conductors. We first reduce the problem to cyclic torsors. We then treat cyclic groups of order prime to the characteristic using Kummer theory and, in characteristic $p > 0$, cyclic groups of order p and p^r, with $r \geq 2$, using Artin–Schreier and Artin–Schreier–Witt theory, respectively. These descriptions provide explicit local equations, which allow us to analyze the relevant valuations after blowing up and prove that the ramification is at most tame. To prove geometric class field theory for curves, we utilize Takeuchi's construction of the compactification $\tilde{C}_{\mathfrak{m}}^{(d)}$ of $U^{(d)}$ via the blowing up of $C^{(d)}$.

We now introduce the notation needed to state geometric class field theory, its standard degree-d reduction, and the ramification theorem proved in this thesis.

Let k be a perfect field, and let C be a projective smooth geometrically connected curve over k, with genus g.

Fix a modulus $\mathfrak{m}$, i.e. an effective Cartier divisor of C and let U be its complement in C. The pairs $(\mathcal{L}, \alpha)$, where $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module and $\alpha : \mathcal{O}_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}}$ is a rigidification of $\mathcal{L}$ along $\mathfrak{m}$, are parametrized by a k-group scheme $\mathrm{Pic}_{C,\mathfrak{m}}$, called the rigidified Picard scheme. We write $\mathrm{Pic}_{C,\mathfrak{m}}^{d}$ for its degree-d component. The Abel-Jacobi morphism

$$\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$$

is the morphism which sends a section x of U to the pair $(\mathcal{O}_C(x), 1)$.

Geometric class field theory asserts that rank-one local systems on U, with ramification bounded by $\mathfrak{m}$, arise uniquely as pullbacks of multiplicative rank-one local systems on $\mathrm{Pic}_{C,\mathfrak{m}}$. For a finite commutative coefficient ring Λ, the $\Lambda^\times$-torsor associated with such a multiplicative local system is an isogeny onto $\mathrm{Pic}_{C,\mathfrak{m}}$ with finite constant kernel $\Lambda^\times$. The isogeny formulation will not play a direct role in our setting, which instead focuses on the degree-d descent statement introduced below.

The fundamental result of GCFT can be formulated as:

Theorem 1 (Geometric Class Field Theory)**.** *Let Λ be a finite commutative ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Then, there exists a unique (up to isomorphism) multiplicative[1] étale sheaf of Λ-modules $\mathcal{G}$ on $\mathrm{Pic}_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}$ by Φ is isomorphic to $\mathcal{F}$.*

Let d be a positive integer. We denote by $U^{(d)}$ the d-th symmetric power of U over k. For an étale sheaf $\mathcal{F}$ on $U_{\text{ét}}$, we denote by $\mathcal{F}^{[d]}$ the d-th symmetric power of $\mathcal{F}$ on $U^{(d)}$. The degree-d Abel–Jacobi morphism is

$$\Phi_d : U^{(d)} \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}^{d},$$

[1]Here multiplicative means that the local system is equipped with an isomorphism compatible with the group law on $\mathrm{Pic}_{C,\mathfrak{m}}$, satisfying the usual associativity and unit conditions; see [Gui19].

sending an effective divisor $D = x_1 + \cdots + x_d$ on U to the pair

$$\big(\mathcal{O}_C(D), 1\big).$$

A standard symmetric-power argument[2] reduces Theorem 1 to the following problem: for every sufficiently large d, construct a rank-one local system $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$ whose pullback along Φ_d is isomorphic to $\mathcal{F}^{[d]}$. In our setting, pullback along Φ_d is fully faithful, so any such local system is automatically unique up to isomorphism. Moreover, the canonical compatibility of the symmetric powers $\mathcal{F}^{[d]}$ under the addition of effective divisors descends uniquely to the family $\{\mathcal{G}_d\}$. The resulting multiplicative family determines the multiplicative local system on $\mathrm{Pic}_{C,\mathfrak{m}}$ appearing in Theorem 1. Consequently, it suffices to prove the following reduced statement:

Theorem 2. *Let Λ be a finite commutative ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{ét}$, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) étale sheaf of Λ-modules $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}_d$ by Φ_d is isomorphic to $\mathcal{F}^{[d]}$.*

The main technical input in the proof of Theorem 2 is the following local ramification theorem:

Theorem 3. *Assume that k is algebraically closed. Let Λ be a finite commutative ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{ét}$, with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a single-point submodulus of $\mathfrak{m}$. Then $\mathcal{F}^{[\deg \mathfrak{n}]}$ is tamely ramified at the generic point $\eta_{\mathfrak{n}}$ of the exceptional divisor $E_{\mathfrak{n}}$ of the blowup $X_{\mathfrak{n}}$ of $C^{(\deg \mathfrak{n})}$ at the closed point $\mathfrak{n}$.*

As a final methodological note, for convenience we carry out the ramification calculations in the language of G-torsors, with $G = \Lambda^\times$, rather than rank-one local systems. By Proposition 9, the two formulations are equivalent, and the results translate directly between them.

The thesis is organized as follows:

Chapter 1 - General Preliminaries establishes the equivalence between rank-one local systems and $G = \Lambda^\times$-torsors and develops the torsor operations, finite étale covers, and symmetric-power constructions used throughout the thesis. It also collects the basic geometric facts about symmetric powers and blowups needed in the subsequent ramification calculations.

Chapter 2 - Ramification After Blowup Is Tame proves Theorem 3. After recalling the necessary ramification theory for discrete valuations and finite abelian torsors, it reduces the problem to cyclic torsors on $\mathbb{P}^1$ and treats the Kummer, Artin–Schreier, and Artin–Schreier–Witt cases through explicit calculations at the exceptional divisor.

Chapter 3 - Geometric Class Field Theory deduces Theorem 2 from the ramification theorem proved in Chapter 2. It recalls the rigidified Picard scheme and the Abel–Jacobi morphism, introduces Takeuchi's compactification, and combines the single-point calculations to establish tameness along the full boundary H. It then extends $\mathcal{F}^{[d]}$ across the boundary and descends the resulting local system to $\mathrm{Pic}^d_{C,\mathfrak{m}}$.

Appendix A *Cyclic Torsor Theories and Witt Vectors* collects the group-scheme constructions, classification results, normal forms, and proofs supporting the three cyclic ramification theories used in Chapter 2.

[2] See the last page of [Gui19], Section 8.3 of [Ten15], or the proof of Theorem 1.2 in [Tak18] for details on this reduction.

Appendix B *A Failure of an Artin–Schreier Induction Strategy* examines a tempting attempt to deduce the cyclic p-power calculation inductively from the degree-p Artin–Schreier case, and explains how wild base change obstructs that argument.

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Use of Artificial Intelligence. In preparing this thesis, I used AI-assisted tools including Codex with GPT-5.6 Sol, Claude Code with Claude Opus 4.8 and Claude Fable, and Google Gemini with Gemini 3.1 Pro. These tools assisted with phrasing, editing, restructuring, drafting portions of the exposition, and formulating mathematical statements and arguments. They also assisted in developing and checking parts of proofs, particularly the Artin–Schreier–Witt calculation for cyclic groups of order p^r, as well as several particular cases in Kummer and Artin–Schreier theory that were independently verified by hand. More broadly, these tools supported the organization and completion of the thesis as a whole. Every AI-generated suggestion or draft incorporated into the thesis was fully reviewed and mathematically checked by the author, revised where necessary, and adapted to the notation, presentation, and style of the thesis. No AI-generated material was accepted without human verification, and the author assumes full responsibility for all mathematical and expository content.

Chapter 1

General Preliminaries

This chapter collects the constructions and conventions used in both main parts of the thesis. Rank-one étale local systems are the natural objects in the statement of geometric class field theory. For the geometric and ramification-theoretic arguments, however, it is more convenient to work with torsors: they are represented by finite étale covers, admit concrete contracted products and quotients, and make ramification accessible through extensions of local fields.

If Λ is a finite commutative ring and $G = \Lambda^\times$, the sheaf of frames of a rank-one Λ-local system is a G-torsor, and conversely a G-torsor determines an associated rank-one local system. We record this equivalence once below. Thereafter, the main results are stated in the language of local systems, while the geometric and ramification calculations are carried out with the corresponding torsors.

1.1 Torsors and Rank-One Local Systems

In what follows, we largely adhere to the treatment of torsors and group objects found in published notes of Alex Youcis [You20]. Let $\mathscr{C} = (\mathcal{C}, J)$ be a site and let $\mathcal{E} = Sh(\mathscr{C})$ be the associated topos. Let $\mathcal{G}$ be a group object in $\mathcal{E}$. We denote by $\mathcal{G}\mathcal{E}$ the category of objects in $\mathcal{E}$ endowed with a left $\mathcal{G}$-action.

Definition 4. A **$\mathcal{G}$-torsor** in $\mathcal{E}$ is an object $\mathcal{P}$ of $\mathcal{G}\mathcal{E}$ satisfying the following conditions:

1. The structural morphism $\mathcal{P} \to 1$ is an epimorphism in $\mathcal{E}$ (i.e., $\mathcal{P}$ is locally non-empty).
2. The map $\mathcal{G} \times \mathcal{P} \to \mathcal{P} \times \mathcal{P}$ defined by $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism in $\mathcal{E}$ (i.e., $\mathcal{G}$ acts simply transitively on $\mathcal{P}$).

Since $\mathcal{E}$ is the topos of sheaves on a site $\mathscr{C}$, the definition can be reformulated in terms of covers. A $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if:

1. For every object $X \in \mathcal{C}$, there exists a covering $\{U_i \to X\} \in J$ such that $\mathcal{P}(U_i) \neq \emptyset$ for all i.
2. For any $X \in \mathcal{C}$ where $\mathcal{P}(X)$ is non-empty, the action of $\mathcal{G}(X)$ on $\mathcal{P}(X)$ is simply transitively.

A fundamental property of torsors is their local triviality: a $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if and only if it is locally isomorphic to the trivial torsor. Specifically, for every $X \in \mathcal{C}$, there must exist a cover

$\{U_i \to X\}$ such that the restriction $\mathcal{P}|_{U_i}$ is isomorphic, as a $\mathcal{G}|_{U_i}$-sheaf, to $\mathcal{G}|_{U_i}$ acting on itself by left multiplication.

A **morphism of $\mathcal{G}$-torsors** $f : \mathcal{P}_1 \to \mathcal{P}_2$ is a morphism of sheaves that is equivariant with respect to the $\mathcal{G}$-action.

Proposition 5. *Every morphism of $\mathcal{G}$-torsors is an isomorphism. Consequently, the category of $\mathcal{G}$-torsors in $\mathcal{E}$ forms a groupoid.*

Proof. Locally on a cover trivializing both torsors, such a morphism is a $\mathcal{G}$-equivariant endomorphism of the trivial torsor $\mathcal{G}$, i.e. right translation by a section of $\mathcal{G}$, hence an isomorphism; and being an isomorphism may be checked on a cover. □

Definition 6. We denote the groupoid of $\mathcal{G}$-torsors in $\mathcal{E}$ by $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. The set of isomorphism classes of $\mathcal{G}$-torsors is denoted by $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$.

Torsors exhibit functoriality with respect to the group object:

Definition 7. Let $\varphi : \mathcal{G}_1 \to \mathcal{G}_2$ be a morphism of group sheaves on $\mathscr{C}$, and let $\mathcal{P}$ be a $\mathcal{G}_1$-torsor. We define the **contracted product** $\mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$ as the quotient sheaf $\mathcal{G}_1 \backslash (\mathcal{G}_2 \times \mathcal{P})$, where $\mathcal{G}_1$ acts on the product by:

$$g_1 \cdot (g_2, p) = (g_2 \varphi(g_1)^{-1}, g_1 \cdot p)$$

The contracted product inherits a natural left $\mathcal{G}_2$-action given on local sections by $h \cdot [g_2, p] = [hg_2, p]$, which endows it with the structure of a $\mathcal{G}_2$-torsor.

This construction yields a functor:

$$\varphi_* : \mathbf{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathbf{Tors}(\mathcal{E}, \mathcal{G}_2), \quad \mathcal{P} \mapsto \mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$$

There is a canonical φ-equivariant morphism

$$j_\varphi : \mathcal{P} \longrightarrow \varphi_* \mathcal{P}, \qquad p \longmapsto [1, p],$$

meaning that $j_\varphi(g \cdot p) = \varphi(g) \cdot j_\varphi(p)$ for every local section g of $\mathcal{G}_1$. On the level of isomorphism classes, φ_* induces a map of pointed sets $\mathrm{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathrm{Tors}(\mathcal{E}, \mathcal{G}_2)$, sending the class of the trivial $\mathcal{G}_1$-torsor to the class of the trivial $\mathcal{G}_2$-torsor.

When $\mathcal{G}$ is a **sheaf of abelian groups** (an *abelian sheaf*), the pointed set $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ inherits the structure of an abelian group:

Definition 8. Assume $\mathcal{G}$ is abelian sheaf. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be objects of $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. We define the **product** $\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2$ to be the quotient sheaf $(\mathcal{P}_1 \times \mathcal{P}_2)/\mathcal{G}$, where $\mathcal{G}$ acts on T-points by having:

$$g \cdot (f_1, f_2) := (g f_1, g^{-1} f_2)$$

The group object $\mathcal{G}$ then acts on the resulting quotient via its action on the presheaf quotient, which is given on classes by:

$$g \cdot [(f_1, f_2)] = [(g f_1, f_2)] = [(f_1, g f_2)]$$

where the square brackets denote the class in the quotient set.[1]

[1] Equivantly, the sum is obtained as the contracted product of the $\mathcal{G} \times \mathcal{G}$-torsor $\mathcal{P}_1 \times \mathcal{P}_2$ along the multiplication map $m : \mathcal{G} \times \mathcal{G} \to \mathcal{G}$.

Let $[\mathcal{P}_1]$ and $[\mathcal{P}_2]$ be the isomorphism classes of $\mathcal{P}_1$ and $\mathcal{P}_2$ in $\mathrm{Tors}(\mathcal{E},\mathcal{G})$. We define the sum $[\mathcal{P}_1]+[\mathcal{P}_2]$ to be the class $[\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2]$. This structure turns $\mathrm{Tors}(\mathcal{E},\mathcal{G})$ into an abelian group, where the identity is the class of the trivial torsor and the inverse is obtained by the opposite action.

We now record the equivalence used in the thesis. Let Λ be a commutative ring object of $\mathcal{E}$ and let $G=\Lambda^\times$ be its group of units. We write $\mathbf{Pic}(\mathcal{E},\Lambda)$ for the groupoid of locally free Λ-modules of rank one. For $\mathcal{L}$ in this groupoid, its **frame torsor** is

$$\mathrm{Fr}(\mathcal{L}) := \underline{\mathrm{Isom}}_\Lambda(\Lambda,\mathcal{L}).$$

The group G acts on frames by precomposition with multiplication on Λ. Conversely, if $\mathcal{P}$ is a G-torsor, its **associated Λ-module** is the contracted product

$$\mathcal{P}\times^G\Lambda := (\mathcal{P}\times\Lambda)/G, \qquad g\cdot(p,\lambda)=(gp,g^{-1}\lambda),$$

where G acts on Λ by multiplication.

Proposition 9. *The functors*

$$\mathcal{P}\longmapsto \mathcal{P}\times^G\Lambda, \qquad \mathcal{L}\longmapsto \mathrm{Fr}(\mathcal{L})$$

are mutually quasi-inverse equivalences

$$\mathbf{Tors}(\mathcal{E},G)\xrightarrow{\sim}\mathbf{Pic}(\mathcal{E},\Lambda).$$

Proof. There are natural morphisms

$$\mathcal{P}\longrightarrow \mathrm{Fr}(\mathcal{P}\times^G\Lambda), \qquad p\longmapsto\big(\lambda\longmapsto[p,\lambda]\big),$$

and

$$\mathrm{Fr}(\mathcal{L})\times^G\Lambda\longrightarrow\mathcal{L}, \qquad [\varphi,\lambda]\longmapsto\varphi(\lambda).$$

After passing to a cover on which $\mathcal{P}$ is the trivial torsor and $\mathcal{L}$ is free of rank one, both morphisms become the identity. Hence they are isomorphisms. □

When $\mathcal{E}=\mathrm{Sh}(X_{\text{ét}})$ and Λ is a finite constant ring, the objects of $\mathbf{Pic}(\mathcal{E},\Lambda)$ are precisely the rank-one Λ-local systems on X. Henceforth, we identify such a local system with its $G=\Lambda^\times$-torsor of frames and pass between the two descriptions without further comment.[2]

Lastly, for any object $\mathcal{X}\in\mathcal{E}$, there is a canonical identification between the slice category $(\mathcal{GE})/\mathcal{X}$ and the category of group objects $\mathcal{G}(\mathcal{E}/\mathcal{X})$, where $\mathcal{X}$ is viewed as having the trivial $\mathcal{G}$-action.

We denote by $\mathbf{Tors}(\mathcal{X},\mathcal{G})$ the category of G-torsors over $\mathcal{X}$ in $\mathcal{GE}_{/\mathcal{X}}$.

1.2 Étale Torsors, Covers, and Quotients

Fix a scheme X, and let G be a smooth affine X-group scheme. Denote by $X_{\text{Ét}}$ and $X_{\text{ét}}$ the big and small étale sites on X, respectively. The functor of points of G, which we denote by $G(\cdot)=\mathrm{Hom}_X(-,G)$, defines a group sheaf on $X_{\text{ét}}$.

[2] The equivalence is compatible with pullback and with tensor products, hence with the symmetric-power constructions used below. These compatibilities may be checked after a local trivialization. It also preserves ramification bounds: a rank-one local system and its frame torsor determine, on each connected component, the same character $\pi_1^{\text{et}}(X,\bar{x})\to\Lambda^\times$, and therefore have the same restrictions to inertia and to the upper ramification groups.

Definition 10. A *principal G-bundle* (or *principal homogeneous space*) for G is a scheme $f : Y \to X$ equipped with a left G-action $\rho : G \times_X Y \to Y$ satisfying:

1. The morphism $G \times_X Y \to Y \times_X Y$ sending $(g, y) \mapsto (gy, y)$ is an isomorphism of X-schemes.
2. There exists an étale covering $\{U_i \to X\}$ such that $Y_{U_i} \cong G_{U_i}$ as G-schemes, where G acts on itself by left translation.

Remark. Since $G \to X$ is smooth and Y is étale-locally isomorphic to G, the morphism $f : Y \to X$ is smooth and affine. Note that Y is generally not étale over X unless G is an étale group scheme (e.g., a finite constant group). Similarly, if G is finite then $Y \to X$ is finite.

Similarly to the case of G-torsors of a topos, we define a morphism of principal G-bundles to be a morphism of X-schemes commuting with the G-action. We now have:

Theorem 11. *The morphism sending $Y \mapsto \mathrm{Hom}_X(-, Y)$ is an equivalence of categories from the category of principal G-bundles to the category of G-torsors on $X_{\acute{E}t}$ Similarly, the morphism sending Y to $\mathrm{Hom}_X(-, Y)$ to the category of G-torsors on $X_{\acute{e}t}$ is an equivalence.*

Corollary 12. *There is a natural equivalence $\mathbf{Tors}(X_{\acute{e}t}, G) \cong \mathbf{Tors}(X_{\acute{E}t}, G)$ inducing a bijection of pointed sets $\mathrm{Tors}(X_{\acute{e}t}, G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, G)$ which is an isomorphism of abelian groups if G is abelian. And every G-torsor, in each of the above sites can be realized as a scheme, which is a prinicipal G-bundle.*

Torsors and first cohomology

We use the following standard fact without proof [Mil80, III, §4]. Let G be an abelian sheaf of groups on $X_{\text{ét}}$. Then there is a natural group isomorphism

$$\mathrm{Tors}(X_{\text{ét}}, G) \xrightarrow{\cong} H^1_{\text{ét}}(X, G),$$

and the trivial torsor maps to 0. So an element of $H^1_{\text{ét}}(X, G)$ is the same as an isomorphism class of G-torsors. The Kummer, Artin–Schreier, and Artin–Schreier–Witt classifications (Section A.1, Theorem 85) use this fact whenever they write $[T_a]$ for the class of a torsor in H^1.

Constant Finite Group Torsors

Let G be a finite group, within this section, we denote the associated constant group scheme over X by $\underline{G}$ to maintain a clear distinction between the group as a set and the group as a scheme. In subsequent sections, we shall follow the standard practice of identifying G with its constant group scheme and omit the underline for brevity.

$\underline{G}$ is given by $\coprod_{g \in G} X$ with the action shuffeling the X's according to multiplication.

By following the definitions, one sees that if X be a connected scheme and $f : Y \to X$ is a finite Galois cover with Galois group G. Then, $f : Y \to X$ is a principal $\underline{G}$-bundle. (Recall that a *finite Galois cover* is a finite étale surjection $Y \to X$ with Y connected and such that $G = \mathrm{Aut}(Y/X)$ acts transitively on the geometric points of Y lying over any geometric point of X).

On the otherhand if $f : Y \to X$ is a principal $\underline{G}$-bundle with Y connected, then Y is a finite Galois cover with automorphism group G.

However, not all $\underline{G}$-torsors are connected. If $H \subset G$ is a proper subgroup then any connected finite étale cover $f : Y \to X$ with Galois group H gives rise to a non-connected $\underline{G}$-torsor by looking at the induced $\underline{G}$-torsor $\varphi_*(Y)$ under the inclusion $\varphi : \underline{H} \to \underline{G}$. On the otherhand, if we fix a geometric point $\overline{x} \to X$, then to give an homomorphism $\rho \in \mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X,x), G)$ is equivilant to give a connected pointed Galois cover $(Y, \overline{y}) \to (X, \overline{x})$ with Galois group $H = \rho(\pi_1^{\mathrm{et}}(X,x)) \subset G$. Thus, pushing forward to $\underline{G}$ we get a prinicipal $\underline{G}$-bundle. The choice of a different geometric point $\overline{x'} \to X$ differ the homomorphism by an inner automorphism, thus we have:

Theorem 13. *Let X be a connected scheme and $\overline{x}$ a geometric point of X. Suppose in addition that G is a finite abstract group. Define a map*

$$\mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X,\overline{x}), G)/\mathrm{Inn}(G) \to \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.1}$$

by sending a homomorphism $\rho : \pi_1^{\mathrm{et}}(X,\overline{x}) \to G$ to the principal $\underline{G}$-bundle $\varphi_*(Y)$ where Y is the principal $\underline{\rho(\pi_1^{\mathrm{et}}(X,\overline{x}))}$-bundle obtained above and φ is the inclusion $\rho(\pi_1^{\mathrm{et}}(X,\overline{x})) \hookrightarrow G$. Then, the map is a bijection of pointed sets where the trivial homomorphisms (which is the only element of its $\mathrm{Inn}(G)$-orbit) is the distinguished element of the left hand side.

If G is abelian then $\mathrm{Inn}(G)$ is trivial, and we obtain:

Corollary 14. *Let G be a finite abelian group, X a connected scheme, and $\overline{x}$ a geometric point of X. Then, the map from Theorem 13 induces an isomorphism of abelian groups*

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\acute{e}t}(X,\overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.2}$$

For $G = \Lambda^\times$, the character in Corollary 14 is also the monodromy character of the associated rank-one Λ-local system. In particular, the torsor and the local system have the same restrictions to inertia and to the upper ramification groups. Thus, one has ramification bounded by a given modulus, or is tamely ramified along a given divisor, if and only if the other does.

Quotients and Decomposition of Torsors

Let $\mathcal{H}$ be a sheaf of groups acting on a sheaf $\mathcal{P}$ on a site $\mathscr{C}$. The **quotient sheaf** $\mathcal{P}/\mathcal{H}$ is the coequalizer in $\mathrm{Sh}(\mathscr{C})$

$$\mathcal{H} \times \mathcal{P} \rightrightarrows \mathcal{P} \longrightarrow \mathcal{P}/\mathcal{H},$$

where the parallel arrows are the action and the projection. Equivalently, it is the sheaf associated with the presheaf of orbits $U \mapsto \mathcal{H}(U)\backslash\mathcal{P}(U)$. We use the notation $\mathcal{P}/\mathcal{H}$ even though our torsor actions are written on the left. In particular, the quotient always exists as a sheaf. It is characterized by the following universal property: for every sheaf $\mathcal{F}$, regarded as an $\mathcal{H}$-sheaf with the trivial action, composition with $\mathcal{P} \to \mathcal{P}/\mathcal{H}$ identifies morphisms $\mathcal{P}/\mathcal{H} \to \mathcal{F}$ with the $\mathcal{H}$-equivariant morphisms $\mathcal{P} \to \mathcal{F}$, equivalently with morphisms f satisfying $f(hp) = f(p)$ on local sections. A quotient sheaf need not in general be representable by a scheme; representability in the finite-torsor situation used here is part of the next proposition.

Let G be a finite abelian group, let $H \subseteq G$ be a subgroup, and let $\pi : G \to G/H$ be the quotient homomorphism.

Proposition 15. *Let $\mathcal{P} \to X$ be a G-torsor in $X_{\acute{e}t}$. The quotient sheaf $\mathcal{P}/H$ is represented by a finite étale X-scheme, and there is a canonical isomorphism of G/H-torsors*

$$\mathcal{P}/H \xrightarrow{\sim} \pi_*\mathcal{P} = (G/H) \times^G \mathcal{P}.$$

Consequently, $\mathcal{P} \to X$ admits a canonical factorization

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}/H \xrightarrow{\psi} X,$$

where ϕ is an H-torsor and ψ is a G/H-torsor. This factorization is unique up to a unique isomorphism commuting with the projections.

Proof. The canonical morphism

$$\mathcal{P} \longrightarrow \pi_*\mathcal{P}, \qquad p \longmapsto [1, p],$$

is H-invariant because $\pi(H) = 1$. The universal property of the quotient sheaf therefore gives a canonical morphism

$$\mathcal{P}/H \longrightarrow \pi_*\mathcal{P}.$$

Choose an étale cover $U \to X$ that trivializes $\mathcal{P}$. Over U this morphism identifies with the identity of $(G/H) \times U$, since

$$(G \times U)/H \cong (G/H) \times U \qquad \text{and} \qquad \pi_*\mathcal{P}|_U \cong (G/H) \times U.$$

It is therefore an isomorphism of sheaves. The contracted product $\pi_*\mathcal{P}$ is a G/H-torsor and, by Theorem 11, is represented by a finite étale X-scheme. Hence the same is true of $\mathcal{P}/H$.

Under the same local trivialization, the morphism $\mathcal{P} \to \mathcal{P}/H$ is the product of the quotient map with the identity,

$$G \times U \longrightarrow (G/H) \times U,$$

and is thus an H-torsor. Finally, suppose $\mathcal{P} \to \mathcal{Q} \to X$ is another such factorization. Since an H-torsor is locally split, every H-invariant morphism out of $\mathcal{P}$ descends uniquely through $\mathcal{Q}$. Thus $\mathcal{Q}$ satisfies the same universal property as $\mathcal{P}/H$, which gives the unique isomorphism compatible with the projections. □

1.3 Symmetric Powers of Schemes and Torsors

This section reviews the construction of quotients for schemes and torsors under finite group actions, specifically focusing on symmetric powers. To ensure these quotients exist as schemes, we utilize the framework of admissible actions from [SGA1]. Our treatment here closely follows the exposition in [Gui19] The definitions and results presented below are adapted from their work, the difference being that we ask Γ to act on the left, to make the later construction of symmetric powers more natural. This foundation provides the necessary criteria for admissibility and base change required to define the symmetric powers of a scheme X and a G-torsor $\mathcal{P}$ over X.

Let S be a scheme.

Definition 16 (([SGA1], V.1.7).)**.**

- Let T be an object of a category $\mathcal{C}$ endowed with a right action of a group Γ. We say that **the quotient T/Γ exists** in $\mathcal{C}$ if the covariant functor

$$\begin{aligned} \mathcal{C} &\to \text{Sets} \\ U &\mapsto \text{Hom}_{\mathcal{C}}(T, U)^{\Gamma} \end{aligned}$$

  is representable by an object of $\mathcal{C}$.

- Let T be an S-scheme. An action of a finite group Γ on T is **admissible** if there exists an affine Γ-invariant morphism $f : T \to T'$ such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^\Gamma$.

Proposition 17. *The following holds:*

1. ***([SGA1] V.1.3).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ. If $f : T \to T'$ is an affine Γ-invariant morphism such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^\Gamma$, then the quotient T/Γ exists and is isomorphic to T'.*

2. ***([SGA1], V.1.8).*** *Let T be an S-scheme endowed with a right action of a finite group Γ. Then, the action of Γ on T is admissible if and only if T is covered by Γ-invariant affine open subsets.*

3. ***([SGA1], V.1.9).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ, and let S' be a flat S-scheme. Then, the action of Γ on the S'-scheme $T \times_S S'$ is admissible, and the canonical morphism*

$$(T \times_S S')/\Gamma \to (T/\Gamma) \times_S S'$$

is an isomorphism.

Now let X be an S-scheme and let $d \geq 0$ be an integer. The group S_d of permutations of $[\![1, d]\!]$ acts on the right on the S-scheme $X^{\times_S d} = X \times_S \cdots \times_S X$ by the formula

$$(x_i)_{i\in[\![1,d]\!]} \cdot \sigma = (x_{\sigma(i)})_{i\in[\![1,d]\!]}.$$

Proposition 18 ([Gui19] Proposition 2.27)**.** *If X is a scheme, Zariski locally quasi-projective over S, then the right action of the symmetric group S_d on the d-fold fiber product $X^{\times_S d}$ is admissible. Consequently, the quotient $\mathrm{Sym}^d_S(X) = X^{\times_S d}/S_d$ exists as a scheme over S.*

Definition 19. Under the hypotheses of the proposition above, we define the **relative symmetric product** of X over S of degree d as the quotient

$$\mathrm{Sym}^d_S(X) \coloneqq X^{\times_S d}/S_d.$$

When the base scheme S is clear from the context, we shall denote this quotient by $X^{(d)}$.

Guignard shows that when $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ then $\mathrm{Sym}^d_S(X)$ is representable by an affine S-scheme (See [Gui19, Remark 2.28]).

Proposition 20 ([Gui19] Proposition 2.28)**.** *If X is flat and Zariski-locally quasi-projective over S, then $Sym^d_S(X)$ is flat over S. Moreover, for any S-scheme S', the canonical morphism*

$$Sym^d_{S'}(X \times_S S') \to Sym^d_S(X) \times_S S'$$

is an isomorphism.

Geometric Properties of Symmetric Powers

We record the geometric properties of symmetric powers that will be used in both main parts of the thesis.

Theorem 21. *Let X and Y be integral schemes over a field k. If X is geometrically integral, then $X \times_k Y$ is integral. If both X and Y are geometrically integral, then $X \times_k Y$ is geometrically integral.*

Proof. By [Stacks, Tag 038K], a geometrically integral scheme is both geometrically irreducible and geometrically reduced. Since X is geometrically irreducible, the projection

$$X \times_k Y \longrightarrow Y$$

induces a bijection on irreducible components by [Stacks, Tag 038F]. Because Y is integral, it follows that $X \times_k Y$ is irreducible. Moreover, X is geometrically reduced and Y is reduced, so $X \times_k Y$ is reduced by [Stacks, Tag 035Z]. Hence it is integral.

If both X and Y are geometrically integral, the same argument applies after every field extension k'/k. Thus

$$(X \times_k Y)_{k'} \cong X_{k'} \times_{k'} Y_{k'}$$

is integral for every k'/k, and therefore $X \times_k Y$ is geometrically integral. □

Proposition 22. *Let $f : X \to S$ be a smooth, Zariski-locally quasi-projective morphism of relative dimension one. Then the relative symmetric power $X_S^{(d)} = \mathrm{Sym}_S^d(X)$ is smooth over S of relative dimension d.*

Proof. Since $f : X \to S$ is smooth, the d-fold product X_S^d is smooth over S. By Proposition 20, the quotient $X_S^{(d)}$ is flat over S. Smoothness is a fiberwise property for flat morphisms of finite presentation (cf. [Stacks, Tag 01V8]); thus, it suffices to verify the smoothness of the fibers. For every $s \in S$, the fiber of the symmetric power is given by:

$$(\mathrm{Sym}_S^d(X))_s \cong \mathrm{Sym}_{\kappa(s)}^d(X_s)$$

Consequently, we may reduce to the case where $S = \mathrm{Spec}(k)$ for a field k. Since smoothness is preserved under base change to the algebraic closure, we may further assume $k = \bar{k}$. Let $z = \sum_{i=1}^r d_i \cdot x_i \in X^{(d)}$ be a point represented by an effective cycle of degree d, where the points $x_i \in X(k)$ are distinct and $\sum d_i = d$. Choose pairwise disjoint Zariski-open subsets $U_i \subset X$ such that $x_i \in U_i$. Let $W \subset X^{(d)}$ be the open subset consisting of cycles whose support is contained in $\bigcup U_i$ and which meet each U_i with degree exactly d_i. There is a canonical isomorphism:

$$W \cong \prod_{i=1}^r U_i^{(d_i)}$$

Thus, it suffices to show that each $U_i^{(d_i)}$ is smooth of dimension d_i at the point $d_i \cdot x_i$.

We may therefore restrict our attention to the "worst" case: the diagonal point $z = d \cdot x$. Let $R = \mathcal{O}_{X,x}$ be the local ring of the curve at x. Since X is a smooth curve over k, R is a regular local ring of dimension 1, i.e., a discrete valuation ring. Its completion is $\hat{R} \cong k[[t]]$. The completed local ring of the product X^d at the point $(x, \ldots, x)$ is:

$$\hat{\mathcal{O}}_{X^d,(x,\ldots,x)} \cong \hat{R} \hat{\otimes}_k \ldots \hat{\otimes}_k \hat{R} \cong k[[t_1, \ldots, t_d]]$$

where t_i is the uniformizer for the i-th copy of X. The symmetric group S_d acts on this power series ring by permuting the variables t_i. By the *Fundamental Theorem of Symmetric Polynomials*, the ring of invariants is:

$$\big(k[[t_1,\dots,t_d]]\big)^{S^d} = k[[e_1,\dots,e_d]]$$

where e_j is the j-th elementary symmetric power series in the variables t_i.

The ring $k[[e_1,\dots,e_d]]$ is a regular local ring of dimension d. Regularity of a Noetherian local ring is equivalent to regularity of its completion by [Stacks, Tag 07NY]. Hence $\mathcal{O}_{X^{(d)},z}$ is a regular local ring of dimension d. Since k is algebraically closed and $X^{(d)}$ is locally of finite type over k, regularity at z is equivalent to smoothness at z. This completes the proof. □

Corollary 23. *Let C be a smooth, projective, geometrically connected curve over a field k. For every $d \geq 0$, the scheme $C^{(d)}$ is smooth and geometrically integral. Moreover, every finite product of symmetric powers of C is geometrically integral.*

Proof. The smoothness follows from Proposition 22. The curve C is geometrically integral by [Stacks, Tag 0366]. After any field extension, C^d is integral and its image under the quotient map $C^d \to C^{(d)}$ is irreducible. Since $C^{(d)}$ is smooth, it is geometrically reduced, and hence geometrically integral. The assertion about finite products now follows inductively from Theorem 21. □

A Symmetric-Polynomial Calculation

Let $e_1,\dots,e_d$ be the elementary symmetric polynomials in $x_1,\dots,x_d$, put $y_i = x_i^{-1}$, and write

$$\kappa := k(e_1/e_d,\dots,e_{d-1}/e_d).$$

Lemma 24 (Symmetric regularity)**.** *If $\Phi \in k[y_1,\dots,y_d]^{S_d}$ and every monomial of Φ has total degree strictly less than d, then*

$$\Phi \in k(e_1/e_d,\dots,e_{d-1}/e_d) = \kappa.$$

Proof. The identity

$$e_j(y_1,\dots,y_d) = \frac{e_{d-j}(x_1,\dots,x_d)}{e_d(x_1,\dots,x_d)}$$

puts $e_1(y),\dots,e_{d-1}(y)$ in κ. By the fundamental theorem of symmetric polynomials, each homogeneous part of Φ is a polynomial in $e_1(y),\dots,e_d(y)$ with their natural degrees. Since every homogeneous part has degree $< d$, the generator $e_d(y)$ cannot occur. Hence $\Phi \in \kappa \subset \kappa[[e_d]]$. □

Symmetric Powers of Torsors

Let G be a finite abelian group, let $\mathcal{P}$ be a G-torsor over an S-scheme X in $S_{\text{Ét}}$.

Proposition 25 ([[SGA1]], IX.5.8)**.** *Assume that $\mathcal{P}$ and X are endowed with right actions from a finite group Γ such that the morphism $\mathcal{P} \to X$ is Γ-equivariant, and that the following properties hold:*

(a) The right Γ-action on $\mathcal{P}$ commutes with the left G-action.

(b) The right Γ-action on X is admissible, and the quotient morphism $X \to X/\Gamma$ is finite.

(c) For any geometric point $\bar{x}$ of X, the action of the stabilizer $\Gamma_{\bar{x}}$ of $\bar{x}$ in Γ on the fiber $\mathcal{P}_{\bar{x}}$ of $\mathcal{P}$ at $\bar{x}$ is trivial.

Then the action of Γ on $\mathcal{P}$ is admissible, and $\mathcal{P}/\Gamma$ is a G-torsor over X/Γ in $S_{Ét}$.

By Theorem 11, $\mathcal{P}$ is representable by a finite étale X-scheme. For each $i \in [\![1, d]\!]$ let $p_i : X^{\times_S d} \to X$ be the projection on i-th factor, and let us consider the G-torsor

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} = K_G \backslash \mathcal{P}^{\times_S d}$$

over $X^{\times_S d}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \to G$ (the dependence on d is suppressed in the notation). Equivalently, this is the contraction of the G^d-torsor $p_1^{-1}\mathcal{P} \times \cdots \times p_d^{-1}\mathcal{P}$ along the multiplication map $m : G^d \to G$. Under the identification $G^d/K_G \simeq G$ induced by m, the residual G-action is given by

$$g \cdot [(u_1, \ldots, u_d)] = [(g_1 u_1, \ldots, g_d u_d)],$$

where $(g_1, \ldots, g_d)$ is any lift of g, so that $g_1 \cdots g_d = g$. This is independent of the chosen lift because any two lifts differ by an element of K_G. The object $K_G \backslash \mathcal{P}^{\times_S d}$ of $S_{\text{Ét}}$ is too representable by an S-scheme which is finite étale over $X^{\times_S d}$. The group S_d acts on the right on $K_G \backslash \mathcal{P}^{\times_S d}$ by the formula

$$(p_i)_{i \in [\![1,d]\!]} \cdot \sigma = (p_{\sigma(i)})_{i \in [\![1,d]\!]}.$$

This action of S_d commutes with the left action of G on $K_G \backslash \mathcal{P}^{\times_S d}$.

Proposition 26 ([Gui19] Proposition 2.32.)**.** *If X is Zariski-locally quasi-projective on S, then the right action of S_d on $K_G \backslash \mathcal{P}^{\times_S d}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G \backslash \mathcal{P}^{\times_S d}$ by S_d exists as a scheme over S. Moreover, the canonical morphism $\mathcal{P}^{[d]} \to \mathrm{Sym}^d_S(X)$ is a G-torsor, and the morphism*

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} \to r^{-1}\mathcal{P}^{[d]}$$

where $r : X^{\times_S d} \to \mathrm{Sym}^d_S(X)$ is the canonical projection, is an isomorphism of G-torsors over $X^{\times_S d}$.

Definition 27 (Symmetric powers of rank-one local systems)**.** Let Λ be a finite commutative ring, let $G = \Lambda^\times$, and let $\mathcal{F}$ be a rank-one Λ-local system on X. If $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$, we define

$$\mathcal{F}^{[d]} := \mathcal{P}^{[d]} \times^G \Lambda$$

on $X^{(d)}$. Thus there is a canonical identification

$$\mathrm{Fr}\big(\mathcal{F}^{[d]}\big) \cong \mathcal{P}^{[d]}.$$

By Proposition 26, its pullback along $r : X^{\times_S d} \to X^{(d)}$ satisfies

$$r^{-1}\mathcal{F}^{[d]} \cong p_1^{-1}\mathcal{F} \otimes_\Lambda \cdots \otimes_\Lambda p_d^{-1}\mathcal{F}.$$

Thus this definition agrees with the usual descent construction of the symmetric power of a rank-one local system.

Proposition 28 (Semidirect-product description)**.** *Let $\Xi_G := K_G \rtimes S_d$, where S_d acts on K_G by permuting coordinates. Then Ξ_G acts on $\mathcal{P}^{\times_S d}$ by*

$$((g_1, \ldots, g_d), \sigma) \cdot (p_1, \ldots, p_d) = (g_1 p_{\sigma^{-1}(1)}, \ldots, g_d p_{\sigma^{-1}(d)}),$$

and there is a canonical isomorphism

$$\mathcal{P}^{[d]} \cong \Xi_G \backslash \mathcal{P}^{\times_S d}.$$

Moreover, the formation of $\mathcal{P}^{[d]}$ commutes with flat base change on S. If, in addition, X is flat over S, then it commutes with arbitrary base change; see [Gui19, Proposition 2.33].

Proof. Because G is abelian, the condition $\prod_{i=1}^{d} g_i = 1$ defining K_G is invariant under permutation of the coordinates. Thus the S_d-action normalizes K_G and defines the semidirect product Ξ_G. By Proposition 26,

$$\mathcal{P}^{[d]} = (K_G \backslash \mathcal{P}^{\times_S d})/S_d.$$

The universal properties of the two quotients identify this iterated quotient canonically with $\Xi_G \backslash \mathcal{P}^{\times_S d}$. Formation of the torsor quotient $K_G \backslash \mathcal{P}^{\times_S d}$ commutes with base change, and the final admissible S_d-quotient commutes with flat base change by [SGA1, p. V.1.9]. If X is flat over S, compatibility with arbitrary base change is [Gui19, Proposition 2.33]. □

We record one further compatibility of symmetric powers. It will allow us later to replace a torsor by the connected torsor under its monodromy subgroup.

Lemma 29 (Symmetric powers and induced torsors)**.** *Let $\iota : H \hookrightarrow G$ be an injection of finite abelian groups, and let $\mathcal{Q}$ be an H-torsor on U. There is a canonical isomorphism of G-torsors over $U^{(d)}$*

$$(\iota_* \mathcal{Q})^{[d]} \cong \iota_* (\mathcal{Q}^{[d]}).$$

Proof. Let

$$j : \mathcal{Q} \longrightarrow \iota_* \mathcal{Q} = G \times^H \mathcal{Q}$$

be the canonical morphism ($q \mapsto [1, q]$). It is H-equivariant, where H acts on the target through ι. Applying j to each factor gives

$$j^{\times d} : \mathcal{Q}^{\times d} \longrightarrow (\iota_* \mathcal{Q})^{\times d}, \qquad (q_1, \ldots, q_d) \longmapsto (j(q_1), \ldots, j(q_d)).$$

This map is compatible with permutations of the factors. Moreover, for $(h_1, \ldots, h_d) \in H^d$,

$$j^{\times d}(h_1 q_1, \ldots, h_d q_d) = (\iota(h_1) j(q_1), \ldots, \iota(h_d) j(q_d)).$$

Put $K_H := \ker(H^d \to H)$. If $(h_1, \ldots, h_d) \in K_H$, then

$$\iota(h_1) \cdots \iota(h_d) = \iota(h_1 \cdots h_d) = 1,$$

so $(\iota(h_1), \ldots, \iota(h_d)) \in K_G$. Hence $j^{\times d}$ descends to a morphism

$$\mathcal{Q}^{[d]} = (K_H \rtimes S_d) \backslash \mathcal{Q}^{\times d} \longrightarrow (K_G \rtimes S_d) \backslash (\iota_* \mathcal{Q})^{\times d} = (\iota_* \mathcal{Q})^{[d]}.$$

The induced map between the residual groups $H^d/K_H \simeq H$ and $G^d/K_G \simeq G$ is ι. Therefore this morphism is H-equivariant, where H acts on the target by $h \cdot x := \iota(h) \cdot x$. By the universal property of induction, this extends uniquely to a G-equivariant morphism

$$\iota_* (\mathcal{Q}^{[d]}) \longrightarrow (\iota_* \mathcal{Q})^{[d]}.$$

Both sides are G-torsors over $U^{(d)}$, so Proposition 5 shows that it is an isomorphism. □

Additivity of Symmetric Powers

For $d \geq 0$, write

$$X_S^d := \underbrace{X \times_S \cdots \times_S X}_{d \text{ factors}}, \qquad X_S^0 := S.$$

The construction of symmetric powers is compatible with a natural addition law. For any integers $d_1, d_2 \geq 0$, concatenating the factors gives a canonical isomorphism

$$c_{d_1,d_2} : X_S^{d_1} \times_S X_S^{d_2} \xrightarrow{\sim} X_S^{d_1+d_2}.$$

It is equivariant for the inclusion $S_{d_1} \times S_{d_2} \hookrightarrow S_{d_1+d_2}$ and therefore induces the canonical **addition morphism**

$$+_{d_1,d_2} : X^{(d_1)} \times_S X^{(d_2)} \longrightarrow X^{(d_1+d_2)}.$$

Proposition 30. *Let $\mathcal{P}$ be a G-torsor over X. There is a canonical isomorphism of G-torsors over $X^{(d_1)} \times_S X^{(d_2)}$:*

$$+_{d_1,d_2}^{-1}\left(\mathcal{P}^{[d_1+d_2]}\right) \cong r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}.$$

Proof. Let $\pi = r_{d_1} \times r_{d_2}$. By the commutativity of the following diagram

$$\begin{array}{ccc} X_S^{d_1} \times_S X_S^{d_2} & \xrightarrow[\sim]{c_{d_1,d_2}} & X_S^{d_1+d_2} \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle r_{d_1+d_2}} \\ X^{(d_1)} \times_S X^{(d_2)} & \xrightarrow{+_{d_1,d_2}} & X^{(d_1+d_2)} \end{array}$$

and applying Proposition 26, we obtain canonical isomorphisms:

$$\pi^{-1}\left(+_{d_1,d_2}^{-1}\mathcal{P}^{[d_1+d_2]}\right) \cong r_{d_1+d_2}^{-1}\mathcal{P}^{[d_1+d_2]} \cong \bigotimes_{i=1}^{d_1+d_2,\,G} p_i^{-1}\mathcal{P}$$

and

$$\pi^{-1}(r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}) \cong \left(\bigotimes_{i=1}^{d_1,\,G} p_i^{-1}\mathcal{P}\right) \otimes^G \left(\bigotimes_{j=d_1+1}^{d_1+d_2,\,G} p_j^{-1}\mathcal{P}\right).$$

Since both pullbacks identify with

$$\bigotimes_{k=1}^{d_1+d_2,\,G} p_k^{-1}\mathcal{P}$$

in an $S_{d_1} \times S_{d_2}$-equivariant manner, the isomorphism descends to the quotient $X^{(d_1)} \times_S X^{(d_2)}$ by Proposition 20. □

1.4 Basic Geometric Lemmas

We conclude with two elementary facts about codimension-one points and blowups. They are placed here because they enter the geometric arguments in both later chapters.

Proposition 31. *Let $f : X \to Y$ be a finite flat morphism between integral schemes of finite type over a field k. If $Z \subset X$ is a prime divisor with generic point η_Z, then $f(Z) \subset Y$ is a prime divisor, and its generic point is $f(\eta_Z)$.*

Proof. f is finite hence proper hence closed so $f(Z)$ is closed subset of Y, it is irreducible as the image of an irreducible. Since $Z = \overline{\{\eta_Z\}}$ we get:

$$\{f(\eta_Z)\} \subset f(Z) = f(\overline{\{\eta_Z\}}) \subseteq \overline{f(\{\eta_Z\})} = \overline{\{f(\eta_Z)\}}$$

And since $f(Z)$ is closed we get $f(Z) = \overline{\{f(\eta_Z)\}}$.

For flat map of integral schemes we have for every $x \in X$, $y = f(x)$ the dimension formula:

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X_y,x})$$

And since $\dim(\mathcal{O}_{X_y,x}) = 0$ we get $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$ concluding that $f(Z)$ is a prime divisor as well. □

Blowups

Theorem 32 ([Stacks, Tag 0805])**.** *Let $X_1 \to X_2$ be a flat morphism of schemes, let $Z_2 \subset X_2$ be a closed subscheme, and let Z_1 be its inverse image in X_1. If X_i' is the blowup of X_i along Z_i, then the diagram*

$$\begin{array}{ccc} X_1' & \longrightarrow & X_2' \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & X_2 \end{array}$$

is cartesian.

Proposition 33. *Let X be a smooth k-scheme of pure dimension $d \geq 1$, let $z \in X(k)$, and let*

$$\pi : \widetilde{X} = \mathrm{Bl}_z(X) \longrightarrow X$$

be the blowup at z. Then the exceptional divisor $E = \pi^{-1}(z)$ is isomorphic to $\mathbb{P}_k^{d-1}$. In particular, E is irreducible and has codimension 1 in $\widetilde{X}$. Consequently, it has a unique generic point η_E, and $\overline{\{\eta_E\}} = E$.

Proof. Because X is smooth of dimension d at the k-rational point z, the étale local description of smooth morphisms [Stacks, Tag 054L] shows that, after replacing X by a sufficiently small open neighborhood of z, there is an étale morphism

$$f : X \longrightarrow \mathbb{A}_k^d$$

that sends z to the origin 0. We may shrink this neighborhood further so that z is the scheme-theoretic inverse image of 0. Since an étale morphism is flat, Theorem 32 reduces the assertion to the blowup of $\mathbb{A}_k^d$ at the origin.

Write $x_1, \ldots, x_d$ for the affine coordinates on $\mathbb{A}_k^d$ and $[u_1 : \cdots : u_d]$ for the homogeneous coordinates on $\mathbb{P}_k^{d-1}$. The blowup of the origin is the closed subscheme

$$\mathrm{Bl}_0(\mathbb{A}_k^d) = \left\{((x_1, \ldots, x_d), [u_1 : \cdots : u_d]) \;\middle|\; x_i u_j = x_j u_i \text{ for all } i, j\right\} \subset \mathbb{A}_k^d \times_k \mathbb{P}_k^{d-1}.$$

Over the origin all the x_i vanish, so the displayed equations impose no condition on $[u_1 : \cdots : u_d]$. Hence the exceptional fiber is

$$\{0\} \times_k \mathbb{P}_k^{d-1} \cong \mathbb{P}_k^{d-1}.$$

It is therefore irreducible of dimension $d-1$. The blowup has dimension d (for example, each standard chart $u_i \neq 0$ has d affine coordinates), so the exceptional divisor has codimension 1. □

Theorem 34 ([Stacks, Tag 01OF], Lemma 31.33.9)**.** *If X is integral and $Z \subsetneq X$ is a proper closed subscheme, then the blowup $\mathrm{Bl}_Z(X)$ is integral.*

Chapter 2

Ramification After Blowup Is Tame

Throughout this chapter, let C be a smooth, projective, geometrically connected curve over k. Fix a modulus

$$\mathfrak{m} = \sum_i n_i P_i$$

on C, and set $U = C \setminus \mathfrak{m}$.

For each single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, set $d_\mathfrak{n} = \deg(\mathfrak{n})$. We identify the effective divisor $\mathfrak{n}$ with the corresponding k-rational closed point of $C^{(d_\mathfrak{n})} = \mathrm{Sym}_k^{d_\mathfrak{n}}(C)$.

Let

$$\pi_\mathfrak{n} \colon X_\mathfrak{n} := \mathrm{Bl}_\mathfrak{n}\big(C^{(d_\mathfrak{n})}\big) \longrightarrow C^{(d_\mathfrak{n})}$$

be the blowup at this point. We denote its exceptional divisor by

$$E_\mathfrak{n} := X_\mathfrak{n} \times_{C^{(d_\mathfrak{n})}} \{\mathfrak{n}\}.$$

By Propositions 22 and 33, $E_\mathfrak{n} \cong \mathbb{P}_k^{d_\mathfrak{n}-1}$; in particular, it is irreducible and has codimension 1 in $X_\mathfrak{n}$. We denote its unique generic point by $\eta_\mathfrak{n}$, so that $\overline{\{\eta_\mathfrak{n}\}} = E_\mathfrak{n}$. Thus we have the Cartesian diagram

$$\begin{array}{ccc} E_\mathfrak{n} = \overline{\{\eta_\mathfrak{n}\}} & \hookrightarrow & X_\mathfrak{n} \\ \downarrow & & \downarrow{\scriptstyle \pi_\mathfrak{n}} \\ \{\mathfrak{n}\} & \hookrightarrow & C^{(d_\mathfrak{n})}. \end{array} \tag{2.1}$$

The purpose of this chapter is to prove Theorem 3. Let $\mathcal{F}$ be the rank-one Λ-local system appearing there and put $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$. The theorem becomes the following torsor statement.[1]

[1] Here one uses $\mathrm{Fr}(\mathcal{F}^{[d_\mathfrak{n}]}) \cong \mathcal{P}^{[d_\mathfrak{n}]}$ and the fact that a local system and its frame torsor have the same inertia character, as recorded in a footnote in Chapter 1.

Theorem 35. *Assume that k is algebraically closed. Let G be a finite abelian group, and let $\mathcal{P} \to U$ be a G-torsor whose ramification is bounded by $\mathfrak{m}$. For every single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, the symmetric-power torsor $\mathcal{P}^{[d_\mathfrak{n}]}$ is tamely ramified at $\eta_\mathfrak{n}$.*

After recalling the necessary local ramification theory, we make two reductions. First, the general finite abelian case is reduced to cyclic groups. Second, formal–local invariance reduces the geometric calculation on C to a calculation on $\mathbb{G}_m \subset \mathbb{P}^1$. These reductions motivate the Kummer, Artin–Schreier, and Artin–Schreier–Witt theories used in the cyclic calculations that follow.

2.1 Ramification of Discrete Valuations

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [Stacks, Tag 0EXQ] and [Ser79].

Definition 36 ([Stacks, Tag 09E4])**.** We say that $A \to B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K, let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L. Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i. We have

$$[L : K] = \sum\nolimits_{i=1,\ldots,n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n = 1$.

Definition 37 ([Stacks, Tag 09E9])**.** Let A be a discrete valuation ring with fraction field K. Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i = 1, \ldots, n$ as above, we say the extension L/K is

1. unramified with respect to A if $e_i = 1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i,
2. tamely ramified with respect to A if, for every i, the residue field extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable and the ramification index e_i is prime to the residue characteristic. Thus, when

$\mathrm{char}(\kappa_A) = 0$, the condition on e_i is vacuous (and the residue field extensions are automatically separable), whereas when $\mathrm{char}(\kappa_A) = p > 0$, each e_i must be prime to p, and

3. totally ramified with respect to A if $n = 1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

We will use two standard stability properties of tame ramification. First, ramification indices and residue-field extensions are unchanged by passage to the completion; in particular, tameness can be checked after completion [Ser79, Chapter II, §2]. Second, finite tame extensions of a complete discretely valued field are closed under taking Galois closures, finite composita, and subextensions [Ser79, Chapter IV, §2].

For the remainder of this section, assume that A is a discrete valuation ring of equal characteristic $p > 0$, complete with respect to its maximal-ideal-adic topology (cf. [Stacks, Tag 00IE] and [Stacks, Tag 0324]), with uniformizer π and perfect residue field κ. Thus, $\mathrm{char}(A) = \mathrm{char}(\kappa) = p$. The Cohen structure theorem ([Stacks, Tag 032A]; see also [Stacks, Tag 0C0S]) provides a coefficient field $k \subset A$ mapping isomorphically onto κ. Consequently, $A \cong k[[\pi]] \cong k[[t]]$, and its fraction field is $K \cong k((\pi))$. Let L/K be a finite Galois extension with Galois group G. The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

The *lower numbering ramification filtration* of G, denoted by $(G_i)_{i \geq -1}$, is defined as:

$$G_i = \{\sigma \in G \mid v_B(\sigma(x) - x) \geq i + 1 \text{ for all } x \in B\}$$

where v_B is the valuation on L associated with B. In this indexing, $G_{-1} = G$ and G_0 corresponds to the *inertia group* of the extension L/K. Note that L/K is unramified if and only if $G_0 = \{1\}$, and it is tamely ramified if and only if $G_1 = \{1\}$.

A standard result in the theory of local fields (complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B. Specifically, if we define the function

$$i_K^L(\sigma) = v_B(\sigma(\Pi) - \Pi)$$

for $\sigma \in G$, then $G_i = \{\sigma \in G \mid i_K^L(\sigma) \geq i + 1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i.

Consider a tower of fields $K \subset E \subset L$, and let $H = \mathrm{Gal}(L/E) \subset G$. The lower numbering is compatible with subgroups:

$$G_i \cap H = H_i \quad \text{for all } i \geq -1$$

which follows from the identity $i_E^L = i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G, the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_iH/H = (G/H)_j$, where the index j is given by the formula:

$$j = \frac{1}{e_{L/E}} \sum_{\tau \in H} \min(i_K^L(\tau), i + 1) - 1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1, \infty) \to [-1, \infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma \in G} \min(i_K^L(\sigma), i + 1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^i H/H = (G/H)^i$$

for all $i \geq -1$.

We record, for orientation, the structure of the inertia group and of the graded pieces of the filtration; these facts are not used in the sequel, but they explain the dichotomy — prime-to-p versus p-power — that shapes the whole chapter.

Proposition 38 ([Ser79], IV Corollary 3)**.** *The quotients G_i/G_{i+1}, $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p. The group G_1 is a p-group.*

Proposition 39 ([Ser79], IV Corollary 4)**.** *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p.*

2.2 Ramification of G-Torsors

Generally speaking, assume we are given a locally Noetherian scheme X, a dense open $U \subset X$, a finite étale morphism $f : Y \to U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \mathrm{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{array}{ccc} \mathrm{Spec}(K_\xi) & \longrightarrow & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathcal{O}_{X,\xi}) & \longrightarrow & X \end{array}$$

of schemes. In particular, $Y \times_U \mathrm{Spec}(K_\xi)$ is the spectrum of a finite étale K_ξ-algebra L_ξ. Write its canonical decomposition into fields as

$$L_\xi \cong \prod_{j=1}^{r} L_{\xi,j},$$

where each $L_{\xi,j}/K_\xi$ is a finite separable field extension.

Definition 40. We say:

1. Y is *unramified* over X at (Z, ξ) resp. Y is *tamely ramified* over X at (Z, ξ) if every field extension $L_{\xi,j}/K_\xi$ is unramified, resp. tamely ramified, with respect to $\mathcal{O}_{X,\xi}$ (Definition 37)
2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1, if the preceding condition holds for every (Z, ξ) as above.

Now back to our original assumptions, C is a projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \to U$ is actually a G-torsor $\mathcal{P} \to U$ in $U_{\text{Ét}}$ for G a finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\mathrm{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction is a finite étale scheme over the field $\widehat{K}_P$ and therefore, by [Stacks, Tag 02GL], decomposes into a finite disjoint union of spectra of finite separable field extensions:

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \coprod_{i=1}^{N} \mathrm{Spec}(F_i).$$

These extensions have the following properties:

1. The extensions $F_i/\widehat{K}_P$ are mutually isomorphic and Galois. Since $\widehat{K}_P$ is complete, every F_i is complete for the unique extension of the discrete valuation on $\widehat{K}_P$ [Ser79, Chapter II, §2, Proposition 3].
2. If F denotes any one of the fields F_i, then the Galois group $H = \mathrm{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of the corresponding connected component of the pullback $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$, which is a subgroup of G.
3. The number of components is $N = [G : H]$.

Definition 41. We say that the extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. Here H^r denotes the upper-numbering ramification group of $H = \mathrm{Gal}(F/\widehat{K}_P)$ associated with the extension $F/\widehat{K}_P$; it is not a filtration inherited from the ambient group G. This condition is independent of the chosen connected component of $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 42. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. We say the G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for each point P_i, the local restriction $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i.

Fix an algebraic closure $\bar{k}$ of k and write $\bar{s} = \mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$ for the corresponding geometric point.

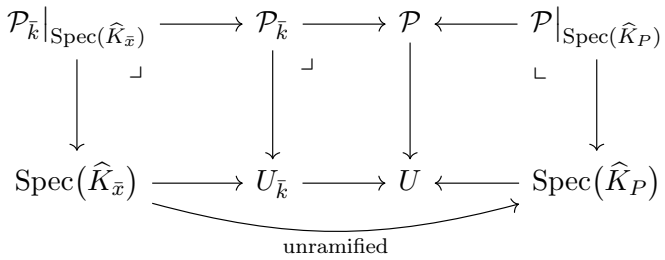
Definition 43 ([Gui19, Definition 5.2])**.** Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. The G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for every geometric point $\bar{x}$ of $\mathfrak{m}$ with image $\bar{s}$ in $\mathrm{Spec}(k)$, the restriction of $\mathcal{P}_{\bar{k}}$ to the punctured formal neighborhood

$$\mathrm{Spec}\big(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}\big) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{k}}$ at $\bar{x}$ in the sense of Definition 41.

To compare the two definitions, fix a point $P = P_i$ in the support of $\mathfrak{m}$, let $\bar{x}$ be a geometric point of $C_{\bar{k}}$ lying over P, and abbreviate $\widehat{K}_{\bar{x}} := \mathrm{Frac}\big(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}\big)$. Since $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}$ is a complete discrete valuation ring whose closed point lies over $\bar{x} \notin U_{\bar{k}}$ while its generic point lies over the generic point of $C_{\bar{k}}$, the punctured formal neighborhood of Definition 43 is $\mathrm{Spec}(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}) \times_{C_{\bar{k}}} U_{\bar{k}} \cong \mathrm{Spec}(\widehat{K}_{\bar{x}})$. The local restriction of Definition 42 (right-hand column) and that of Definition 43 (left-hand column) then

fit into a single commutative diagram



in which every square is cartesian. Since cartesian squares compose, the outer rectangle exhibits

$$\mathcal{P}_{\bar{k}}|_{\mathrm{Spec}(\widehat{K}_{\bar{x}})} \cong \mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \times_{\mathrm{Spec}(\widehat{K}_P)} \mathrm{Spec}(\widehat{K}_{\bar{x}}), \tag{2.2}$$

i.e. the geometric restriction is the base change of the closed-point restriction along the bottom bent arrow, which is an unramified extension of complete discretely valued fields. This is the content of the following proposition.

Proposition 44. *The closed-point formulation in Definition 42 and the geometric formulation in Definition 43 are equivalent.*

Proof. Let $P = P_i$, let $\bar{x}$ lie over P, and choose a uniformizer t at P. Because k is perfect, the base change $C_{\bar{k}} \to C$ is ind-étale, so the pullback of t is again a uniformizer at $\bar{x}$. Since $C_{\bar{k}}$ is a smooth curve over the algebraically closed field $\bar{k}$, it follows that $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}} \cong \bar{k}[[t]]$. Thus

$$\widehat{K}_P \cong \kappa(P)((t)) \longrightarrow \widehat{K}_{\bar{x}} \cong \bar{k}((t))$$

is unramified, as claimed in the diagram.

By (2.2), the geometric restriction is obtained from the closed-point restriction by this unramified base change. Ramification bounds are unchanged under unramified extensions of complete discretely valued fields, so the two restrictions have the same ramification bound.

Finally, as $\kappa(P)/k$ is separable the fiber of P_i in $C_{\bar{k}}$ is reduced, so $\mathfrak{m}_{\bar{k}}$ has multiplicity n_i at every $\bar{x}$ above P_i. Thus the geometric condition at $\bar{x}$ is precisely the closed-point condition at P_i. □

Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 40.

There is another equivalent formulation through the language of characters. The group G, being a finite abelian group over k, is étale. Consequently, the correspondence between G-torsors and the fundamental group allows us to describe G-torsors via characters. Specifically, there is an isomorphism (Corollary 14):

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\text{ét}}(U, \overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(U_{\text{ét}}, G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \mathrm{Gal}(\widehat{K}_P^{\mathrm{sep}}/\widehat{K}_P)$. Thus, the restricted G-torsor $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \to G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r-th ramification group in the upper numbering:

$$\rho(G^r_{\widehat{K}_P}) = \{1\}$$

Basic Properties of Ramification of G-Torsors

We now prove some elementary lemmas regarding the ramification of G-torsors.

Lemma 45. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k. Let $U \subset X$ be a dense open subset and let (Z, ξ) be a prime divisor in the complement $X \setminus U$. If $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G-torsors on $U_{\acute{e}t}$ that are tamely ramified at (Z, ξ), then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at (Z, ξ).*

Proof. Let $A = \mathcal{O}_{X,\xi}$ and $K = \mathrm{Frac}(A)$. Tameness may be checked after completing A, so we may assume that K is complete. For $i = 1, 2$, let $L_{i,j}/K$ be the field factors of

$$\mathcal{P}_i \times_U \mathrm{Spec}(K).$$

These extensions are tame by assumption. Let M/K be a finite Galois extension containing the Galois closures of all the $L_{i,j}/K$. Finite tame extensions of a complete discretely valued field are closed under Galois closure and finite compositum, so M/K is tame.

Both $\mathcal{P}_1$ and $\mathcal{P}_2$ become trivial over M. Hence their contracted product also becomes trivial over M. Consequently, every field factor of

$$(\mathcal{P}_1 \otimes^G \mathcal{P}_2) \times_U \mathrm{Spec}(K)$$

is isomorphic to an intermediate extension of M/K: indeed, after tensoring such a factor with M, one obtains a product having M as a factor, and hence a K-embedding of that field factor into M. Since subextensions of a finite tame extension are tame, every field factor is tame. The result follows from Definition 40. □

Lemma 46. *Let $A \to B$ be a local extension of discrete valuation rings. Assume that B/A is weakly unramified and that the residue-field extension κ_B/κ_A is separable. Write $K = \mathrm{Frac}(A)$ and $L = \mathrm{Frac}(B)$. Let $\mathcal{P}$ be a finite G-torsor over $\mathrm{Spec}(K)$, and put*

$$\mathcal{P}_L := \mathcal{P} \times_{\mathrm{Spec}(K)} \mathrm{Spec}(L).$$

Then $\mathcal{P}$ is unramified with respect to A if and only if $\mathcal{P}_L$ is unramified with respect to B. Likewise, $\mathcal{P}$ is tamely ramified with respect to A if and only if $\mathcal{P}_L$ is tamely ramified with respect to B.

Proof. Write the finite étale K-algebra underlying $\mathcal{P}$ as

$$\prod_i E_i,$$

where every E_i/K is finite separable. For every i, write

$$E_i \otimes_K L \cong \prod_j F_{i,j}$$

as a product of finite separable extensions of L. The standard base-change properties of extensions of discrete valuation rings give, for every i and j,

$$e(F_{i,j}/L) = e(E_i/K),$$

and show that $\kappa_{F_{i,j}}/\kappa_L$ is separable if and only if κ_{E_i}/κ_K is separable.

The two assertions now follow immediately from Definition 37: unramifiedness means residue separability and ramification index 1, while tameness means residue separability and ramification index prime to the residue characteristic. □

Corollary (Étale form of Lemma 46)**.** *Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$. Let (Z_X, ξ_X) and (Z_Y, ξ_Y) be prime divisors contained in $X \setminus U_X$ and $Y \setminus U_Y$, respectively, and suppose that $f(\xi_X) = \xi_Y$. Assume that $\mathcal{O}_{X,\xi_X}$ and $\mathcal{O}_{Y,\xi_Y}$ are discrete valuation rings and that f is étale at ξ_X. If $\mathcal{P}$ is a finite G-torsor over U_Y, then $f^{-1}\mathcal{P}$ is unramified (resp. tamely ramified) at ξ_X if and only if $\mathcal{P}$ is unramified (resp. tamely ramified) at ξ_Y.*

Proof. Set $A = \mathcal{O}_{Y,\xi_Y}$ and $B = \mathcal{O}_{X,\xi_X}$. Since f is étale at ξ_X, the induced local homomorphism $A \to B$ is étale. In particular, $\mathfrak{m}_A B = \mathfrak{m}_B$, so B/A is weakly unramified, and the residue-field extension $\kappa(\xi_X)/\kappa(\xi_Y)$ is finite separable. The restriction of $f^{-1}\mathcal{P}$ to $\operatorname{Spec}(\operatorname{Frac}(B))$ is the base change of the restriction of $\mathcal{P}$ to $\operatorname{Spec}(\operatorname{Frac}(A))$. The two equivalences now follow from Lemma 46. □

Lemma 47. *Let G_1, G_2 be finite abelian groups. Let X be a locally Noetherian scheme over a field k, let $U \subset X$ be a dense open subscheme, and let (Z, ξ) be a prime divisor in $X \setminus U$. For $i = 1, 2$, let $\mathcal{P}_i$ be a G_i-torsor on $U_{\text{ét}}$. If both $\mathcal{P}_1$ and $\mathcal{P}_2$ are tamely ramified at (Z, ξ), then the $(G_1 \times G_2)$-torsor $\mathcal{P}_1 \times_U \mathcal{P}_2$ is tamely ramified at (Z, ξ).*

Proof. Let $A = \mathcal{O}_{X,\xi}$ and $K = \operatorname{Frac}(A)$. Tameness may be checked after completing A, so we may assume that K is complete. As in the proof of Lemma 45, there exists a finite tame Galois extension M/K that splits both $\mathcal{P}_1$ and $\mathcal{P}_2$. Their fiber product is therefore split over M. Consequently, every field factor of

$$(\mathcal{P}_1 \times_U \mathcal{P}_2) \times_U \operatorname{Spec}(K)$$

embeds into M. Since subextensions of a finite tame extension are tame, every such field factor is tame. The result follows from Definition 40. □

Lemma 48. *Let A be a discrete valuation ring with fraction field K, let $\iota : H \hookrightarrow G$ be an injective homomorphism of finite abelian groups, and let $\mathcal{Q}$ be an H-torsor over $\operatorname{Spec}(K)$. Let*

$$\iota_* \mathcal{Q} := G \times^H \mathcal{Q}$$

be the induced G-torsor, where H acts on $G \times \mathcal{Q}$ by $h \cdot (g, q) = (g\iota(h)^{-1}, h \cdot q)$. Then, as a K-scheme, $\iota_* \mathcal{Q}$ is noncanonically isomorphic to a disjoint union of $[G : H]$ copies of $\mathcal{Q}$. Consequently:

1. *$\iota_* \mathcal{Q}$ is tamely ramified with respect to A if and only if $\mathcal{Q}$ is;*
2. *$\iota_* \mathcal{Q}$ is unramified with respect to A if and only if $\mathcal{Q}$ is;*
3. *if A is complete and its residue field is perfect, then, for every r, $\iota_* \mathcal{Q}$ has ramification bounded by r if and only if $\mathcal{Q}$ does.*

Proof. As a K-scheme, $\iota_* \mathcal{Q}$ is noncanonically isomorphic to a disjoint union of $[G : H]$ copies of $\mathcal{Q}$. Thus it has the same field factors as $\mathcal{Q}$, each repeated $[G : H]$ times, and the assertions follow immediately from the definitions. □

2.3 Reduction to Cyclic Torsors on $\mathbb{P}^1$

Before introducing explicit equations, we explain the strategy of the proof and isolate two reductions that simplify the situation. The theorem concerns a torsor under an arbitrary finite abelian group on an arbitrary smooth curve, whereas the calculation we ultimately wish to perform is local and cyclic. We show that both the structure group and the geometry may be simplified without changing the ramification problem relevant to the exceptional divisor.

The first reduction is algebraic: using the decomposition of a finite abelian group into cyclic factors, together with the compatibility of the symmetric-power construction and ramification bounds with products and quotients, we reduce the theorem to cyclic groups of order prime to p and cyclic groups of order p^r. This explains why the three cyclic theories recalled in the next section suffice. The second reduction is geometric and formal–local: the ramification of the induced torsor at the generic point of the exceptional divisor depends only on the restriction of the original torsor to the punctured formal neighbourhood of the point supporting the modulus. It therefore tells us exactly what the cyclic theories must accomplish: realize the resulting local cyclic torsor by an explicit equation on $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Together, these reductions transform the original problem into three parallel explicit calculations, corresponding to Kummer, Artin–Schreier, and Artin–Schreier–Witt theory. The following section recalls precisely the conclusions from these theories that provide the required realizations.

2.3.1 Reduction from Finite Abelian Groups to Cyclic Groups

Fix a single-point submodulus $\mathfrak{n}$ and write $d = \deg(\mathfrak{n})$. The first reduction concerns the structure group and is independent of the geometry of the curve.

Proposition 49 (Reduction to cyclic groups)**.** *Suppose that the conclusion of Theorem 35 is known whenever the structure group is cyclic of order prime to p, and whenever the structure group is a cyclic p-group. Then it holds for every finite abelian group G.*

Proof. By the structure theorem for finite abelian groups, choose a decomposition

$$G \cong \prod_{i=1}^{s} \mathbb{Z}/e_i\mathbb{Z} \times \prod_{j=1}^{t} \mathbb{Z}/p^{r_j}\mathbb{Z}, \qquad (e_i, p) = 1.$$

Let $\rho : \pi_1^{\text{ét}}(U, \bar{x}) \to G$ be the character corresponding to $\mathcal{P}$. Composing ρ with the projections to the displayed cyclic factors gives torsors $\mathcal{P}_i$ under $\mathbb{Z}/e_i\mathbb{Z}$ and torsors $\mathcal{Q}_j$ under $\mathbb{Z}/p^{r_j}\mathbb{Z}$. Since the character of each factor is a quotient of ρ, the ramification bound on $\mathcal{P}$ passes to every $\mathcal{P}_i$ and $\mathcal{Q}_j$.

The original torsor is recovered as their fiber product over U:

$$\mathcal{P} \cong \mathcal{P}_1 \times_U \cdots \times_U \mathcal{P}_s \times_U \mathcal{Q}_1 \times_U \cdots \times_U \mathcal{Q}_t.$$

The construction of the symmetric-power torsor commutes with this product. Indeed, after pullback to U^d, both sides are obtained by taking the contracted product of the d pullbacks along multiplication in the structure group. Hence

$$\mathcal{P}^{[d]} \cong \mathcal{P}_1^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{P}_s^{[d]} \times_{U^{(d)}} \mathcal{Q}_1^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{Q}_t^{[d]}.$$

By the assumed cyclic cases, every factor on the right is tamely ramified at $\eta_{\mathfrak{n}}$. The fiber product is therefore tamely ramified there by Lemma 47. This proves the assertion for G. □

Thus it remains to understand cyclic groups. Their prime-to-p part is described by Kummer theory, while cyclic groups of order p and p^r are described by Artin–Schreier and Artin–Schreier–Witt theory, respectively.

2.3.2 Formal–Local Reduction from C to $\mathbb{P}^1$

We argue that it is enough to prove Theorem 35 for $C = \mathbb{P}^1_k$, $\mathfrak{n} = d \cdot 0$, and $\mathcal{P} \to \mathbb{G}_m = \mathbb{P}^1_k \setminus \{0, \infty\}$ a G-torsor given by an explicit global equation. Informally, this is because *the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is a formal–local invariant of $\mathcal{P}$ at P*: it depends only on the completed local ring $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to the punctured formal disc $\mathrm{Spec}(k((x)))$, since all of the constructions involved — self products, quotients by finite groups, and blowups — commute with the flat base change to the completion; for blowups, this is Theorem 32. We emphasize that this is not a general principle that can simply be quoted: it is a statement about the particular constructions at hand, and it is proved by checking each construction separately. This is the content of Lemma 50 and Corollary 51 below. The second ingredient of the reduction is Lemma 56: every local torsor class appearing in Theorem 35 is realized by an explicit torsor over $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Throughout this section we assume that k is algebraically closed; in particular every closed point of C is k-rational, and $\mathfrak{n} = dP$ with $P \in C(k)$ and $d = \deg \mathfrak{n}$.

Remark. The assumption that k is algebraically closed is harmless in our situation, for the following reasons.

1. **(Base change to $\bar{k}$.)** Assume k is perfect, so that $\bar{k} = k^{sep}$. The formation of $C^{(d)}$ (a quotient by a finite group), of the blowup $X_{\mathfrak{n}}$ at $\mathfrak{n}$, and of $\mathcal{P}^{[d]}$ (Proposition 28) commutes with the flat base change $\mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$. Since $E_{\mathfrak{n}} \cong \mathbb{P}^{\deg \mathfrak{n}-1}_k$ is geometrically irreducible, $(E_{\mathfrak{n}})_{\bar{k}}$ is irreducible and its generic point $\bar{\eta}$ lies over $\eta_{\mathfrak{n}}$; moreover $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{(X_{\mathfrak{n}})_{\bar{k}},\bar{\eta}}$ is a weakly unramified extension of discrete valuation rings with separable residue field extension. Therefore, by Lemma 46, $(\mathcal{P}_{\bar{k}})^{[d]} = (\mathcal{P}^{[d]})_{\bar{k}}$ is unramified (resp. tamely ramified) at $\bar{\eta}$ if and only if $\mathcal{P}^{[d]}$ is unramified (resp. tamely ramified) at $\eta_{\mathfrak{n}}$. Thus both properties may be checked after base change to $\bar{k}$.

2. **(Rationality of P — a caveat.)** If P is a closed point with $\kappa(P) \neq k$, then after base change to $\bar{k}$ the point $\mathfrak{n}$ corresponds to the divisor $\sum_{\bar{P} \mapsto P} n\bar{P}$ on $C_{\bar{k}}$, whose support consists of $[\kappa(P) : k]$ distinct points, whereas the computation below treats a modulus supported at a single rational point. We therefore prove Theorem 35 only under the standing assumption that k is algebraically closed. This is the version used later: at the beginning of the proof of Theorem 62, all data are first base changed to $\bar{k}$. The modulus then decomposes into its rational-point-supported summands; the present theorem is applied to each summand, and Corollary 64 combines them before tameness is descended back to k.

We now formulate the formal–local invariance precisely. Fix $P \in C(k)$ lying in the support of $\mathfrak{m}$, set $\mathfrak{n} = dP$, and fix a uniformizer x of $\mathcal{O}_{C,P}$; it induces an isomorphism $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. The scheme $\mathrm{Spec}(k[[x]])$ has two points — the closed point, mapping to P, and the generic point, mapping to the generic point of C — so the punctured formal disc $\mathrm{Spec}(k((x)))$ maps to U, and we denote by

$$\mathcal{P}_x := \mathcal{P} \times_U \mathrm{Spec}(k((x)))$$

the corresponding restriction of $\mathcal{P}$; it is a G-torsor over $\mathrm{Spec}(k((x)))$. Let $e_1, \dots, e_d \in k[[x_1, \dots, x_d]]$

be the elementary symmetric polynomials in $x_1, \dots, x_d$, and set

$$V = \operatorname{Spec}\Big(k[[e_1, \dots, e_d]][e_d^{-1}]\Big), \qquad W = \operatorname{Spec}\Big(k[[x_1, \dots, x_d]][(x_1 \cdots x_d)^{-1}]\Big).$$

The inclusion $k[[e_1, \dots, e_d]] \subset k[[x_1, \dots, x_d]]$ induces a finite flat morphism $\hat{r} : W \to V$ (note that inverting $e_d = x_1 \cdots x_d$ inverts each x_i), and for each i we let

$$q_i : W \to \operatorname{Spec}(k((x))), \qquad x \mapsto x_i.$$

The group S_d acts on W over V by permuting $x_1, \dots, x_d$, and $q_{\sigma(i)} = q_i \circ \sigma$. Recall from Proposition 28 that

$$K_G := \ker\big(G^d \xrightarrow{m} G\big), \qquad \Xi_G := K_G \rtimes S_d,$$

where $m(g_1, \dots, g_d) = g_1 \cdots g_d$ and S_d acts on K_G by permuting coordinates. Thus quotienting by Ξ_G means first contracting the G^d-torsor along multiplication, by quotienting by K_G, and then descending from the ordered product to the symmetric product by quotienting by S_d.

Lemma 50 (Formal–local invariance)**.** *In the above notation:*

1. *The choice of x induces an isomorphism $\widehat{\mathcal{O}_{C^{(d)},\mathfrak{n}}} \cong k[[e_1, \dots, e_d]]$ carrying the maximal ideal to $(e_1, \dots, e_d)$, such that the induced morphism $c : \operatorname{Spec}(k[[e_1, \dots, e_d]]) \to C^{(d)}$ is flat and satisfies $c^{-1}(U^{(d)}) = V$; we write $c_V : V \to U^{(d)}$ for the restriction.*
2. *There is an induced isomorphism $\widehat{\mathcal{O}_{X_\mathfrak{n},\eta_\mathfrak{n}}} \cong \kappa[[e_d]]$, where $\kappa = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d})$. In particular $\widehat{K}_{\eta_\mathfrak{n}} := \operatorname{Frac}(\widehat{\mathcal{O}_{X_\mathfrak{n},\eta_\mathfrak{n}}}) \cong \kappa((e_d))$, and the canonical morphism $\operatorname{Spec}(\widehat{K}_{\eta_\mathfrak{n}}) \to U^{(d)}$ factors as $\operatorname{Spec}(\kappa((e_d))) \to V \xrightarrow{c_V} U^{(d)}$, where the first map does not depend on $(C, \mathcal{P})$.*
3. *There is a canonical isomorphism of G-torsors over V*

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \;\cong\; \Xi_G \backslash \Big(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x \Big).$$

Consequently, the isomorphism class of the G-torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_\mathfrak{n}})}$ depends only on d and on the isomorphism class of the G-torsor $\mathcal{P}_x$ over $k((x))$, and not otherwise on the pair $(C, \mathcal{P})$. In particular, whether $\mathcal{P}^{[d]}$ is tamely ramified or unramified at $\eta_\mathfrak{n}$ depends only on $(\mathcal{P}_x, d)$.

Proof. **(1)** The chosen parameter gives $\widehat{\mathcal{O}_{C^d,(P,\dots,P)}} \cong k[[x_1, \dots, x_d]]$. We use the standard facts that completion is compatible with finite morphisms and that the completed local ring of a finite quotient is the invariant ring of the completed local ring. Since $(P, \dots, P)$ is the unique point of C^d above dP, this gives

$$\widehat{\mathcal{O}_{C^{(d)},dP}} \cong k[[x_1, \dots, x_d]]^{S_d} = k[[e_1, \dots, e_d]],$$

where the last equality is the formal-power-series version of the fundamental theorem of symmetric functions. The maximal ideal is thus $(e_1, \dots, e_d)$, and the resulting map c is flat by flatness of completion.

It remains only to identify the boundary. Near dP, the only component of $C^{(d)} \setminus U^{(d)}$ that occurs is the locus of divisors containing P. Its inverse image in the ordered formal neighbourhood is

$$\bigcup_i V(x_i) = V(x_1 \cdots x_d) = V(e_d).$$

Since the ordered-to-symmetric map is finite and surjective, the boundary in $\operatorname{Spec} k[[e_1,\dots,e_d]]$ is also $V(e_d)$. Hence $c^{-1}(U^{(d)}) = V$ (and similarly the corresponding ordered inverse image is W).

(2) We use that blowups commute with flat base change. Thus the pullback of $X_{\mathfrak{n}}$ along c is the blowup of $\operatorname{Spec} k[[e_1,\dots,e_d]]$ at its closed point. Completion preserves the exceptional divisor and its residue field, so it is enough to compute the completed local ring at the generic point of the exceptional divisor in this standard blowup.

On the chart where e_d is the exceptional parameter, put $t_i = e_i/e_d$ for $i < d$. The exceptional divisor is given by $e_d = 0$, and its generic point has residue field

$$\kappa = k(t_1,\dots,t_{d-1}) = k\left(\frac{e_1}{e_d},\dots,\frac{e_{d-1}}{e_d}\right).$$

The usual local calculation for the blowup of a regular local scheme at its closed point therefore gives $\widehat{\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}}} \cong \kappa[[e_d]]$ and $\widehat{K}_{\eta_{\mathfrak{n}}} \cong \kappa((e_d))$. Since e_d is invertible in the latter field, its map to the completed symmetric power factors through V, giving the asserted factorization through c_V.

(3) Let $h : \mathcal{P}^{\times_k d} \to U^{(d)}$ denote the canonical morphism; it is finite, being the composition of the finite étale morphism $\mathcal{P}^{\times_k d} \to U^d$ with the finite morphism $U^d \to U^{(d)}$. By Proposition 28, $\mathcal{P}^{[d]} = \Xi_G\backslash\mathcal{P}^{\times_k d} = \underline{\operatorname{Spec}}_{U^{(d)}}\left((h_*\mathcal{O}_{\mathcal{P}^{\times_k d}})^{\Xi_G}\right)$. Pushforward along the affine morphism h commutes with the flat base change $c_V : V \to U^{(d)}$, and the formation of Ξ_G-invariants is a kernel, hence also commutes with flat base change; therefore

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi_G\backslash\left(\mathcal{P}^{\times_k d} \times_{U^{(d)}} V\right).$$

Completing the finite map $C^d \to C^{(d)}$ at the unique point $(P,\dots,P)$ above dP gives the Cartesian square defined by $k[[e_1,\dots,e_d]] \subset k[[x_1,\dots,x_d]]$. Restricting this square to $e_d = x_1\cdots x_d \neq 0$ gives directly $U^d \times_{U^{(d)}} V \cong W$, and hence $\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \cong \mathcal{P}^{\times_k d} \times_{U^d} W$.

The i-th component of the resulting map $W \to U^d$ is $W \xrightarrow{q_i} \operatorname{Spec}(k((x))) \to U$: indeed, it is induced in coordinates by $x \mapsto x_i$, and x_i is invertible on W. Since $\mathcal{P}^{\times_k d} = p_1^{-1}\mathcal{P} \times_{U^d} \cdots \times_{U^d} p_d^{-1}\mathcal{P}$, compatibility of fiber products with base change now gives

$$\mathcal{P}^{\times_k d} \times_{U^d} W \cong q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x,$$

which is equivariant for the actions of G^d (in particular of K_G) and of S_d — the latter acting on W by permuting the x_i, compatibly with $q_{\sigma(i)} = q_i \circ \sigma$. Passing to Ξ_G-quotients yields the displayed isomorphism.

For the final statement: the schemes V and W, the maps q_i and $\hat{r}$, the Ξ_G-action, and the point $\operatorname{Spec}(\kappa((e_d))) \to V$ of (2) are all constructed from d alone; by (3), the torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ is the pullback along $\operatorname{Spec}(\kappa((e_d))) \to V$ of a torsor constructed from $\mathcal{P}_x$ and these data. Since tame ramification and unramifiedness at $\eta_{\mathfrak{n}}$ are, by definition, determined by this restricted torsor together with the discrete valuation ring $\kappa[[e_d]]$ (Definition 40), both properties depend only on $(\mathcal{P}_x, d)$. □

Corollary 51 (Reduction to $\mathbb{P}^1$)**.** *Let C, C' be smooth projective geometrically connected curves over the algebraically closed field k, let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ and $\mathcal{P}' \to U' = C' \setminus \mathfrak{m}'$ be G-torsors, let $P \in \operatorname{Supp}(\mathfrak{m})(k)$ and $P' \in \operatorname{Supp}(\mathfrak{m}')(k)$, and let x, x' be uniformizers at P, P' respectively. Assume that $\mathcal{P}_x \cong \mathcal{P}'_{x'}$ as G-torsors, compatibly with the isomorphism $k((x)) \cong k((x'))$, $x \mapsto x'$. Then, for every $d \geq 1$, $\mathcal{P}^{[d]}$ is unramified (resp. tamely ramified) at η_{dP} if and only if $\mathcal{P}'^{[d]}$ is unramified (resp. tamely ramified) at $\eta_{dP'}$.*

Proof. Immediate from Lemma 50: both $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP}})}$ and $\mathcal{P}'^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP'}})}$ are identified, compatibly with the discrete valuation ring $\kappa[[e_d]]$, with the pullback along $\mathrm{Spec}(\kappa((e_d))) \to V$ of $\Xi_G\backslash(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x)$. □

The corollary reduces the geometry to $\mathbb{P}^1$ once the local cyclic torsor is realized by a global equation on $\mathbb{G}_m$. The next section introduces the three cyclic theories that provide precisely these equations and control their ramification.

2.4 Three Parallel Theories of Cyclic Torsors

The first reduction of the preceding section explains why cyclic theories suffice: it reduces the theorem from finite abelian groups to cyclic groups of order prime to p and cyclic groups of order p^r. The second reduction explains exactly what these theories must accomplish: they must realize a local cyclic torsor over $k((x))$ by an explicit equation on $\mathbb{G}_m \subset \mathbb{P}^1_k$, with the required control of its ramification. We therefore recall the three theories in the parallel form in which they will be used:

theory	H	$\varphi : H \to H$	$\ker(\varphi)$
Kummer	$\mathbb{G}_m$	$z \mapsto z^e$	$\mu_e \cong \mathbb{Z}/e\mathbb{Z}$
Artin–Schreier	$\mathbb{G}_a$	$z \mapsto z^p - z$	$\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$
Artin–Schreier–Witt	$\mathbb{W}_r$	$F - \mathrm{id}$	$W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$.

For e invertible, and in characteristic p for the latter two rows, the corresponding exact sequences of étale sheaves are

$$1 \longrightarrow \mu_e \longrightarrow \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \longrightarrow 1,$$
$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{G}_a \xrightarrow{z \mapsto z^p - z} \mathbb{G}_a \longrightarrow 0,$$
$$0 \longrightarrow W_r(\mathbb{F}_p) \longrightarrow \mathbb{W}_r \xrightarrow{F-\mathrm{id}} \mathbb{W}_r \longrightarrow 0.$$

Here the identification in the Kummer row is made after choosing a primitive e-th root of unity. In every row, pulling back φ along a section a produces the torsor given by the equation $\varphi(X) = a$, and changing a by an element of $\varphi(H)$ does not change its class. The kernel action, pullback construction, and resulting cohomology class are detailed in Section A.1.

Artin–Schreier–Witt theory contains Artin–Schreier theory as the case $r = 1$. We nevertheless treat the Artin–Schreier case separately: it displays, without the additional Witt-vector bookkeeping, the basic principle later used in the general case—choose a reduced representative of the local torsor, control its pole order by the ramification break, and use the same equation to realize the torsor on $\mathbb{G}_m$.

Only the conclusions used in Chapter 2 are stated here. The group-scheme constructions, classification arguments, normal-form results, and proofs are collected in Chapter A.

Throughout this section, let K be a complete discrete valuation field with perfect residue field κ of characteristic $p > 0$.

2.4.1 Kummer theory: order prime to p

Classes and equations. Assume that K contains μ_n and that $(n,p)=1$. Write

$$H^1(K,\mu_n) := H^1_{\text{ét}}(\operatorname{Spec} K, \mu_n);$$

equivalently, this is the first continuous Galois cohomology group of K with coefficients in μ_n. Its elements classify μ_n-torsors over K. The Kummer sequence gives a natural isomorphism

$$H^1(K,\mu_n) \cong K^\times/(K^\times)^n.$$

Under this isomorphism, the coset of $a \in K^\times$ corresponds to the torsor

$$\operatorname{Spec} K[T]/(T^n - a),$$

on which μ_n acts by $T \mapsto \zeta T$. In particular, replacing a by ab^n, with $b \in K^\times$, does not change the isomorphism class of the torsor. If the class of a has order n, adjoining a solution of $T^n = a$ gives a cyclic field extension of degree n; conversely, every cyclic extension of degree n arises in this way.

When $K = k((x))$ with k algebraically closed, every unit in $k[[x]]$ is an nth power by Hensel's lemma. Thus valuation gives

$$K^\times/(K^\times)^n \cong \mathbb{Z}/n\mathbb{Z}, \qquad [x] \longmapsto 1. \tag{2.3}$$

In particular, a class has order n exactly when it is represented by x^a with $\gcd(a,n)=1$.

Theorem 52 (Ramification in Kummer Extensions, [Koc97, Proposition 1.83])**.** *Assume K contains the n-th roots of unity μ_n and $(n,p)=1$. Let L/K be the extension given by $X^n = a$, for $a \in K^\times$. Then:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in κ is an n-th power, the extension is trivial.*
2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in κ is not an n-th power, the extension is unramified.*
3. *If $v_K(a) \notin n\mathbb{Z}$, the ramification index is $n/\gcd(n, v_K(a))$; in particular, if $\gcd(n, v_K(a)) = 1$, the extension is totally and tamely ramified of degree n.*

2.4.2 Artin–Schreier theory: order p

Classes and equations. Assume that K has characteristic p, and put $\wp(z) = z^p - z$. Write

$$H^1(K,\mathbb{Z}/p\mathbb{Z}) := H^1_{\text{ét}}(\operatorname{Spec} K, \mathbb{Z}/p\mathbb{Z}).$$

Its elements classify $\mathbb{Z}/p\mathbb{Z}$-torsors over K. The Artin–Schreier sequence gives a natural isomorphism

$$H^1(K,\mathbb{Z}/p\mathbb{Z}) \cong K/\wp(K).$$

Under this isomorphism, the coset $[a] \in K/\wp(K)$ corresponds to the $\mathbb{Z}/p\mathbb{Z}$-torsor

$$\operatorname{Spec} K[X]/(X^p - X - a),$$

on which $c \in \mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ acts by $X \mapsto X + c$. In particular, replacing a by $a + (b^p - b)$, with $b \in K$, does not change the isomorphism class of the torsor. If $[a] \neq 0$, adjoining a solution of $X^p - X = a$ gives a cyclic field extension of degree p; conversely, every cyclic extension of degree p arises in this way. When $K = k((x))$ with k algebraically closed, every class has a representative in $x^{-1}k[x^{-1}]$ whose nonzero pole order is prime to p (Proposition 89, case $r = 1$).

Theorem 53 (Ramification in Artin–Schreier Extensions, [Tho05, Proposition 2.1])**.** *Let K be a complete discretely valued field of characteristic p with perfect residue field κ, and let L/K be given by $X^p - X = a$. Then:*

1. *If $v_K(a) > 0$, or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension is trivial.*
2. *If $v_K(a) = 0$ and $a \notin \wp(K)$, the extension is cyclic of degree p and unramified.*
3. *If $v_K(a) = -m < 0$ with $p \nmid m$, the extension is cyclic, totally ramified of degree p, and has unique upper ramification break m.*

2.4.3 Artin–Schreier–Witt theory: order p^r

Classes and equations. For an $\mathbb{F}_p$-algebra A, write $W_r(A) = A^r$ for the ring of p-typical Witt vectors of length r, with Witt addition and subtraction denoted by $\oplus$ and $\ominus$, and put $\wp = F - \mathrm{id}$. For a field K of characteristic p, write

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) := H^1_{\text{ét}}(\operatorname{Spec} K, \mathbb{Z}/p^r\mathbb{Z}).$$

Its elements classify $\mathbb{Z}/p^r\mathbb{Z}$-torsors over K. The Artin–Schreier–Witt sequence gives a natural isomorphism

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) \cong W_r(K)/\wp W_r(K).$$

Under this isomorphism, the coset $[\vec{g}] \in W_r(K)/\wp W_r(K)$ corresponds to an affine solution scheme as follows. Let $\vec{X} = (X_0, \dots, X_{r-1})$ be the universal Witt vector on $\mathbb{W}_{r,K} = \operatorname{Spec} K[X_0, \dots, X_{r-1}]$. Then

$$T_{\vec{g}} := \operatorname{Spec}\Big(K[X_0, \dots, X_{r-1}]/\big(\wp\vec{X} \ominus \vec{g}\big)\Big),$$

where the parenthesized Witt vector denotes the ideal generated by its r coordinate polynomials. Equivalently, $T_{\vec{g}}$ is the fiber of $\wp : \mathbb{W}_{r,K} \to \mathbb{W}_{r,K}$ over the K-point $\vec{g}$. Its first defining equation is $X_0^p - X_0 - g_0 = 0$, and each subsequent equation has the triangular form

$$X_j^p - X_j - g_j + C_j(X_0, \dots, X_{j-1}; g_0, \dots, g_{j-1}) = 0.$$

Here C_j is the universal Witt correction polynomial determined by the addition law and depends only on the preceding coordinates. This is a finite étale K-scheme of degree p^r, on which $\vec{c} \in W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ acts by Witt translation $\vec{X} \mapsto \vec{X} \oplus \vec{c}$. In particular, replacing $\vec{g}$ by $\vec{g} \oplus \wp\vec{h}$, with $\vec{h} \in W_r(K)$, gives an isomorphic torsor. If $[\vec{g}]$ has order p^s, its associated character cuts out a cyclic field extension of degree p^s; thus $T_{\vec{g}}$ is connected precisely when $[\vec{g}]$ has order p^r. Conversely, every cyclic extension of degree dividing p^r arises in this way. The Witt-vector group scheme is constructed in Section A.3. The Artin–Schreier–Witt operator, the proof that the displayed equation defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor, and the classification theorem—including its description of the degree and connected components and of induction from a torsor under the subgroup $\mathbb{Z}/p^s\mathbb{Z} \subseteq \mathbb{Z}/p^r\mathbb{Z}$—are given in Section A.4.

Definition 54. Let $K = k((x))$. For $g \in K$, put $m(g) := \max(0, -v_x(g))$. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(K)$ is **reduced** if, for every j, either $m(g_j) = 0$ or $p \nmid m(g_j)$.

Every class in $W_r(K)/\wp W_r(K)$ admits a reduced representative. The stronger reduced polynomial normal form for algebraically closed k is proved in the appendix (Proposition 89).

Theorem 55 (Ramification in Artin–Schreier–Witt Extensions (Schmid–Witt)). *Let $K = k((x))$, with k perfect of characteristic p, and let L/K be a cyclic totally ramified extension of degree p^r whose Artin–Schreier–Witt class is represented by a reduced vector $\vec{g} = (g_0, \dots, g_{r-1})$, with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K is*

$$u = \max_{0 \le j \le r-1} p^{r-1-j} m_j.$$

The largest upper break is the last index at which an upper-numbering ramification group is nontrivial. The formula reads this break from the pole orders of the coordinates of a reduced Witt vector. The jth coordinate has pole order m_j, and the formula assigns it the value $p^{r-1-j}m_j$. For example, when $r = 2$ it says $u = \max\{pm_0, m_1\}$. For any bound d, the inequality $u < d$ is therefore equivalent to $p^{r-1-j}m_j < d$ for every j, as required in (2.4) below.

The word "reduced" matters here. Already for $r = 1$, the elements x^{-p} and x^{-1} represent the same Artin–Schreier class, since $x^{-p} - x^{-1} = \wp(x^{-1})$. The first has pole order p, but the reduced representative x^{-1} shows that the break is 1. In this case the formula agrees with Theorem 53.

We use the Schmid–Witt formula as a known theorem. For finite residue fields, it is due to Kanesaka–Sekiguchi [KS79] and Thomas [Tho05], building on Schmid [Sch37] for $r = 1$; for arbitrary perfect residue fields, it is due to Elder–Keating [EK25].

2.4.4 The common realization conclusion

The three theories give the following single conclusion, which is the form needed after the formal–local reduction to $\mathbb{P}^1_k$.

Lemma 56 (Parallel realization over $\mathbb{G}_m$). *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{Q}$ be a G-torsor over* $\operatorname{Spec} k((x))$.

1. *If $G = \mathbb{Z}/e\mathbb{Z}$ with $(e, p) = 1$ and $\mathcal{Q}$ is connected, then $\mathcal{Q}$ is represented by*

$$T^e = x^a$$

for some a prime to e. The same equation defines a G-torsor on $\mathbb{G}_m$ that is tamely ramified at 0 and ∞.

2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification of $\mathcal{Q}$ is bounded by d, then $\mathcal{Q}$ is represented by*

$$X^p - X = f(x), \qquad f \in x^{-1}k[x^{-1}],$$

where either $f = 0$, or $m := \deg_{x^{-1}} f < d$ and $p \nmid m$. The same equation defines a torsor on $\mathbb{G}_m$ extending étale across ∞.

3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$, the torsor $\mathcal{Q}$ is connected, and its ramification is bounded by d, then $\mathcal{Q}$ is represented by*

$$\wp\vec{X} = \vec{f},$$

where $\vec{f} = (f_0, \dots, f_{r-1})$ is reduced, each $f_j \in x^{-1}k[x^{-1}]$, and, writing $m_j := \deg_{x^{-1}} f_j$ (with $m_j = 0$ if $f_j = 0$),

$$p^{r-1-j}m_j < d \qquad (0 \le j < r). \tag{2.4}$$

The same equation defines a torsor on $\mathbb{G}_m$ extending étale across ∞.

Proof. For (1), choose $\mu_e \cong G$. By Kummer theory, $\mathcal{Q}$ is represented by an equation $T^e = b$ for some $b \in k((x))^\times$. By (2.3), $[b] = [x^a]$ in $k((x))^\times/(k((x))^\times)^e$ for some integer a. Thus $b = x^a c^e$ for some $c \in k((x))^\times$, and replacing T by T/c gives the equation $T^e = x^a$. Connectedness means that the class $[x^a]$ has order e; under (2.3), this is equivalent to $\gcd(a, e) = 1$. The equation $T^e = x^a$ is étale over $k[x, x^{-1}]$ and defines the required torsor on $\mathbb{G}_m$. At 0 and ∞ the valuations of x^a are a and $-a$, respectively, so Theorem 52 shows that its ramification at both boundary points is tame.

For (2), Artin–Schreier theory represents $\mathcal{Q}$ as $\operatorname{Spec}\big(k((x))[X]/(X^p - X - b)\big)$ for some $b \in k((x))$, with $\mathbb{F}_p \cong G$ acting by $X \mapsto X + \lambda$. The length-one construction in the proof of Proposition 89 gives an $f \in x^{-1}k[x^{-1}]$ with $[f] = [b]$ in $k((x))/\wp(k((x)))$, where either $f = 0$, or $m := \deg_{x^{-1}} f$ is prime to p. Thus $b - f = c^p - c$ for some $c \in k((x))$, and replacing X by $X - c$ gives the equation $X^p - X = f$ for $\mathcal{Q}$. If $f = 0$, then $\mathcal{Q}$ is trivial; otherwise, Theorem 53 identifies m with its unique upper ramification break, so the assumed ramification bound is equivalent to $m < d$. The same equation defines a torsor over $\mathbb{G}_m$: it is finite étale over $k[x, x^{-1}]$ because its derivative with respect to X is -1, and since $f \in k[x^{-1}]$, it extends étale over $\operatorname{Spec} k[x^{-1}] = \mathbb{P}^1_k \setminus \{0\}$.

For (3), Artin–Schreier–Witt theory represents $\mathcal{Q}$ by an equation $\wp\vec{X} = \vec{g}$ for some $\vec{g} \in W_r(k((x)))$, with $W_r(\mathbb{F}_p) \cong G$ acting by Witt translation $\vec{X} \mapsto \vec{X} \oplus \vec{c}$. By Proposition 89, the class $[\vec{g}]$ has a reduced representative $\vec{f} = (f_0, \dots, f_{r-1})$ with each $f_j \in x^{-1}k[x^{-1}]$. Thus $\vec{g} = \vec{f} \oplus \wp\vec{h}$ for some $\vec{h} \in W_r(k((x)))$, and the change of variables $\vec{Y} = \vec{X} \ominus \vec{h}$ gives $\wp\vec{Y} = \vec{f}$; renaming $\vec{Y}$ as $\vec{X}$ gives the equation in the statement. Connectedness means that $[\vec{f}]$ has order p^r, so Corollary 86 gives $f_0 \neq 0$. Since k is algebraically closed, the corresponding degree-p^r extension is totally ramified. With the convention for m_j in the statement, Theorem 55 therefore gives

$$\max_j p^{r-1-j} m_j < d,$$

which is equivalent to (2.4). Finally, Lemma 84 applied over $k[x, x^{-1}]$ shows that the same equation defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$ whose restriction is $\mathcal{Q}$. Since every f_j lies in $k[x^{-1}]$, applying the same lemma over $k[x^{-1}]$ extends this torsor étale across ∞. □

Remark. The weighted pole bound in part (3) has the following immediate consequence. If $p^{r-1-j} \geq d$, then m_j cannot be positive, so $f_j = 0$. Thus sufficiently early Witt coordinates must vanish. In particular, since connectedness requires $f_0 \neq 0$, a connected $\mathbb{Z}/p^r\mathbb{Z}$-torsor necessarily has $d > p^{r-1}$. This observation is only a sanity check explaining the meaning of the weights and will not be used below.

2.4.5 The common contracted-product calculation

The explicit equations above enter the symmetric-power construction through one contracted-product calculation. We record it once for all three cyclic theories.

Let S be a scheme, let H be a commutative S-group scheme, and let $\varphi : H \to H$ be a finite étale surjective homomorphism with constant kernel G. For a section $a : X \to H$, write

$$\mathcal{T}_a := X \times_{a,H,\varphi} H.$$

Thus $\mathcal{T}_a$ is a G-torsor on X. Equivalently, for every X-scheme U,

$$\mathcal{T}_a(U) = \{h \in H(U) : \varphi(h) = a|_U\}.$$

In other words, a point of $\mathcal{T}_a$ is an element of H lying over a under φ. We write $\star$ for the group law of H: multiplication on $\mathbb{G}_m$, ordinary addition on $\mathbb{G}_a$, and Witt addition on $\mathbb{W}_r$.

Proposition 57 (Compatibility with contracted products)**.** *Let $a_i : X \to H$, $1 \le i \le d$, be sections. There is a canonical isomorphism of G-torsors*

$$\mathcal{T}_{a_1} \otimes^G \cdots \otimes^G \mathcal{T}_{a_d} \cong \mathcal{T}_{a_1 \star \cdots \star a_d}.$$

On local sections it is induced by

$$[(h_1, \ldots, h_d)] \longmapsto h_1 \star \cdots \star h_d, \qquad h_i \in \mathcal{T}_{a_i}.$$

In other words, the construction $a \mapsto \mathcal{T}_a$ is compatible with the group law on $H(X)$ and the contracted-product group law on G-torsors. Moreover, if a group Γ acts on this construction by permuting the indexed factors, the isomorphism is Γ-equivariant and hence descends whenever the relevant quotients exist.

Proof. By the definition of the contracted product, the left-hand side is

$$K_G \backslash \prod_i \mathcal{T}_{a_i}, \qquad K_G = \ker(G^d \xrightarrow{m} G),$$

where K_G acts coordinatewise by

$$(g_1, \ldots, g_d) \cdot (h_1, \ldots, h_d) = (g_1 \star h_1, \ldots, g_d \star h_d).$$

Consider the map

$$F : \prod_i \mathcal{T}_{a_i} \longrightarrow H, \qquad (h_1, \ldots, h_d) \longmapsto h_1 \star \cdots \star h_d.$$

Since $\varphi(h_i) = a_i$ and φ is a homomorphism,

$$\varphi\big(F(h_1, \ldots, h_d)\big) = a_1 \star \cdots \star a_d,$$

so F takes values in $\mathcal{T}_{a_1 \star \cdots \star a_d}$. Moreover, F is K_G-invariant because, for $(g_1, \ldots, g_d) \in K_G$, the product $g_1 \star \cdots \star g_d$ is the identity. It therefore descends to the contracted product. The descended map is G-equivariant, and hence is an isomorphism of G-torsors by Proposition 5. Finally, commutativity of $\star$ gives the asserted Γ-equivariance. □

2.5 The Cyclic Exceptional–Divisor Calculation and Final Assembly

We are now ready to prove the main result. The previous sections reduce the problem to three cases, according to whether the cyclic group has order prime to p, order p, or order p^r. The following theorem proves these three cases, which together imply Theorem 35.

Theorem 58 (The three cyclic cases)**.** *Assume that k is algebraically closed. Let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ be a G-torsor with G cyclic, and let $\mathfrak{n} = dP \subset \mathfrak{m}$ be a single-point submodulus.*

1. *If $G = \mathbb{Z}/e\mathbb{Z}$ with $(e, p) = 1$, then $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*
2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification of $\mathcal{P}$ at P is bounded by d, then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*
3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$, $r \ge 2$, and the ramification of $\mathcal{P}$ at P is bounded by d, then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*

Proof. Since k is algebraically closed, P is k-rational. Choose a uniformizer x at P. It identifies the completed local ring at P with $k[[x]]$ and its fraction field with $k((x))$. Let $\mathcal{P}_x$ be the restriction of $\mathcal{P}$ to the punctured formal disc $\operatorname{Spec} k((x))$. By Corollary 14, the isomorphism class of $\mathcal{P}_x$ corresponds to a unique continuous homomorphism

$$\rho : \operatorname{Gal}(k((x))^{\mathrm{sep}}/k((x))) \longrightarrow G.$$

Put $H := \operatorname{im}(\rho)$. Under the construction of Theorem 13, the surjection onto H determines a connected H-torsor $\mathcal{Q}$ over $\operatorname{Spec} k((x))$, and, for the inclusion $\iota : H \hookrightarrow G$, one has $\mathcal{P}_x \cong \iota_*\mathcal{Q}$. If H is trivial, then $\mathcal{P}_x$ is trivial. Formal–local invariance (Lemma 50) then shows that $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$, which proves all three conclusions in this case.

Assume $H \neq 0$. In parts (2) and (3), the assumed ramification bound passes from $\mathcal{P}_x$ to $\mathcal{Q}$ by Lemma 48(3), because $k[[x]]$ is complete and its residue field k is perfect.

The realization lemma (Lemma 56) gives an explicit connected H-torsor $\mathcal{Q}'$ on $\mathbb{G}_m$ whose restriction to $\operatorname{Spec} k((x))$ is $\mathcal{Q}$. Hence the induced G-torsor $\iota_*\mathcal{Q}'$ has the same punctured-formal-disc restriction as $\mathcal{P}$. The reduction to $\mathbb{P}^1$ (Corollary 51) therefore identifies the relevant ramification of $\mathcal{P}^{[d]}$ with that of $(\iota_*\mathcal{Q}')^{[d]}$. The compatibility of symmetric powers with induced torsors gives

$$(\iota_*\mathcal{Q}')^{[d]} \cong \iota_*(\mathcal{Q}'^{[d]})$$

by Lemma 29. Parts (1) and (2) of Lemma 48 then show that tameness or unramifiedness of $\mathcal{Q}'^{[d]}$ is equivalent to the same property for the induced torsor. Thus it is enough to treat the connected explicit model

$$C = \mathbb{P}^1_k, \qquad U = \mathbb{G}_m, \qquad \mathfrak{n} = d \cdot 0.$$

For the rest of the calculation, we rename the subgroup H as G and the realized torsor $\mathcal{Q}'$ as $\mathcal{P}$. In Case (1), write $n := |G|$ for the order of this renamed subgroup. Since it is a subgroup of the original group $\mathbb{Z}/e\mathbb{Z}$, one has $n \mid e$, and hence $(n, p) = 1$.

Write $x_1, \dots, x_d$ for the coordinates on $\mathbb{G}_m^d$ and $e_1, \dots, e_d$ for their elementary symmetric polynomials. By Lemma 50, the completed local ring and field at the generic point $\eta = \eta_{d\cdot 0}$ of the exceptional divisor are

$$\widehat{\mathcal{O}}_\eta = \kappa[[e_d]], \qquad \widehat{K}_\eta = \kappa((e_d)), \qquad \kappa = k(e_1/e_d, \dots, e_{d-1}/e_d). \tag{2.5}$$

Each of the three calculations follows the same plan. We write the local torsor in a standard form, combine the d equations using the relevant group law, and check the resulting coefficient at the exceptional valuation. The calculation has the following shape:

	$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p^r\mathbb{Z}$
local equation	$T^n = x^a$	$X^p - X = f$	$\wp\vec{X} = \vec{f}$
d-fold coordinate	$\prod_i T_i$	$\sum_i X_i$	$\bigoplus_i \vec{X}_i$
coefficient after descent	e_d^a	$\sum_i f(x_i)$	$\bigoplus_i \vec{f}(x_i)$
at the exceptional valuation	$v(e_d^a) = a$	regular	componentwise regular
conclusion	tame	unramified	unramified.

Case (1): Kummer. Put

$$R := k[x, x^{-1}], \qquad U = \mathbb{G}_m = \operatorname{Spec} R.$$

By Lemma 56(1), the connected local torsor $\mathcal{Q}$ is represented by

$$T^n = x^a$$

for some integer a with $(a, n) = 1$, and the realized torsor $\mathcal{P} = \mathcal{Q}' \to U$ is defined by the same equation. Equivalently, because k is algebraically closed, we may choose an identification $\mu_n \cong G$, and then $\mathcal{P}$ fits into the Cartesian square

$$\begin{array}{ccc} \mathcal{P} & \longrightarrow & \mathbb{G}_m \\ \downarrow & & \downarrow{\scriptstyle [n]} \\ U & \xrightarrow[x^a]{} & \mathbb{G}_m, \end{array}$$

where $[n](T) = T^n$ and the lower horizontal arrow is the morphism corresponding to the section $x^a \in \mathbb{G}_m(U) = R^\times$. Thus $\mathcal{P}$ is the pullback of $[n]$ along the section x^a. Put

$$S := R[T]/(T^n - x^a).$$

Thus $\mathcal{P} \cong \operatorname{Spec} S$ as a G-torsor over U. After pulling it back along the d projections $U^d \to U$, the product torsor has coordinate algebra $S^{\otimes_k d}$ over $R^{\otimes_k d}$. Contracting the product action G^d along $G^d \to G$ means taking invariants under $K_G = \ker(G^d \to G)$. Write T_i for the pullback of the coordinate T along the i-th projection from the product torsor. By Proposition 57, the contracted product of the d torsors

$$T_i^n = x_i^a$$

is obtained by multiplying their defining sections. It therefore has the equation

$$Y^n = \prod_{i=1}^{d} x_i^a = (x_1 \cdots x_d)^a = e_d^a.$$

Here Y is the coordinate on the resulting torsor. Under the isomorphism of Proposition 57, its pullback to the product torsor is $\prod_{i=1}^d T_i$. Consequently,

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[Y]/(Y^n - e_d^a).$$

The group S_d permutes the tensor factors and fixes both Y and e_d^a. The algebra on the right is free over $R^{\otimes_k d}$ with the S_d-invariant basis $1, Y, \ldots, Y^{n-1}$, so taking S_d-invariants acts only on its coefficients. Since

$$(R^{\otimes_k d})^{S_d} = k[x_1^{\pm 1}, \ldots, x_d^{\pm 1}]^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}],$$

Proposition 28 identifies the descended torsor with

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \ldots, e_d, e_d^{-1}][Y]}{(Y^n - e_d^a)}.$$

After base change to $\widehat{K}_\eta = \kappa((e_d))$, we obtain

$$\mathcal{P}^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \cong \operatorname{Spec} \frac{\widehat{K}_\eta[Y]}{(Y^n - e_d^a)}.$$

Since e_d is a uniformizer of $\widehat{\mathcal{O}}_\eta = \kappa[[e_d]]$, we have

$$v_\eta(e_d^a) = a.$$

Together with $\gcd(a,n)=1$, Theorem 52 shows that the Kummer extension defined by $Y^n = e_d^a$ is totally and tamely ramified. Hence $\mathcal{P}^{[d]}$ is tamely ramified at η, proving part (1).

Case (2): Artin–Schreier. With $R = k[x,x^{-1}]$ and $U = \mathbb{G}_m = \operatorname{Spec} R$ as in Case (1), Lemma 56(2) presents the realized torsor $\mathcal{P} \to U$ as

$$X^p - X = f(x), \qquad 0 \neq f \in x^{-1}k[x^{-1}], \qquad m := \deg_{x^{-1}} f < d, \quad p \nmid m.$$

Equivalently, $\mathcal{P}$ is the pullback of the finite étale homomorphism

$$\wp : \mathbb{G}_a \longrightarrow \mathbb{G}_a, \qquad X \longmapsto X^p - X,$$

whose kernel is $\mathbb{F}_p \cong G$, along the section $f \in \mathbb{G}_a(U) = R$. Put

$$S := R[X]/(X^p - X - f(x)).$$

Then $\mathcal{P} \cong \operatorname{Spec} S$. As in Case (1), the contracted product has coordinate algebra $(S^{\otimes_k d})^{K_G}$, where $K_G := \ker(G^d \to G)$. Write X_i for the pullback of X from the i-th factor. Since the group law on $\mathbb{G}_a$ is addition, Proposition 57 shows that the defining sections $f(x_i)$ add, giving the equation

$$Y^p - Y = \alpha, \qquad \alpha := \sum_{i=1}^{d} f(x_i).$$

Here Y is the coordinate on the contracted product, and its pullback to the product torsor is $\sum_{i=1}^d X_i$. Consequently,

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[Y]/(Y^p - Y - \alpha).$$

As in Case (1), S_d fixes Y and α, and taking invariants acts only on the coefficients. Thus Proposition 28 gives

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1,\dots,e_d,e_d^{-1}][Y]}{(Y^p - Y - \alpha)}.$$

Put $y_i = x_i^{-1}$. The polynomial α is symmetric in $y_1,\dots,y_d$, and all its monomials have total degree at most $m < d$. Hence Lemma 24 gives

$$\alpha \in \kappa \subset \widehat{\mathcal{O}}_\eta.$$

Thus, although the individual $f(x_i)$ have poles at $x_i = 0$, their symmetric sum has no pole at the exceptional valuation. Set $A := \widehat{\mathcal{O}}_\eta$ and

$$B := A[Y]/(Y^p - Y - \alpha).$$

The defining polynomial is monic and has derivative -1, so B/A is finite étale. After base change to $\widehat{K}_\eta = \kappa((e_d))$, we obtain

$$\mathcal{P}^{[d]}\big|_{\operatorname{Spec}\widehat{K}_\eta} \cong \operatorname{Spec}\big(B \otimes_A \widehat{K}_\eta\big) = \operatorname{Spec} \frac{\widehat{K}_\eta[Y]}{(Y^p - Y - \alpha)}.$$

Every field factor is therefore unramified over $\widehat{K}_\eta$. Hence $\mathcal{P}^{[d]}$ is unramified at η, proving part (2).

Case (3): Artin–Schreier–Witt. We use a direct Artin–Schreier–Witt calculation; Appendix B explains why induction through the degree-p layers does not suffice after the intervening wild base change.

The common reduction replaces the original group $\mathbb{Z}/p^r\mathbb{Z}$ by the subgroup $H = \mathrm{im}(\rho)$, of order p^s for some $1 \le s \le r$, and renames H as G. Case (2) treats $s = 1$, so we may assume that $G = \mathbb{Z}/p^s\mathbb{Z}$ with $s \ge 2$.

With $R = k[x, x^{-1}]$ and $U = \mathbb{G}_m = \operatorname{Spec} R$ as above, Lemma 56(3) presents the realized torsor $\mathcal{P} \to U$ as

$$\wp\vec{X} = \vec{f} = (f_0, \dots, f_{s-1}), \qquad f_j \in x^{-1}k[x^{-1}],$$

where, on setting $m_j = \deg_{x^{-1}} f_j$ for $f_j \neq 0$ and $m_j = 0$ for $f_j = 0$, one has

$$p^{s-1-j} m_j < d \qquad (0 \le j < s).$$

Equivalently, $\mathcal{P}$ is the pullback of the finite étale homomorphism

$$\wp = F - \mathrm{id} : \mathbb{W}_s \longrightarrow \mathbb{W}_s,$$

whose kernel is $W_s(\mathbb{F}_p) \cong G$, along the section $\vec{f} \in \mathbb{W}_s(U) = W_s(R)$. Put

$$S := R[\vec{X}]/(\wp\vec{X} \ominus \vec{f}).$$

Then $\mathcal{P} \cong \operatorname{Spec} S$. As in the previous cases, the contracted product has coordinate algebra $(S^{\otimes_k d})^{K_G}$, where $K_G := \ker(G^d \to G)$. Write $\vec{X}_i$ for the pullback of $\vec{X}$ from the i-th factor. Since the group law on $\mathbb{W}_s$ is Witt addition, Proposition 57 shows that the defining sections $\vec{f}(x_i)$ add, giving the equation

$$\wp\vec{Y} = \vec{\alpha}, \qquad \vec{\alpha} := \bigoplus_{i=1}^{d} \vec{f}(x_i).$$

Here $\vec{Y}$ is the Witt-vector coordinate on the contracted product, and its pullback to the product torsor is $\bigoplus_{i=1}^{d} \vec{X}_i$. Consequently,

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha}).$$

The S_d-descent follows the same pattern as in Cases (1) and (2): the group S_d fixes $\vec{Y}$ and $\vec{\alpha}$ componentwise, and Lemma 84 supplies an S_d-invariant basis over $R^{\otimes_k d}$, so taking invariants acts only on the coefficients. Thus Proposition 28 gives

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]}{(\wp\vec{Y} \ominus \vec{\alpha})}.$$

Put $y_i = x_i^{-1}$. Since each $f_j(x_i)$ is a polynomial in y_i, and Witt addition is given by universal polynomials, every component α_n is a polynomial in $y_1, \dots, y_d$. Moreover, commutativity of Witt addition makes α_n symmetric in these variables. We now show that every monomial of α_n has total degree strictly less than d. Since Witt addition is not componentwise, we control these degrees using the Witt weights: assign $f_j(x_i)$ weight p^j. By Lemma 90, every nonzero monomial

$$\prod_{i,j} f_j(x_i)^{a_{ij}}$$

appearing in α_n satisfies

$$\sum_{i,j} a_{ij} p^j = p^n.$$

Put $A_j := \sum_i a_{ij}$. Since $f_j(x_i)$ has degree m_j in y_i, every monomial obtained after expansion has total degree at most $\sum_j A_j m_j$. The pole bounds therefore give

$$\sum_j A_j m_j < d \sum_j A_j p^{j-(s-1)} = d\, p^{n-(s-1)} \le d.$$

The first inequality is strict because a nonzero monomial contains some nonzero f_j, whose pole bound is strict. Thus every monomial in α_n has total degree strictly less than d, and Lemma 24 gives

$$\alpha_n \in \kappa \qquad (0 \le n < s),$$

and therefore

$$\vec{\alpha} \in W_s(\kappa) \subset W_s(\widehat{\mathcal{O}}_\eta).$$

Set $A := \widehat{\mathcal{O}}_\eta$ and

$$B := A[\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha}).$$

By Lemma 84, the algebra B is finite étale over A. After base change to $\widehat{K}_\eta = \kappa((e_d))$, we obtain

$$\mathcal{P}^{[d]}\big|_{\operatorname{Spec}\widehat{K}_\eta} \cong \operatorname{Spec}\big(B \otimes_A \widehat{K}_\eta\big) = \operatorname{Spec}\frac{\widehat{K}_\eta[\vec{Y}]}{(\wp\vec{Y} \ominus \vec{\alpha})}.$$

Every field factor is therefore unramified over $\widehat{K}_\eta$. Hence $\mathcal{P}^{[d]}$ is unramified at η, proving part (3). □

Proof of Theorem 35. Part (1) of Theorem 58 proves the conclusion for cyclic groups of order prime to p, while parts (2) and (3) prove it for cyclic p-groups. The result for every finite abelian group G now follows directly from Proposition 49. □

Chapter 3

Geometric Class Field Theory

The ramification theorem of Chapter 2 is the key input for the proof of Theorem 2. Retain the notation and hypotheses of that theorem: C is a smooth, projective, geometrically connected curve over the perfect field k, $\mathfrak{m}$ is a modulus, $U = C \setminus \mathfrak{m}$, Λ is a finite commutative ring whose cardinality is invertible in k, and $\mathcal{F}$ is a rank-one Λ-local system on U whose ramification is bounded by $\mathfrak{m}$. Our goal is to descend $\mathcal{F}^{[d]}$ along

$$\Phi_d \colon U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}$$

for all sufficiently large d.

Set $G = \Lambda^\times$ and let $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$ be the frame torsor of $\mathcal{F}$. By Proposition 9, together with its compatibility with pullback, symmetric powers, and ramification, it is enough to construct a G-torsor $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$ such that

$$\Phi_d^* \mathcal{Q}_d \cong \mathcal{P}^{[d]}.$$

We therefore work with G-torsors throughout the proof and return to rank-one local systems only in the final step.

When $\mathfrak{m} = 0$, the projective-space fibers of Φ_d give the descent directly. When $\mathfrak{m} > 0$, Φ_d is not proper and its fibers are affine, so we instead use Takeuchi's extension to a projective space bundle

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}},$$

where $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of the blowup of $C^{(d)}$ along the locus obtained by adding the modulus.

The single-point calculation of Chapter 2, combined with compatibility under products and addition of divisors, first shows that $\mathcal{P}^{[d]}$ is tamely ramified along the boundary

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}.$$

Passing to the geometric generic fiber, we then show that $\mathcal{P}^{[d]}$ extends uniquely across H; the extension then descends along the projective-space bundle $\widetilde{\Phi}_d$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$.

The remainder of the chapter makes these steps precise. Section 3.1 recalls the rigidified Picard scheme and the Abel–Jacobi morphism. Section 3.2 introduces Takeuchi's compactification. Section 3.3 proves tameness along its boundary, and Section 3.4 extends the torsor across that boundary. Finally, Section 3.5 completes the descent and returns from torsors to rank-one local systems.

3.1 The Rigidified Picard Scheme and the Abel–Jacobi Map

Our presentation of the rigidified Picard scheme and the Abel–Jacobi morphism follows Guignard [Gui19, §4] and Takeuchi [Tak18, §3]. For background on the classical theory of Jacobians and relative Picard schemes, we refer respectively to Milne [Mil08] and Bosch–Lütkebohmert–Raynaud [BLR90, Chapter 9]. Precise references for the individual results are given where they are used.

3.1.1 The rigidified Picard scheme

Although the main theorem is over the fixed perfect field k, the rigidified Picard construction is most naturally stated over an arbitrary base scheme S. Let S be a scheme and let $f : C \to S$ be a projective smooth morphism whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C, defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \to T$ by pr for any S-scheme T.

Let d be an integer. For an S-scheme T, we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T.
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $\alpha\colon \mathcal{L} \to \mathcal{L}'$ such that the following diagram commutes:

$$\begin{array}{ccc} & \mathcal{O}_{\mathfrak{m}_T} & \\ {\scriptstyle\psi}\swarrow & & \searrow{\scriptstyle\psi'} \\ \mathcal{L}|_{\mathfrak{m}_T} & \xrightarrow[\alpha|_{\mathfrak{m}_T}]{} & \mathcal{L}'|_{\mathfrak{m}_T} \end{array}$$

We define the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ on $\mathrm{Sch}\,/S$ by assigning to T the set of isomorphism classes of such pairs. Let $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ denote the étale sheafification of this presheaf.

We use the following structural facts, recorded in [Tak18, §3, fundamental properties (1)–(4)]. When $\mathfrak{m}$ is everywhere positive, representability and smoothness of the full rigidified Picard presheaf are proved in [Gui19, Proposition 4.8]; when $\mathfrak{m} = 0$, the corresponding results for the ordinary relative Picard scheme are given in [BLR90, §9.3, Theorem 1].

1. $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is represented by an S-scheme. If $\mathfrak{m}$ is faithfully flat over S, the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ is already an étale sheaf.
2. $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$ is a smooth commutative group S-scheme with geometrically connected fibers.
3. For any d, $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is a $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$-torsor.

We call $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$ the *generalized Jacobian* of C with modulus $\mathfrak{m}$. When $\mathfrak{m} = 0$, it is the ordinary Jacobian, and $\mathrm{Pic}^{d}_{C,\mathfrak{m}} = \mathrm{Pic}^{d}_{C}$.

The rigidified Picard schemes are naturally related as the modulus varies. Indeed, let $\mathfrak{m}_1 \subset \mathfrak{m}_2$. Restriction of rigidifications defines a natural morphism of étale sheaves

$$\rho_{\mathfrak{m}_2,\mathfrak{m}_1} \colon \operatorname{Pic}^d_{C,\mathfrak{m}_2} \longrightarrow \operatorname{Pic}^d_{C,\mathfrak{m}_1}.$$

When $\mathfrak{m}_2$ is finite over S, this morphism is an epimorphism [Tak18, §3]. Taking $\mathfrak{m}_1 = 0$ gives, for every modulus $\mathfrak{m}$, a natural epimorphism of étale sheaves

$$\rho_{\mathfrak{m},0} \colon \operatorname{Pic}^d_{C,\mathfrak{m}} \longrightarrow \operatorname{Pic}^d_{C}.$$

3.1.2 The Abel–Jacobi Morphism and Its Fibers

Let $U = C \setminus \mathfrak{m}$ be the complement of the modulus in C. The effective Cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\operatorname{Sym}^d_S(U) = U^{(d)}$ over S (See [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\operatorname{supp}(D) \cap \operatorname{supp}(\mathfrak{m}) = \emptyset$. Its restriction is a nowhere-vanishing section on $\mathfrak{m}$, thereby determining a trivialization $\psi_D : \mathcal{O}_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_C(D)|_{\mathfrak{m}}$. This is functorial in families and yields a morphism from the symmetric power to the rigidified Picard scheme over S:

$$\Phi_d \colon U^{(d)} \to \operatorname{Pic}^d_{C,\mathfrak{m}}, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)], \tag{3.1}$$

The results of Guignard and Takeuchi used below require the degree to be sufficiently large.[1] Let g denote the genus of C, viewed as a locally constant function on S. From this point on, we impose the standing condition

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \tag{3.2}$$

This is Takeuchi's condition [Tak18, §3, (3.1)] and is, in particular, strong enough for the application of [Gui19, Proposition 4.14] below. If S is quasi-compact, the locally constant functions g and $\deg \mathfrak{m}$ take only finitely many values; hence a single integer d satisfying this condition exists.

Suppose that d satisfies (3.2). When $\mathfrak{m} = 0$ and $C \to S$ admits a section, the Abel–Jacobi morphism

$$C^{(d)} \longrightarrow \operatorname{Pic}^d_C$$

is a projective space bundle. In particular, it is proper and smooth with geometrically connected fibers.

When $\mathfrak{m} > 0$, Guignard [Gui19, Proposition 4.14] proves the stronger statement that

$$\Phi_d \colon U^{(d)} \longrightarrow \operatorname{Pic}^d_{C,\mathfrak{m}}$$

is, for the fppf topology, a torsor under the additive group scheme of a vector bundle of rank $d - \deg \mathfrak{m} - g + 1$ over $\operatorname{Pic}^d_{C,\mathfrak{m}}$. In particular, Φ_d is fppf-locally isomorphic over $\operatorname{Pic}^d_{C,\mathfrak{m}}$ to an affine space of relative dimension $d - \deg \mathfrak{m} - g + 1$; it is smooth and surjective, with geometrically connected fibers.

[1] The cohomological reason for this bound is that, for a degree-d line bundle $\mathcal{L}$, it ensures that both $\mathcal{L}$ and $\mathcal{L}(-\mathfrak{m})$ have fiberwise degree at least $2g - 1$. Their first cohomology then vanishes by Serre duality and a degree argument; Riemann–Roch together with cohomology and base change implies that the corresponding pushforwards are locally free of the expected ranks and commute with base change. See [Gui19, Proposition 4.3]. The additional inequality $d \geq \deg \mathfrak{m}$ ensures that $d - \deg \mathfrak{m}$ is nonnegative, which is relevant when $g = 0$.

We now return to the standing situation $S = \operatorname{Spec}(k)$, which remains in force for the rest of the chapter. From Guignard's result, we retain only the following fiberwise consequence, stated together with the classical unramified case.

Theorem 59. *Assume that $S = \operatorname{Spec}(k)$ and that d satisfies (3.2). Let $\bar{s} = \operatorname{Spec}\Omega \to \operatorname{Pic}^d_{C,\mathfrak{m}}$ be a geometric point, where Ω is separably closed. Then the geometric fiber of the Abel–Jacobi morphism*

$$\Phi_d \colon U^{(d)} \longrightarrow \operatorname{Pic}^d_{C,\mathfrak{m}}$$

at $\bar{s}$ is isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg\mathfrak{m}-g+1}_{\Omega} & \text{if } \mathfrak{m} > 0, \\ \mathbb{P}^{d-g}_{\Omega} & \text{if } \mathfrak{m} = 0. \end{cases}$$

Thus every geometric fiber is an affine space when $\mathfrak{m} > 0$ and a projective space when $\mathfrak{m} = 0$.

Proof. See [Ten15] Propositions 3.13-3.14, or [Tót11] Proposition 2.1.4.

□

3.2 Takeuchi's Compactification of the Abel–Jacobi Morphism

We recall that our objective is to descend the G-torsor $\mathcal{P}^{[d]}$ from $U^{(d)}$ to $\operatorname{Pic}^d_{C,\mathfrak{m}}$ along the Abel–Jacobi map Φ_d:

$$\begin{array}{ccc} \mathcal{P}^{[d]} & & \\ \downarrow & & \\ U^{(d)} & \xrightarrow{\Phi_d} & \operatorname{Pic}^d_{C,\mathfrak{m}} \end{array}$$

(Here, the purple arrow emphasizes that $\mathcal{P}^{[d]}$ is a torsor on the étale site of $U^{(d)}$.)

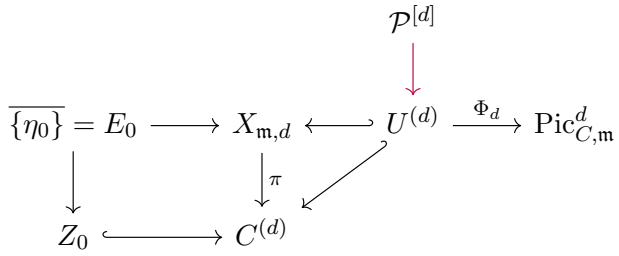
In the ramified case $\mathfrak{m} > 0$, Theorem 59 identifies the geometric fibers of Φ_d with affine spaces. Since Φ_d is not proper, the proper-smooth fundamental-group argument available in the unramified case cannot be applied directly. Takeuchi [Tak18, §3] overcomes this difficulty by extending Φ_d to a projective-space bundle. We now recall the ambient blowup and the properties of the resulting compactification that will be used below.

Let $\mathfrak{m} = \sum_{i=1}^{s} n_i P_i$, where the P_i are distinct closed points of C, be a modulus on C, and let d satisfy (3.2). Takeuchi [Tak18, §3, before (3.6)] defines $Z_0 = Z_0(\mathfrak{m}, d)$ as the closed subscheme of $C^{(d)}$ given by the image of the closed immersion $C^{(d-\deg\mathfrak{m})} \to C^{(d)}$ that adds $\mathfrak{m}$. He also defines $X_{\mathfrak{m},d}$ as the blowup of $C^{(d)}$ along Z_0. Let $E_0 = E_{\mathfrak{m},d} = Z_0(\mathfrak{m}, d) \times_{C^{(d)}} X_{\mathfrak{m},d}$ be the exceptional divisor. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathfrak{m},d}$ be its generic point.

Diagrammatically:

$$\begin{array}{ccc} \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ Z_0 & \hookrightarrow & C^{(d)} \end{array}$$

Incorporating $U^{(d)}$, the torsor $\mathcal{P}^{[d]}$, and the Abel–Jacobi map, we have:



We will not need Takeuchi's explicit fiber-product definition of the compactification. The properties required below are summarized in the following existence and structure statement, which follows from [Tak18, §3, (3.9), Proposition 3.7, and Corollary 3.8].

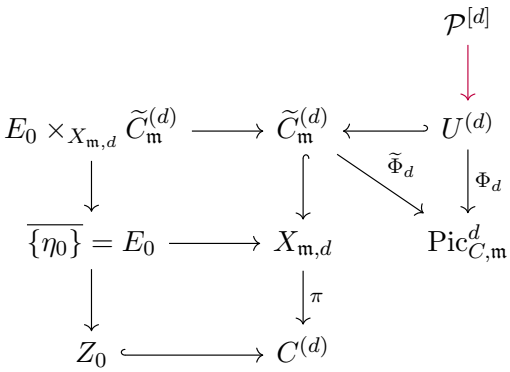
Theorem 60 (Takeuchi)**.** *Let d satisfy (3.2). There exists an open subscheme $\widetilde{C}^{(d)}_{\mathfrak{m}} \subset X_{\mathfrak{m},d}$ containing $U^{(d)}$ and a morphism*

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}$$

extending $\Phi_d \colon U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$, such that $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Furthermore, the complement of $U^{(d)}$ in $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is the scheme-theoretic intersection

$$E_0 \times_{X_{\mathfrak{m},d}} \widetilde{C}^{(d)}_{\mathfrak{m}} \cong Z_0 \times_{C^{(d)}} \widetilde{C}^{(d)}_{\mathfrak{m}}.$$

Diagrammatically we have:



Here the boundary may equivalently be written as $Z_0 \times_{C^{(d)}} \widetilde{C}^{(d)}_{\mathfrak{m}}$, since $E_0 = Z_0 \times_{C^{(d)}} X_{\mathfrak{m},d}$.

Takeuchi's description of the compactification identifies its boundary as a relative hyperplane; see [Tak18, Lemmas 3.2 and 3.3(1), Theorem 3.4, (3.9), and Corollary 3.8]. This gives the following fiberwise description.

Corollary 61 (The boundary is a relative hyperplane)**.** *The boundary*

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$$

is a smooth relative hyperplane divisor in the projective-space bundle $\widetilde{\Phi}_d : \widetilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$. Consequently, if $\bar{s} = \operatorname{Spec} \Omega \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ is any geometric point and $r = d - \deg \mathfrak{m} - g + 1$, then there is an

isomorphism of pairs[2]

$$\Big((\widetilde{C}^{(d)}_{\mathfrak{m}})_{\bar{s}}, H_{\bar{s}}\Big) \cong \Big(\mathbb{P}^r_\Omega, \mathbb{P}^{r-1}_\Omega\Big),$$

and hence $U^{(d)}_{\bar{s}} \cong \mathbb{A}^r_\Omega$.

Proof. The formation of a projective bundle and of its relative hyperplane commutes with arbitrary base change. Hence, after base change to $\bar{s}$, the projective-space bundle $\widetilde{C}^{(d)}_{\mathfrak{m}}$ becomes $\mathbb{P}^r_\Omega$ and H becomes a hyperplane $\mathbb{P}^{r-1}_\Omega$. Its complement is therefore $\mathbb{A}^r_\Omega$. □

3.3 Tame Ramification Along the Full Boundary

The ramification calculation of Chapter 2 proves the required tameness when the modulus is supported at a single point and the degree is the degree of that modulus. We now remove these two restrictions. First, we pass from the point-supported components of $\mathfrak{m}$ to the full modulus in degree $\deg \mathfrak{m}$. We then compare the blowup at the point $\mathfrak{m} \in C^{(\deg \mathfrak{m})}$ with the blowup in degree d along the locus of divisors containing $\mathfrak{m}$. This proves the following result.

Theorem 62. *Let G be a finite abelian group, and let $\mathcal{P}$ be a G-torsor on U whose ramification is bounded by $\mathfrak{m}$. For every sufficiently large integer d, the torsor $\mathcal{P}^{[d]}$ is tamely ramified along*

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = E_0 \times_{X_{\mathfrak{m},d}} \widetilde{C}^{(d)}_{\mathfrak{m}} \cong Z_0 \times_{C^{(d)}} \widetilde{C}^{(d)}_{\mathfrak{m}}.$$

To prove the theorem, we retain the blowup notation of Chapter 2. For every nonzero effective submodulus $\mathfrak{n} \leq \mathfrak{m}$, we identify $\mathfrak{n}$ with the corresponding k-point of $C^{(\deg \mathfrak{n})}$, and put

$$X_{\mathfrak{n}} = \mathrm{Bl}_{\mathfrak{n}}\big(C^{(\deg \mathfrak{n})}\big), \qquad E_{\mathfrak{n}} = X_{\mathfrak{n}} \times_{C^{(\deg \mathfrak{n})}} \{\mathfrak{n}\}.$$

We write $\eta_{\mathfrak{n}}$ for the generic point of $E_{\mathfrak{n}}$. For the full modulus in degree d, the center is

$$Z_0 = Z_0(\mathfrak{m}, d) = \mathfrak{m} + C^{(d-\deg \mathfrak{m})} \subset C^{(d)},$$

where the right-hand side denotes the image of the closed immersion $D \mapsto \mathfrak{m} + D$. The corresponding blowup, exceptional divisor, and generic point are denoted by $X_{\mathfrak{m},d}$, $E_0 = E_{\mathfrak{m},d}$, and $\eta_0 = \eta_{\mathfrak{m},d}$, as in Section 3.2.

From point-supported moduli to the full modulus.

We first show that if the relevant torsors are tame for two coprime submoduli, then the corresponding torsor is tame for their sum. Iterating this step over the point-supported components of $\mathfrak{m}$ yields the required statement for the full modulus in degree $\deg \mathfrak{m}$.

To establish this implication, we use the following general comparison. Let X and Y be smooth integral schemes over a field k, and let $x \in X(k)$ and $y \in Y(k)$. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$. Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

Tame ramification is preserved under the contracted product of torsors after pullback to a product, as follows.

[2]For pairs (X, D) and (Y, E), an isomorphism of pairs is an isomorphism $f\colon X \xrightarrow{\sim} Y$ such that $D = X \times_Y E$ as closed subschemes of X. No canonical choice of f is asserted here.

Proposition 63. *Let G be a finite abelian group. Suppose $\mathcal{P}_X$ and $\mathcal{P}_Y$ are G-torsors defined on dense open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ disjoint from the respective exceptional divisors. There exist a dense open subset*

$$\widetilde{U} \subset \mathrm{Bl}_{(x,y)}(X \times_k Y)$$

containing $\eta_{X\times Y}$ and morphisms

$$r_X : \widetilde{U} \longrightarrow \mathrm{Bl}_x(X), \qquad r_Y : \widetilde{U} \longrightarrow \mathrm{Bl}_y(Y),$$

which send $\eta_{X\times Y}$ to η_X and η_Y, respectively. On the dense open subset

$$W = r_X^{-1}(U_X) \cap r_Y^{-1}(U_Y) \subset \widetilde{U},$$

define

$$\mathcal{P}_{X\times Y} := r_X^{-1}\mathcal{P}_X \otimes^G r_Y^{-1}\mathcal{P}_Y.$$

If $\mathcal{P}_X$ is tamely ramified at η_X and $\mathcal{P}_Y$ is tamely ramified at η_Y, then $\mathcal{P}_{X\times Y}$ is tamely ramified at $\eta_{X\times Y}$.

Proof. We first construct r_X and r_Y; the torsors enter only after these morphisms have been defined on the original blowups. Shrinking X and Y around x and y does not change the exceptional divisors or their local rings at the generic points. The ètale local description of smooth morphisms [Stacks, Tag 054L] therefore gives ètale morphisms

$$a : X \longrightarrow \mathbb{A}^n_k, \qquad b : Y \longrightarrow \mathbb{A}^m_k$$

carrying x and y to the origins; after shrinking further, we may assume that x and y are the scheme-theoretic inverse images of the respective origins. We write $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$ for the corresponding standard coordinates and also for their pullbacks to X and Y.

Write

$$B_n = \mathrm{Bl}_0(\mathbb{A}^n_k), \qquad B_m = \mathrm{Bl}_0(\mathbb{A}^m_k), \qquad B_{n,m} = \mathrm{Bl}_{(0,0)}(\mathbb{A}^n_k \times_k \mathbb{A}^m_k),$$

and denote their exceptional divisors by E_n, E_m, and $E_{n,m}$. The blowup $B_{n,m}$ is the subscheme of $(\mathbb{A}^n_k \times_k \mathbb{A}^m_k) \times_k \mathbb{P}^{n+m-1}_k$ defined by

$$\begin{cases} x_i w_j = x_j w_i & 1 \le i, j \le n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \le k, l \le m \\ x_i w_{n+j} = y_j w_i & 1 \le i \le n,\ 1 \le j \le m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}_k$. The exceptional divisor $E_{n,m}$ is isomorphic to $\mathbb{P}^{n+m-1}_k$.

Both $B_n \times_k B_m$ and $B_{n,m}$ are birational to $\mathbb{A}^n_k \times_k \mathbb{A}^m_k$; their common dense open is

$$(\mathbb{A}^n_k \setminus \{0\}) \times_k (\mathbb{A}^m_k \setminus \{0\}).$$

Let

$$q_{n,m} \colon B_{n,m} \longrightarrow \mathbb{P}^{n+m-1}_k$$

be the second projection. Define

$$\Omega_n = \bigcup_{i=1}^{n} D_+(w_i), \qquad \Omega_m = \bigcup_{j=1}^{m} D_+(w_{n+j}),$$

and

$$\widetilde{U}_{n,m} := q_{n,m}^{-1}(\Omega_n \cap \Omega_m) = \bigcup_{\substack{1\le i\le n\\ 1\le j\le m}} q_{n,m}^{-1}\big(D_+(w_i w_{n+j})\big).$$

There is a morphism

$$\begin{aligned} f_{n,m}\colon \widetilde{U}_{n,m} &\longrightarrow B_n \times_k B_m,\\ ((u,v),[w_1:\cdots:w_{n+m}]) &\longmapsto \big((u,[w_1:\cdots:w_n]),(v,[w_{n+1}:\cdots:w_{n+m}])\big). \end{aligned}$$

Thus we have a diagram

$$\begin{array}{ccc} & \widetilde{U}_{n,m} & \\ {\scriptstyle f_{n,m}}\swarrow & & \searrow \\ B_n \times_k B_m & & B_{n,m}. \end{array}$$

The restriction of $q_{n,m}$ induces an isomorphism $E_{n,m} \xrightarrow{\sim} \mathbb{P}_k^{n+m-1}$. Since neither $V_+(w_1,\dots,w_n)$ nor $V_+(w_{n+1},\dots,w_{n+m})$ contains the generic point of $\mathbb{P}_k^{n+m-1}$, we have $\eta_{n,m} \in \widetilde{U}_{n,m}$, where $\eta_{n,m}$ is the generic point of $E_{n,m}$. On the exceptional divisors, the restriction of $f_{n,m}$ is the coordinate-block morphism

$$\begin{aligned} E_{n,m} \cap \widetilde{U}_{n,m} &\longrightarrow E_n \times_k E_m,\\ [w_1:\cdots:w_{n+m}] &\longmapsto \big([w_1:\cdots:w_n],[w_{n+1}:\cdots:w_{n+m}]\big). \end{aligned}$$

This morphism is surjective and hence dominant. It therefore sends $\eta_{n,m}$ to the generic point (η_n,η_m) of $E_n \times_k E_m$.

We now return to the original schemes. By Theorem 32, there are canonical identifications

$$\begin{aligned} \mathrm{Bl}_x(X) &\cong X \times_{\mathbb{A}_k^n} B_n,\\ \mathrm{Bl}_y(Y) &\cong Y \times_{\mathbb{A}_k^m} B_m,\\ B := \mathrm{Bl}_{(x,y)}(X \times_k Y) &\cong (X \times_k Y) \times_{\mathbb{A}_k^n \times_k \mathbb{A}_k^m} B_{n,m}. \end{aligned}$$

Set

$$\widetilde{U} := (X \times_k Y) \times_{\mathbb{A}_k^n \times_k \mathbb{A}_k^m} \widetilde{U}_{n,m}.$$

It is a dense open neighborhood of $\eta_{X\times Y}$ in B. Base change of $f_{n,m}$ gives a morphism

$$f : \widetilde{U} \longrightarrow \mathrm{Bl}_x(X) \times_k \mathrm{Bl}_y(Y)$$

that sends $\eta_{X\times Y}$ to (η_X,η_Y). If p_X and p_Y are the two projections from its target, set

$$r_X = p_X \circ f, \qquad r_Y = p_Y \circ f.$$

These are the morphisms asserted in the statement. They fit into the commutative diagram

$$\begin{array}{ccccc} & & \mathrm{Bl}_{(x,y)}(X \times_k Y) & & \\ & & \big\uparrow & & \\ & & \widetilde{U} & & \\ & {\scriptstyle r_X}\swarrow & \big\downarrow{\scriptstyle f} & \searrow{\scriptstyle r_Y} & \\ \mathrm{Bl}_x(X) & \xleftarrow[p_X]{} & \mathrm{Bl}_x(X) \times_k \mathrm{Bl}_y(Y) & \xrightarrow[p_Y]{} & \mathrm{Bl}_y(Y). \end{array}$$

Over the common dense open $(X \setminus \{x\}) \times_k (Y \setminus \{y\})$, the morphism f is the evident identification; in particular, it is dominant. Hence

$$W = r_X^{-1}(U_X) \cap r_Y^{-1}(U_Y) = f^{-1}(U_X \times_k U_Y)$$

is a nonempty open subset of the integral scheme $\widetilde{U}$, and is therefore dense in $\widetilde{U}$ and in B.

It remains to compare the maps induced by r_X and r_Y on the exceptional valuations. Let

$$S = \mathcal{O}_{B,\eta_{X\times Y}}, \qquad R_X = \mathcal{O}_{\mathrm{Bl}_x(X),\eta_X}, \qquad R_Y = \mathcal{O}_{\mathrm{Bl}_y(Y),\eta_Y}.$$

The morphisms r_X and r_Y induce local homomorphisms

$$R_X \longrightarrow S, \qquad R_Y \longrightarrow S.$$

By base change, the exceptional divisors are identified with

$$E_X \cong \mathbb{P}_k^{n-1}, \qquad E_Y \cong \mathbb{P}_k^{m-1}, \qquad E_{X\times Y} \cong \mathbb{P}_k^{n+m-1},$$

and the restrictions of r_X and r_Y are the two coordinate projections

$$[w_1 : \cdots : w_{n+m}] \longmapsto [w_1 : \cdots : w_n], \qquad [w_1 : \cdots : w_{n+m}] \longmapsto [w_{n+1} : \cdots : w_{n+m}].$$

At the generic point of $E_{X\times Y}$ both coordinate blocks are nonzero. On the chart $D_+(w_1 w_{n+1})$, the functions x_1 and y_1 are uniformizers in R_X and R_Y, respectively, and each is also a uniformizer in S. Hence both extensions have ramification index one. Their residue-field extensions are

$$k\left(\frac{x_2}{x_1}, \ldots, \frac{x_n}{x_1}\right) \subset k\left(\frac{x_2}{x_1}, \ldots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \ldots, \frac{y_m}{x_1}\right)$$

and

$$k\left(\frac{y_2}{y_1}, \ldots, \frac{y_m}{y_1}\right) \subset k\left(\frac{x_1}{y_1}, \ldots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \ldots, \frac{y_m}{y_1}\right),$$

respectively. In particular, both are purely transcendental and hence separable.

We now apply this comparison to the torsors. The extensions $R_X \to S$ and $R_Y \to S$ are weakly unramified and have separable residue-field extensions. Hence Lemma 46 shows that the pullbacks of $\mathcal{P}_X$ and $\mathcal{P}_Y$ to $\mathrm{Frac}(S)$ are tamely ramified. By the definition above, their contracted product is the generic fiber of $\mathcal{P}_{X\times Y}$. Therefore Lemma 45 shows that $\mathcal{P}_{X\times Y}$ is tamely ramified at $\eta_{X\times Y}$. □

Corollary 64. *Let G be a finite abelian group, and let $\mathcal{P}$ be a G-torsor on U. Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime nonzero submoduli. Assume that $\mathcal{P}^{[\deg \mathfrak{n}_1]}$ and $\mathcal{P}^{[\deg \mathfrak{n}_2]}$ are tamely ramified at $\eta_{\mathfrak{n}_1}$ and $\eta_{\mathfrak{n}_2}$, respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is tamely ramified at $\eta_{\mathfrak{n}_1+\mathfrak{n}_2}$.*

Proof. Put $d_i = \deg \mathfrak{n}_i$ for $i = 1, 2$, and pull $\mathcal{P}^{[d_i]}$ back to $X_{\mathfrak{n}_i}$ over the inverse image of $U^{(d_i)}$. Applying Proposition 63 to the two blowups $X_{\mathfrak{n}_i} \to C^{(d_i)}$, and writing r_i for the morphisms supplied by that proposition, shows that

$$r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}$$

is tame; i.e., this contracted product has tame ramification at the generic point of the exceptional divisor of

$$\mathrm{Bl}_{(\mathfrak{n}_1,\mathfrak{n}_2)}\big(C^{(d_1)} \times_k C^{(d_2)}\big).$$

By Lemma 65, applied with $\mathfrak{m} = \mathfrak{n}_1 + \mathfrak{n}_2$ and $d = d_1 + d_2$, the addition morphism

$$+_{d_1,d_2} \colon C^{(d_1)} \times_k C^{(d_2)} \longrightarrow C^{(d_1+d_2)}$$

is ètale at $(\mathfrak{n}_1, \mathfrak{n}_2)$ and maps this point to $\mathfrak{n}_1 + \mathfrak{n}_2$. Choose open neighborhoods

$$(\mathfrak{n}_1, \mathfrak{n}_2) \in V \subset C^{(d_1)} \times_k C^{(d_2)}, \qquad \mathfrak{n}_1 + \mathfrak{n}_2 \in W \subset C^{(d_1+d_2)}$$

such that $+_{d_1,d_2}|_V \colon V \to W$ is ètale and

$$V \times_W \{\mathfrak{n}_1 + \mathfrak{n}_2\} = \{(\mathfrak{n}_1, \mathfrak{n}_2)\}$$

scheme-theoretically. Flat base change for blowups gives

$$\mathrm{Bl}_{(\mathfrak{n}_1,\mathfrak{n}_2)}(V) \cong V \times_W \mathrm{Bl}_{\mathfrak{n}_1+\mathfrak{n}_2}(W).$$

Hence the second projection defines an ètale morphism

$$\widetilde{+}_{d_1,d_2} \colon \mathrm{Bl}_{(\mathfrak{n}_1,\mathfrak{n}_2)}(V) \longrightarrow \mathrm{Bl}_{\mathfrak{n}_1+\mathfrak{n}_2}(W) \subset X_{\mathfrak{n}_1+\mathfrak{n}_2}.$$

If η denotes the generic point of the exceptional divisor of the source, then

$$\widetilde{+}_{d_1,d_2}(\eta) = \eta_{\mathfrak{n}_1+\mathfrak{n}_2}.$$

By Proposition 30, on the common dense open,

$$(\widetilde{+}_{d_1,d_2})^{-1}\mathcal{P}^{[d_1+d_2]} \cong r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}.$$

Its tameness therefore descends by Lemma 46. □

From the point center to the degree-d center.

The preceding reduction gives tameness in degree $\deg \mathfrak{m}$, at the exceptional divisor over the point $\mathfrak{m} \in C^{(\deg \mathfrak{m})}$. To pass to degree $d \geq \deg \mathfrak{m}$, we use the addition morphism to relate this point center to

$$Z_0(\mathfrak{m}, d) = \mathfrak{m} + C^{(d-\deg \mathfrak{m})} \subset C^{(d)},$$

the center of the degree-d blowup. After restricting the second factor to $U^{(d-\deg \mathfrak{m})}$ and blowing up, the addition morphism induces a comparison morphism

$$X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \longrightarrow X_{\mathfrak{m},d}.$$

To descend tameness along this morphism, we must show that it is ètale at the generic point of the exceptional divisor in its source. This follows from the case $l = 1$ of the next lemma, adapted from [Tak18, Lemma 4.1].

Lemma 65. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k. Let $\mathfrak{m} = \sum_{i=1}^{s} n_i P_i$ be an effective divisor where $P_1, \ldots, P_s$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \ldots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^{l} \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \ldots, D_l, D_{extra}) \mapsto \sum_{j=1}^{l} D_j + D_{extra}$.

Then π is étale at the generic point of the closed subvariety

$$V = \{\mathfrak{n}_1\} \times_k \cdots \times_k \{\mathfrak{n}_l\} \times_k C^{(d-\deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$.

Review note 1. The proof below should be revised: the fiber-cardinality argument does not handle multiplicities in the divisors $\mathfrak{n}_j$. A cleaner proof should use the local fact that addition of effective divisors is étale when the summands have pairwise disjoint support.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). The morphism π is finite: it is induced by the finite quotient morphism

$$C^d/\big(S_{\deg \mathfrak{n}_1} \times \cdots \times S_{\deg \mathfrak{n}_l} \times S_{d-\deg \mathfrak{m}}\big) \longrightarrow C^d/S_d.$$

Its source and target are smooth k-schemes of pure dimension d. Hence π is flat by miracle flatness, and therefore finite flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \ldots \mathfrak{n}_l + C^{(d-\deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V.) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + P_{s+1} + \cdots + P_{s+d-\deg \mathfrak{m}}$, where $P_1, \ldots, P_{s+d-\deg \mathfrak{m}}$ are distinct points of $U(k)$. □

Corollary 66. *For $d \geq \deg \mathfrak{m}$, the addition morphism*

$$+ \colon C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

is finite flat, and it is ètale at the generic point of

$$\{\mathfrak{m}\} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}.$$

Proof. This is the case $l = 1$ and $\mathfrak{n}_1 = \mathfrak{m}$ of Lemma 65. □

We first spread out the point-center blowup over $C^{(d-\deg \mathfrak{m})}$. Flat base change along $C^{(d-\deg \mathfrak{m})} \to \operatorname{Spec}(k)$ gives the Cartesian diagram

$$\begin{array}{ccc} E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} & \longrightarrow & X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \\ \downarrow & & \downarrow \\ \{\mathfrak{m}\} \times_k C^{(d-\deg \mathfrak{m})} & \hookrightarrow & C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}. \end{array}$$

Let

$$p_1 \colon X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow X_{\mathfrak{m}}$$

be the first projection, and let θ be the generic point of the prime divisor $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$. The pullback $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ is defined on the dense open $U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$, so its ramification at θ is defined in the sense of Definition 40.

Lemma 67. *If $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$, then $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at θ.*

Proof. This is the stability of tame ramification in codimension one under base change ([Stacks, Tag 0EYD]), applied to the projection

$$X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow X_{\mathfrak{m}}.$$

□

Lemma 68. *Let G be a finite abelian group, let X be a scheme over a field k, and let $U \subset X$ be a dense open subscheme. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be G-torsors on U, and let ξ be the generic point of a prime divisor of X. If $\mathcal{P}_1$ is tamely ramified at ξ and $\mathcal{P}_2$ is unramified at ξ, then $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at ξ.*

Proof. This follows from Lemma 45. □

Restrict the second factor to the dense open $U^{(d-\deg \mathfrak{m})} \subset C^{(d-\deg \mathfrak{m})}$. The point θ remains the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$, and Lemma 67 shows that $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified there. If

$$p_2 \colon X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \longrightarrow U^{(d-\deg \mathfrak{m})}$$

is the second projection, then $p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is ètale, and hence unramified at θ. Therefore Lemma 68 shows that

$$p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$$

is tamely ramified at θ.

The addition morphism now transports this conclusion from the product of the point-center blowup with $C^{(d-\deg \mathfrak{m})}$ to the blowup along the degree-d center $Z_0(\mathfrak{m}, d)$.

Corollary 69. *Let $\mathfrak{m}$ be a modulus, let $d \geq \deg \mathfrak{m}$, and let $\mathcal{P}$ be a G-torsor on U whose ramification is bounded by $\mathfrak{m}$. Assume that $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at the generic point $\eta_{\mathfrak{m}}$ of $E_{\mathfrak{m}}$. If p_1 and p_2 denote the projections from $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$, then*

$$p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$$

is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$. Moreover, $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of the exceptional divisor $E_0 = E_{\mathfrak{m},d}$ over $Z_0(\mathfrak{m}, d)$.

Proof. The first assertion is the conclusion of the preceding discussion. To transport it to $X_{\mathfrak{m},d}$, consider the addition morphism

$$+ \colon C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}.$$

Its scheme-theoretic inverse image of $Z_0(\mathfrak{m}, d)$ is $\{\mathfrak{m}\} \times_k U^{(d-\deg \mathfrak{m})}$. Since $+$ is flat, Theorem 32 gives a canonical isomorphism

$$X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \cong \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d}.$$

The second projection therefore defines a morphism

$$\widetilde{+} \colon X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \longrightarrow X_{\mathfrak{m},d}$$

that maps θ to η_0. By Corollary 66, the addition morphism is ètale at the generic point of $\{\mathfrak{m}\} \times_k U^{(d-\deg \mathfrak{m})}$; hence $\widetilde{+}$ is ètale at θ. Finally, Proposition 30 gives

$$+^{-1}\mathcal{P}^{[d]} \cong p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}.$$

The tame ramification of the right-hand side at θ therefore descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by Lemma 46. □

3.3.1 Proof of Theorem 62

Proof. We first reduce to the case where k is algebraically closed. The formations of symmetric powers, Takeuchi's blowup, and its boundary commute with the flat base change $\operatorname{Spec}(\bar{k}) \to \operatorname{Spec}(k)$. At the generic points of the boundary, this base change induces a weakly unramified extension of discrete valuation rings with separable residue-field extension. Hence Lemma 46 shows that tameness over k is equivalent to tameness after base change to $\bar{k}$. We may therefore replace k by $\bar{k}$.

By Proposition 44, the pullback $\mathcal{P}_{\bar{k}}$ has ramification bounded by the base-changed modulus.

Because the original field is perfect, each closed point P_i splits into distinct $\bar{k}$-points, and the base-changed modulus is

$$\mathfrak{m}_{\bar{k}} = \sum_i \sum_{\bar{P} \mapsto P_i} n_i \bar{P}.$$

In the remainder of the proof we write k, C, $\mathfrak{m}$, and $\mathcal{P}$ for these base-changed objects.

For each single-point submodulus $\mathfrak{n} = n_P P$ of $\mathfrak{m}$, Theorem 35 shows that $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$. Repeated application of Corollary 64 shows that $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Therefore Corollary 69 transports this tameness to the degree-d blowup and shows that $\mathcal{P}^{[d]}$ is tamely ramified at the generic point η_0 of the exceptional divisor over $Z_0(\mathfrak{m}, d)$ for every sufficiently large d. Since H is the nonempty open part of E_0 described in Corollary 61, its generic point is η_0. Thus $\mathcal{P}^{[d]}$ is tamely ramified along H. □

3.4 Extension Across the Boundary

The geometry of the Abel–Jacobi fibers upgrades the tameness proved above to extension across the boundary. In positive characteristic, a prime-to-p power argument on the geometric generic fiber forces the boundary inertia representation to be trivial; in characteristic zero, the same conclusion follows from the simple connectedness of affine space.

Lemma 70. *Let G be a finite abelian group, and let $\mathcal{P}^{[d]}$ be a G-torsor on $U^{(d)}$. If $\mathcal{P}^{[d]}$ is tamely ramified at the generic point of*

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)},$$

then it extends uniquely to a G-torsor $\widetilde{\mathcal{P}}^{[d]}$ on $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

Proof. Put $r = d - \deg \mathfrak{m} - g + 1$, let ξ be the generic point of $\operatorname{Pic}^d_{C,\mathfrak{m}}$, let $\Omega = \kappa(\xi)^{\mathrm{sep}}$, and write $\bar{\xi} = \operatorname{Spec} \Omega$. By Corollary 61, the compactified geometric generic Abel–Jacobi fiber and its boundary are

$$\left((\widetilde{C}^{(d)}_{\mathfrak{m}})_{\bar{\xi}}, H_{\bar{\xi}} \right) \cong \left(\mathbb{P}^r_\Omega, \mathbb{P}^{r-1}_\Omega \right),$$

and hence the geometric generic open fiber is

$$U^{(d)}_{\bar{\xi}} \cong \mathbb{A}^r_\Omega.$$

Let

$$\chi \colon \pi_1\left(\mathbb{A}^r_\Omega\right) \longrightarrow G$$

be the character corresponding to the restriction of $\mathcal{P}^{[d]}$ to this fiber.

Let η be the generic point of H, and let $\bar{\eta}$ be the generic point of $H_{\bar{\xi}}$. Because H is a relative hyperplane divisor, the passage from η to $\bar{\eta}$ induces a weakly unramified extension of discrete valuation rings with separable residue-field extension. Hence Lemma 46 allows tameness and triviality of the finite inertia representation at η to be checked after base change at the inertia group

$$I_\infty := I_{\bar{\eta}}$$

of the hyperplane at infinity.

If $\mathrm{char}(k) = 0$, then

$$\pi_1\big(\mathbb{A}^r_\Omega\big) = 1,$$

so χ is trivial [Ten15, Example 4.14].

Suppose now that $\mathrm{char}(k) = p > 0$. Since $\mathcal{P}^{[d]}$ is tame along H, the finite group $\chi(I_\infty)$ has order prime to p. Let N be its exponent. Then $(N, p) = 1$, and the character χ^N is trivial on I_∞.

Consequently, the G-torsor obtained by pushing out along $g \mapsto g^N$, and corresponding to χ^N, is unramified at the generic point of $\mathbb{P}^{r-1}_\Omega$. By Zariski–Nagata purity of the branch locus, it extends to a finite étale G-torsor on $\mathbb{P}^r_\Omega$. Since

$$\pi_1\big(\mathbb{P}^r_\Omega\big) = 1,$$

this extension is constant, and therefore

$$\chi^N = 1.$$

On the other hand, χ factors through

$$\pi_1\big(\mathbb{A}^r_\Omega\big)^{\mathrm{ab}},$$

which is a pro-p group [Ten15, Example 4.15]. Hence the finite group $\mathrm{im}(\chi)$ is a p-group. Since its exponent divides N, and N is prime to p, it follows that

$$\mathrm{im}(\chi) = 1.$$

Thus χ is trivial.

In either characteristic, the restriction of χ to I_∞ is trivial. By the inertia comparison above, the inertia group at η acts trivially on $\mathcal{P}^{[d]}$.

Therefore $\mathcal{P}^{[d]}$ is unramified at the generic point of H. The scheme $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is smooth and geometrically connected because it is a torsor under the smooth geometrically connected group $\mathrm{Pic}^0_{C,\mathfrak{m}}$. Since $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$, it is regular and geometrically connected; similarly, the relative hyperplane H is smooth and geometrically connected, hence integral with a unique generic point.

Let $\overline{\mathcal{P}}^{[d]} \to \widetilde{C}^{(d)}_{\mathfrak{m}}$ be the normalization of $\widetilde{C}^{(d)}_{\mathfrak{m}}$ in the finite étale cover $\mathcal{P}^{[d]} \to U^{(d)}$. This normalization is finite because $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is of finite type over a field. It is étale over $U^{(d)}$ and unramified over the generic point of the only codimension-one boundary component H. Purity of the branch locus ([Stacks, Tag 0BMB]) therefore shows that $\overline{\mathcal{P}}^{[d]} \to \widetilde{C}^{(d)}_{\mathfrak{m}}$ is finite étale. The G-action extends uniquely to the normalization, and the torsor identity, which is an isomorphism over the dense open $U^{(d)}$, remains an isomorphism over $\widetilde{C}^{(d)}_{\mathfrak{m}}$. Thus $\overline{\mathcal{P}}^{[d]}$ is the required G-torsor $\widetilde{\mathcal{P}}^{[d]}$.

The extension is unique because

$$\pi_1\big(U^{(d)}\big) \longrightarrow \pi_1\big(\widetilde{C}^{(d)}_{\mathfrak{m}}\big)$$

is surjective by [Stacks, Tag 0BQI], since $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is integral and normal. □

Remark. The argument does not assert that tame ramification is generally unramified. Tameness is used only to choose N prime to p. The conclusion that the inertia character is trivial then follows from the global geometry of the geometric generic fiber and from the fact that the abelian fundamental group of affine space is pro-p.

3.5 Descent and Reduced Geometric Class Field Theory

The final descent rests on the fundamental groups of the usual and rigidified Abel–Jacobi morphisms.

3.5.1 The Fundamental-Group Input

Proposition 71 ([Stacks, Tag 0C0J])**.** *Let $f\colon X \to S$ be a flat proper morphism of finite presentation whose geometric fibers are connected and reduced. Assume that S is connected, and let $\bar{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\bar{s}}) \longrightarrow \pi_1(X) \longrightarrow \pi_1(S) \longrightarrow 1.$$

Corollary 72. *Let $f\colon X \to S$ be a proper smooth morphism of finite presentation whose geometric fibers are connected. Then the sequence of Proposition 71 is exact.*

Proof. A smooth geometric fiber is reduced, so this is an immediate application of Proposition 71. □

Corollary 73. *Let $f\colon X \to S$ be a projective space bundle over a connected scheme. Then pullback induces an isomorphism*

$$\pi_1^{\mathrm{et}}(X) \xrightarrow{\sim} \pi_1^{\mathrm{et}}(S).$$

Proof. The geometric fibers are projective spaces and therefore have trivial étale fundamental group. The assertion follows from Corollary 72. □

3.5.2 The Descent Argument

We work over $S = \operatorname{Spec} k$, where k is perfect, and continue in the torsor language fixed at the beginning of the chapter.

Theorem 74 (Torsor formulation)**.** *Let Λ be a finite commutative ring whose cardinality is invertible in k, set $G = \Lambda^{\times}$, and let $\mathcal{P}$ be a G-torsor on U whose ramification is bounded by $\mathfrak{m}$. For every sufficiently large integer d, there exists a G-torsor $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, unique up to isomorphism, such that*

$$\Phi_d^* \mathcal{Q}_d \cong \mathcal{P}^{[d]}.$$

Proof. We divide the proof into two cases.

Case 1: $\mathfrak{m} = 0$.

For d sufficiently large, the usual Abel–Jacobi morphism

$$\Phi_d \colon C^{(d)} \longrightarrow \mathrm{Pic}^d_C$$

is proper and smooth with geometrically connected projective-space fibers; this description holds after an étale base change even when C has no k-rational point. Hence the homotopy exact sequence, as in [Tak18, Corollary 3.9], gives

$$\pi_1^{\mathrm{et}}\big(C^{(d)}\big) \xrightarrow{\sim} \pi_1^{\mathrm{et}}\big(\mathrm{Pic}^d_C\big).$$

The G-torsor $\mathcal{P}^{[d]}$ therefore descends uniquely to a G-torsor on Pic^d_C.

Case 2: $\mathfrak{m} > 0$. For d sufficiently large, Theorem 60 extends Φ_d to a projective space bundle

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}\,.$$

By Theorem 62, $\mathcal{P}^{[d]}$ is tamely ramified along $H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$. Hence Lemma 70 gives a unique extension $\widetilde{\mathcal{P}}^{[d]}$ to $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

The scheme $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is connected because it is a torsor under the geometrically connected group $\mathrm{Pic}^0_{C,\mathfrak{m}}$. Therefore Corollary 73 gives

$$\pi_1^{\mathrm{et}}\big(\widetilde{C}^{(d)}_{\mathfrak{m}}\big) \xrightarrow{\sim} \pi_1^{\mathrm{et}}\big(\mathrm{Pic}^d_{C,\mathfrak{m}}\big).$$

This isomorphism induces an equivalence on finite étale covers and hence on finite constant G-torsors. Thus $\widetilde{\mathcal{P}}^{[d]}$ descends uniquely to a G-torsor $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, and restriction to $U^{(d)}$ gives $\Phi_d^*\mathcal{Q}_d \cong \mathcal{P}^{[d]}$. Uniqueness with this property also uses the surjection

$$\pi_1^{\mathrm{et}}(U^{(d)}) \twoheadrightarrow \pi_1^{\mathrm{et}}(\widetilde{C}^{(d)}_{\mathfrak{m}})$$

induced by the dense open immersion into the normal scheme $\widetilde{C}^{(d)}_{\mathfrak{m}}$ ([Stacks, Tag 0BQI]). □

Proof of Theorem 2. Let $\mathcal{F}$ be as in Theorem 2, set $G = \Lambda^\times$, and let $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$. Apply Theorem 74 to obtain $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, and let

$$\mathcal{G}_d = \mathcal{Q}_d \times^G \Lambda.$$

The associated-module construction commutes with pullback and carries $\mathcal{P}^{[d]}$ to $\mathcal{F}^{[d]}$, so $\Phi_d^*\mathcal{G}_d \cong \mathcal{F}^{[d]}$. If $\mathcal{G}'_d$ is another such local system, then $\mathrm{Fr}(\mathcal{G}'_d)$ has the defining property of $\mathcal{Q}_d$. The uniqueness of $\mathcal{Q}_d$, followed by the associated-module construction, gives $\mathcal{G}'_d \cong \mathcal{G}_d$. □

Appendix A

Cyclic Torsor Theories and Witt Vectors

This appendix contains the supporting theory and proofs for the three cyclic torsor descriptions stated in Chapter 2. We begin with the common group-scheme construction and the classical Kummer and Artin–Schreier cases. We then develop the finite-length p-typical Witt-vector algebra needed for Artin–Schreier–Witt theory, followed by its torsor classification and normal-form arguments. The final section isolates the weighted-homogeneity property used in the cyclic symmetric-power calculation. The Schmid–Witt ramification-break formula is stated, referenced, and applied in Chapter 2; we do not reproduce its proof.

A.1 The common group-scheme construction

The three cyclic theories used in Chapter 2 are instances of the following construction.

A.1.1 The kernel torsor construction

Let H be a commutative group scheme over a scheme S, written additively, and let $\varphi : H \to H$ be a homomorphism. Its scheme-theoretic kernel $K := \ker(\varphi)$ is defined by the Cartesian square

$$\begin{array}{ccc} K & \longrightarrow & H \\ \downarrow & & \downarrow{\scriptstyle \varphi} \\ S & \xrightarrow[0_H]{} & H. \end{array} \tag{A.1}$$

Thus, for every S-scheme U,

$$K(U) = \ker(\varphi_U) = \{k \in H(U) : \varphi_U(k) = 0_{H(U)}\}, \tag{A.2}$$

where $H(U) = \mathrm{Hom}_S(U, H)$ and φ_U is induced by φ.

For an S-scheme X and a section $a \in H(X)$, define the fiber

$$T_a := X \times_{a,H,\varphi} H, \tag{A.3}$$

equivalently by the Cartesian square

$$\begin{array}{ccc} T_a & \longrightarrow & H \\ \downarrow & & \downarrow{\scriptstyle \varphi} \\ X & \xrightarrow[a]{} & H. \end{array}$$

For every X-scheme $U \to X$, with a_U the pullback of a,

$$T_a(U) = \{y \in H(U) : \varphi_U(y) = a_U\}.$$

Thus "the torsor $\varphi(Y) = a$" means the scheme of solutions after every base change to X; it has no preferred identity element in general.

Write $H_X := H \times_S X$, $K_X := K \times_S X$, let $\varphi_X : H_X \to H_X$ be the base change of φ, and let $\iota_K : K_X \to H_X$ and $\iota_a : T_a \to H_X$ be the canonical maps. Addition on H_X restricts to the action

$$K_X \times_X T_a \longrightarrow T_a, \qquad (k, y) \longmapsto y + k. \tag{A.4}$$

It is the top arrow in the commutative diagram

$$\begin{array}{ccc} K_X \times_X T_a & \xrightarrow{(k,y)\mapsto y+k} & T_a \\ {\scriptstyle \iota_K \times_X \iota_a}\downarrow & & \downarrow{\scriptstyle \iota_a} \\ H_X \times_X H_X & \xrightarrow[+_X]{} & H_X. \end{array}$$

Indeed,

$$\varphi_U(y + k) = \varphi_U(y) + \varphi_U(k) = a_U + 0_{H(U)} = a_U,$$

so translation by K preserves the fiber T_a. The action is simply transitive: the morphism

$$K_X \times_X T_a \longrightarrow T_a \times_X T_a, \qquad (k, y) \longmapsto (y, y + k), \tag{A.5}$$

is an isomorphism with inverse

$$(y_1, y_2) \longmapsto (y_2 - y_1, y_1).$$

This inverse is well-defined because

$$\varphi_U(y_2 - y_1) = \varphi_U(y_2) - \varphi_U(y_1) = a_U - a_U = 0_{H(U)}.$$

If φ is finite étale and surjective, then its base change $T_a \to X$ is finite étale and surjective. Hence T_a is a K_X-torsor, namely

$$T_a = a^*(H \xrightarrow{\varphi} H)$$

as a pullback of the K-torsor $\varphi : H \to H$.

Suppose that $a' = a + \varphi_X(b)$ for some $b \in H(X)$. Translation by b defines

$$\tau_b : T_a \longrightarrow T_{a'}, \qquad y \longmapsto y + b,$$

because

$$\varphi_U(y + b_U) = a_U + \varphi_U(b_U) = a'_U,$$

and it is K_X-equivariant since $\tau_b(y + k) = y + k + b = \tau_b(y) + k$. Its inverse is τ_{-b}, so T_a depends only on the class of a modulo $\varphi_X(H(X))$.

When φ is finite étale and surjective, regard the group schemes K_X and H_X as the sheaves that they represent on the étale site of X. Then

$$0 \longrightarrow K_X \longrightarrow H_X \xrightarrow{\varphi_X} H_X \longrightarrow 0$$

is a short exact sequence of sheaves. Applying étale cohomology gives the beginning of the associated long exact sequence

$$0 \longrightarrow H^0_{\text{ét}}(X, K_X) \longrightarrow H^0_{\text{ét}}(X, H_X) \xrightarrow{\varphi_X} H^0_{\text{ét}}(X, H_X) \xrightarrow{\delta} H^1_{\text{ét}}(X, K_X).$$

Because H_X is representable, its global sections are

$$H^0_{\text{ét}}(X, H_X) = \operatorname{Hom}_X(X, H_X) = \operatorname{Hom}_S(X, H) = H(X).$$

For $a \in H(X)$, the connecting class $\delta(a)$ is represented by the sheaf of local lifts of a through φ_X. This sheaf is precisely the pullback $T_a = X \times_{a,H,\varphi} H$, with its K_X-action, and hence $\delta(a) = [T_a]$. Exactness gives $\ker(\delta) = \varphi_X(H(X))$, so δ descends to the injection

$$H(X)/\varphi_X(H(X)) \hookrightarrow H^1_{\text{ét}}(X, K_X), \qquad [a] \longmapsto [T_a]. \tag{A.6}$$

The class $[T_a]$ is trivial exactly when $a \in \varphi_X(H(X))$, equivalently when $\varphi(Y) = a$ has a global solution.

For a multiplicative group scheme the same construction is written multiplicatively: $K(U) = \{k \in H(U) : \varphi_U(k) = 1_{H(U)}\}$, the action is $(k, y) \mapsto ky$, and replacing a by $a\varphi_X(b)$ gives the isomorphic torsor $y \mapsto by$.

A.1.2 The three cyclic theories

For an arbitrary S-scheme X, the fiber over a section $a \in H(X)$ is

$$T_a = \varphi_X^{-1}(a) = X \times_{a,H,\varphi} H.$$

For nonaffine X, explicit equations are interpreted using the structure sheaf $\mathcal{O}_X$. In all three applications in Chapter 2, however, the base is affine: first $\operatorname{Spec} k((x))$ and then $\mathbb{G}_m = \operatorname{Spec} k[x, x^{-1}]$. We therefore work affinely from now on and write

$$X = \operatorname{Spec} A,$$

where the capital letter A is the coordinate ring of X, while the lowercase letter a is the section specifying the fiber. More generally, if $S = \operatorname{Spec} R$ and $H = \operatorname{Spec} C$, then

$$T_a = \operatorname{Spec}\big(A \otimes_C C\big),$$

where C acts on A through $a^{\#} : C \to A$ and on the second copy of C through $\varphi^{\#} : C \to C$.

Kummer. Take $H = \mathbb{G}_m$ and $\varphi(z) = z^e$. A section is an element $a \in \mathbb{G}_m(X) = A^\times$, and its fiber is

$$T_a = \operatorname{Spec}\big(A[Y]/(Y^e - a)\big),$$

where Y is automatically a unit because a is a unit. The kernel is

$$\mu_{e,X} = \operatorname{Spec}\big(A[t,t^{-1}]/(t^e-1)\big),$$

and it acts on T_a through a morphism

$$\alpha : \mu_{e,X} \times_X T_a \longrightarrow T_a.$$

Writing

$$B = A[Y]/(Y^e - a) \quad \text{and} \quad D = A[t,t^{-1}]/(t^e-1),$$

the corresponding homomorphism of coordinate rings is

$$\alpha^{\#} : B \longrightarrow D \otimes_A B, \qquad \overline{Y} \longmapsto t \otimes \overline{Y},$$

where $\overline{Y}$ denotes the class of Y in B. This is well-defined because

$$(t \otimes \overline{Y})^e = t^e \otimes \overline{Y}^e = 1 \otimes a.$$

Equivalently, for every X-scheme U,

$$T_a(U) = \{y \in \Gamma(U,\mathcal{O}_U)^\times : y^e = a_U\}, \qquad \mu_e(U) = \{\xi \in \Gamma(U,\mathcal{O}_U)^\times : \xi^e = 1\},$$

and the action sends (ξ, y) to ξy; indeed, $(\xi y)^e = \xi^e y^e = a_U$. If e is invertible on S and a primitive root $\zeta_e \in \mu_e(S)$ is chosen, then $\bar{j} \mapsto \zeta_e^j$ identifies $(\mathbb{Z}/e\mathbb{Z})_S$ with μ_e. Only this identification, not the μ_e-torsor or its action, depends on the choice of ζ_e.

Artin–Schreier. Let A be an $\mathbb{F}_p$-algebra, take $H = \mathbb{G}_a$, and put $\varphi(Y) = Y^p - Y$. A section is an element $a \in \mathbb{G}_a(X) = A$, and its fiber is

$$T_a = \operatorname{Spec}\big(A[Y]/(Y^p - Y - a)\big).$$

The kernel is the constant group scheme $(\mathbb{F}_p)_X$, acting by $(c, Y) \mapsto Y + c$.

Artin–Schreier–Witt. Let A be an $\mathbb{F}_p$-algebra. The third theory takes $H = \mathbb{W}_r$ and

$$\varphi = \wp := F - \mathrm{id}, \qquad \wp\vec{Y} = F\vec{Y} \ominus \vec{Y}.$$

For a section $\vec{a} \in W_r(A)$, the fiber is explicitly

$$T_{\vec{a}} = \operatorname{Spec}\Big(A[Y_0,\dots,Y_{r-1}]/\big(\wp\vec{Y} \ominus \vec{a}\big)\Big), \qquad \wp\vec{Y} = \vec{a}.$$

Here the parenthesized Witt vector denotes the ideal generated by its r coordinate polynomials. Its kernel is $W_r(\mathbb{F}_p)_X \cong (\mathbb{Z}/p^r\mathbb{Z})_X$, acting by Witt translation $(\vec{c}, \vec{Y}) \mapsto \vec{Y} \oplus \vec{c}$.

This paragraph states the affine model in advance for comparison with the two preceding theories. The operations F, $\oplus$, and $\ominus$, and the coordinate polynomials occurring in the displayed quotient, are defined in Section A.3. The fact that this fiber is finite étale and that the displayed kernel action makes it a torsor is proved in Lemma 84.

A.2 Kummer and Artin–Schreier theory

Let K be a complete discretely valued field with residue characteristic $p > 0$.

A.2.1 Kummer theory

If $(e, p) = 1$ and K contains μ_e, the Kummer sequence

$$1 \longrightarrow \mu_e \longrightarrow \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \longrightarrow 1$$

and Hilbert 90 give

$$H^1_{\text{ét}}(\operatorname{Spec} K, \mu_e) \cong K^\times/(K^\times)^e.$$

Equivalently, since the base is a field, this étale cohomology group is the continuous Galois cohomology group

$$H^1_{\text{ét}}(\operatorname{Spec} K, \mu_e) \cong H^1_{\text{cont}}\big(\operatorname{Gal}(K^{\text{sep}}/K), \mu_e(K^{\text{sep}})\big).$$

Under this identification, the coset $[a] \in K^\times/(K^\times)^e$ corresponds to the μ_e-torsor

$$\operatorname{Spec} K[T]/(T^e - a),$$

on which $\zeta \in \mu_e$ acts by $T \mapsto \zeta T$. Thus replacing a by ab^e, for $b \in K^\times$, gives an isomorphic torsor. If $[a]$ has order e, adjoining a solution of $T^e = a$ gives a cyclic field extension of degree e.

A.2.2 Artin–Schreier theory

In characteristic p, the Artin–Schreier sequence

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{G}_a \xrightarrow{\wp} \mathbb{G}_a \longrightarrow 0, \qquad \wp(z) = z^p - z,$$

and additive Hilbert 90 give

$$H^1(K, \mathbb{Z}/p\mathbb{Z}) := H^1_{\text{ét}}(\operatorname{Spec} K, \mathbb{Z}/p\mathbb{Z}) \cong K/\wp(K).$$

Under this identification, the coset $[a] \in K/\wp(K)$ corresponds to the $\mathbb{Z}/p\mathbb{Z}$-torsor

$$\operatorname{Spec} K[X]/(X^p - X - a),$$

on which $c \in \mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ acts by $X \mapsto X + c$. Thus replacing a by $a + (b^p - b)$, for $b \in K$, gives an isomorphic torsor. If $[a] \neq 0$, adjoining a solution of $X^p - X = a$ gives a cyclic field extension of degree p; conversely, every cyclic extension of degree p arises in this way.

Lemma 75 (Classical cyclic equations are étale torsors)**.** *Let A be a ring.*

(i) If n is invertible in A and $a \in A^\times$, then

$$A[T]/(T^n - a)$$

is finite free over A, with basis $1, T, \ldots, T^{n-1}$, and is étale over A. Multiplication on the T-coordinate makes its spectrum a representable $\mu_{n,\operatorname{Spec} A}$-torsor over $\operatorname{Spec} A$ for the étale topology.

(ii) If A is an $\mathbb{F}_p$-algebra and $a \in A$, then

$$A[X]/(X^p - X - a)$$

is finite free over A, with basis $1, X, \dots, X^{p-1}$, and is étale over A. Translation on the X-coordinate makes its spectrum a representable $(\mathbb{Z}/p\mathbb{Z})_{\operatorname{Spec} A}$-torsor over $\operatorname{Spec} A$ for the étale topology.

Proof. Both defining polynomials are monic, giving the stated bases and finite freeness. In the Kummer algebra, T is a unit because $T^n = a \in A^\times$; therefore the derivative nT^{n-1} is a unit. The derivative of $X^p - X - a$ is -1. Thus both algebras are étale over A. They are the pullbacks from Appendix A.1 of $(-)^n : \mathbb{G}_m \to \mathbb{G}_m$ and $\wp : \mathbb{G}_a \to \mathbb{G}_a$, respectively, so translation by the corresponding kernels μ_n and $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ gives the asserted torsor structures. □

A.3 Witt-vector algebra

A.3.1 Ring structure and ghost coordinates

For $n \geq 0$, the n-th **Witt polynomial** in the variables $X_0, \dots, X_n$ is

$$w_n(X_0, \dots, X_n) \;=\; \sum_{j=0}^{n} p^j X_j^{p^{n-j}} \;=\; X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n.$$

Theorem 76 (Witt; [Ser79, II, §6, Theorem 6], [Rab14, §1])**.** *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$, $n \geq 0$, such that*

$$w_n(S_0, \dots, S_n) = w_n(X_0, \dots, X_n) + w_n(Y_0, \dots, Y_n),$$
$$w_n(P_0, \dots, P_n) = w_n(X_0, \dots, X_n) \cdot w_n(Y_0, \dots, Y_n).$$

for all n. For any commutative ring A, the operations

$$\vec{a} \oplus \vec{b} := \big(S_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}, \qquad \vec{a} \odot \vec{b} := \big(P_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}$$

*make the set A^r into a commutative ring $W_r(A)$, the ring of **p-typical Witt vectors of length r over A**, with zero $(0, \dots, 0)$ and unit $(1, 0, \dots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map***

$$w \colon W_r(A) \longrightarrow A^r, \qquad \vec{a} \longmapsto \big(w_0(\vec{a}), w_1(\vec{a}), \dots, w_{r-1}(\vec{a})\big)$$

is a ring homomorphism to the product ring A^r; it is injective when p is a non-zero-divisor in A, and bijective when $p \in A^\times$.

(Equivalently, the universal Witt addition and inverse polynomials make $\mathbb{A}^r_{\mathbb{Z}}$ an affine commutative group scheme. After base change to $\mathbb{F}_p$, this is the additive Witt-vector group scheme $\mathbb{W}_r$ used in Section A.4.)

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma\colon A \to B$ induces the ring homomorphism

$$W_r(\sigma)\colon W_r(A) \to W_r(B), \qquad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component* a_n is σ-invariant. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma \;=\; W_r\big(A^\Gamma\big) \qquad \text{inside } W_r(A). \tag{A.7}$$

The uniqueness statement in Theorem 76 is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 77 (ghost principle)**.** *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*

Proof. Over $\mathbb{Z}[X_0, X_1, \dots]$ the prime p is a non-zero-divisor, so the ghost map is injective by Theorem 76 and $\Phi = \Psi$ there; this is an identity of integral polynomials, which persists under every base change $\mathbb{Z}[X_0, X_1, \dots] \to A$. □

The only structural feature of the addition polynomials S_n that we shall need is their "triangular" shape:

Proposition 78 (shape of Witt addition)**.** *For every $n \geq 0$ there is a polynomial*

$$C_n \in \mathbb{Z}[X_0, \dots, X_{n-1}; Y_0, \dots, Y_{n-1}]$$

involving only the variables of index strictly less *than n, such that*

$$(\vec{a} \oplus \vec{b})_n \;=\; a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \tag{A.8}$$

We refer to C_n as the ***carry polynomial****. The analogous statement holds for $\ominus$.*

Proof. By induction on n, using the recursion defining S_n:

$$p^n S_n \;=\; \sum_{j \leq n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) \;-\; \sum_{j<n} p^j S_j^{p^{n-j}}.$$

The terms with $j = n$ contribute $p^n(X_n + Y_n)$, and all remaining terms involve only variables and polynomials S_j of index $j < n$, which by induction lie in $\mathbb{Z}[X_{<n}; Y_{<n}]$. Since S_n has integer coefficients (Theorem 76), $C_n := S_n - X_n - Y_n$ is an integral polynomial in the variables of index $< n$. The statement for $\ominus$ follows by the same argument applied to the inversion polynomials. □

A.3.2 Frobenius, Verschiebung, and truncation

From now on, A denotes an $\mathbb{F}_p$-algebra; this is the only case we use. Three operators act on the rings $W_r(A)$:

Definition 79. Let A be an $\mathbb{F}_p$-algebra.

1. The **Frobenius** $F\colon W_r(A) \to W_r(A)$ is $F := W_r(\varphi_A)$, where $\varphi_A\colon a \mapsto a^p$ is the absolute Frobenius of A. Explicitly, F acts componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. By functoriality, F is a ring endomorphism of $W_r(A)$.

2. The **Verschiebung** is the shift

$$V\colon W_{r-1}(A) \to W_r(A), \qquad V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2}).$$

   Iterating, $V^j\colon W_{r-j}(A) \to W_r(A)$; in particular $V^{r-1}\colon A = W_1(A) \to W_r(A)$ sends $c \mapsto (0, \dots, 0, c)$.

3. The **truncation** is

$$t\colon W_r(A) \to W_{r-1}(A), \qquad t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2}).$$

Remark. For a general (not necessarily $\mathbb{F}_p$-) algebra A, the Frobenius is defined by universal polynomials characterized by $w_n \circ F = w_{n+1}$, and the componentwise formula above holds precisely in characteristic p; see [Ser79, II, §6] or [Rab14, §2]. We will not need the general case.

We record from the general theory ([Ser79, II, §6], [Rab14, §2]) how multiplication by p interacts with these operators.

Proposition 80. *Let A be an $\mathbb{F}_p$-algebra. Then:*

1. *t is a natural ring homomorphism commuting with F, and its kernel is*

$$\ker\big(t\colon W_r(A) \to W_{r-1}(A)\big) \;=\; V^{r-1}A \;=\; \{(0, \dots, 0, c) : c \in A\}.$$

2. *V is additive, and $FV = VF$.*

3. *Multiplication by p on $W_r(A)$ equals $V \circ F = F \circ V$ (composed with the appropriate truncation); explicitly,*

$$p \cdot (a_0, \dots, a_{r-1}) \;=\; (0,\, a_0^p,\, \dots,\, a_{r-2}^p).$$

   Iterating, $p^j \cdot \vec{a} = (0, \dots, 0, a_0^{p^j}, \dots, a_{r-1-j}^{p^j})$; in particular $p^r = 0$ on $W_r(A)$, and

$$p^{r-1} \cdot (a_0, \dots, a_{r-1}) \;=\; \big(0, \dots, 0,\, a_0^{p^{r-1}}\big). \tag{A.9}$$

Proof. (1) That t is a ring homomorphism follows from the recursive definition of the S_n, P_n: the first $r-1$ components of a sum or product depend only on the first $r-1$ components of the arguments (Proposition 78). Naturality and the commutation with the (componentwise) F are clear, and the kernel is computed componentwise. (2) Additivity of V follows from Lemma 77 and the ghost identity $w_n(V\vec{a}) = p\, w_{n-1}(\vec{a})$ (with $w_{-1} := 0$); the commutation $FV = VF$ is immediate

since F is componentwise. (3) Let $M_n \in \mathbb{Z}[X_0, \dots, X_n]$ be the n-th universal coordinate polynomial for multiplication by p. Since the ghost map is a ring homomorphism, these polynomials satisfy

$$p^n M_n = p\, w_n(\vec{X}) - \sum_{i=0}^{n-1} p^i M_i^{p^{n-i}}. \tag{A.10}$$

We claim that

$$M_0 = pX_0, \qquad M_n \equiv X_{n-1}^p \pmod{p} \quad (n \geq 1).$$

The first identity is clear. For the induction step, use $(u + pv)^{p^m} \equiv u^{p^m} \pmod{p^{m+1}}$. If the claim holds below index n, then, modulo p^{n+1}, the sum in equation (A.10) is

$$\sum_{i=1}^{n-1} p^i X_{i-1}^{p^{n-i+1}};$$

the $i = 0$ term is also 0 modulo p^{n+1}. Cancelling this sum against the corresponding terms of $p\, w_n(\vec{X})$ gives

$$p^n M_n \equiv p^n X_{n-1}^p \pmod{p^{n+1}},$$

which proves the claim. After evaluating in the $\mathbb{F}_p$-algebra A, we therefore obtain

$$p \cdot (a_0, \dots, a_{r-1}) = (0, a_0^p, \dots, a_{r-2}^p) = VF(\vec{a}),$$

with the appropriate truncation. Since $FV = VF$ by (2), this also gives $FV = p$. See [Ser79, II, §6, Proposition 8 ff.] for details. □

Example 81. $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ canonically. Indeed, by (A.9) the unit $\underline{1} = (1, 0, \dots, 0)$ satisfies $p^j \cdot \underline{1} = (0, \dots, 0, 1, 0, \dots, 0)$ (1 in position j), so $\underline{1}$ has additive order p^r; since $|W_r(\mathbb{F}_p)| = p^r$, the additive group is cyclic of order p^r generated by the unit, and $m \mapsto m \cdot \underline{1}$ is a ring isomorphism $\mathbb{Z}/p^r\mathbb{Z} \xrightarrow{\cong} W_r(\mathbb{F}_p)$. We write $\underline{m} \in W_r(\mathbb{F}_p)$ for the image of $m \in \mathbb{Z}/p^r\mathbb{Z}$. (Similarly, the full ring of Witt vectors satisfies $W(\mathbb{F}_p) \cong \mathbb{Z}_p$; we shall not need this.)

A.4 Artin–Schreier–Witt theory

We retain the notation for Witt vectors from the beginning of this appendix. For an $\mathbb{F}_p$-algebra A, Frobenius acts componentwise on $W_r(A)$.

A.4.1 The ASW operator and bookkeeping

Definition 82. The **Artin–Schreier–Witt operator** is

$$\wp := F - \mathrm{id}, \qquad \wp\vec{u} = F\vec{u} \ominus \vec{u}.$$

It is an endomorphism of the additive group of $W_r(A)$ and commutes with truncation and Verschiebung.

Lemma 83 (Witt bookkeeping)**.** *Let A be an $\mathbb{F}_p$-algebra.*

(i) The truncation $t : W_r(A) \to W_{r-1}(A)$ is a ring homomorphism commuting with F, hence with $\wp$, and $\ker(t) = V^{r-1}A$.

(ii) The map $V^{r-1} : A \to W_r(A)$ is additive and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$.

(iii) For every $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) = (a_0, \dots, a_{r-2}, a_{r-1} + c).$$

Proof. The first two assertions follow from Proposition 80. For (iii), consider the two operations

$$\Phi(\vec{X}, C) := \vec{X} \oplus V^{r-1}(C) = \big(S_n(\vec{X}; 0, \dots, 0, C)\big)_{0 \le n \le r-1},$$
$$\Psi(\vec{X}, C) := (X_0, \dots, X_{r-2}, X_{r-1} + C).$$

Both are given by universal polynomials with integer coefficients, as required in Lemma 77. Over $\mathbb{Z}[X_0, \dots, X_{r-1}, C]$, for $n < r-1$ we have

$$w_n\big(\Phi(\vec{X}, C)\big) = w_n(\vec{X}) + w_n\big(V^{r-1}(C)\big) = w_n(\vec{X}) = w_n\big(\Psi(\vec{X}, C)\big).$$

For the final ghost component,

$$\begin{aligned} w_{r-1}\big(\Phi(\vec{X}, C)\big) &= w_{r-1}(\vec{X}) + p^{r-1}C \\ &= \sum_{i=0}^{r-2} p^i X_i^{p^{r-1-i}} + p^{r-1}(X_{r-1} + C) \\ &= w_{r-1}\big(\Psi(\vec{X}, C)\big). \end{aligned}$$

Thus $w \circ \Phi = w \circ \Psi$, so Lemma 77 gives $\Phi = \Psi$ over every commutative ring, and in particular over A. □

The last assertion is special to the top Verschiebung slice. Witt addition is not componentwise in lower positions, and the carry polynomials in (A.8) cannot in general be discarded.

A.4.2 Torsors and classification

The Witt addition polynomials make $\mathbb{A}^r_{\mathbb{F}_p}$ into the additive Witt vector group scheme $\mathbb{W}_r$ over $\mathbb{F}_p$. We distinguish this group scheme from its groups of points: for an $\mathbb{F}_p$-algebra A, its group of A-valued points is $\mathbb{W}_r(A) = W_r(A)$. If X is an $\mathbb{F}_p$-scheme, we write $W_r(\mathbb{F}_p)_X$ for the constant group scheme over X associated with the abstract group $W_r(\mathbb{F}_p)$. The operator $\wp = F - \mathrm{id}$ is an endomorphism of $\mathbb{W}_r$ and fits into an exact sequence of group schemes for the étale topology

$$0 \longrightarrow W_r(\mathbb{F}_p)_{\mathbb{F}_p} \cong (\mathbb{Z}/p^r\mathbb{Z})_{\mathbb{F}_p} \longrightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \longrightarrow 0. \tag{A.11}$$

The following is the Witt-vector analogue of Lemma 75. Its coordinate proof requires the triangular form of Witt addition because, unlike the two classical equations, Witt subtraction is not componentwise.

Lemma 84 (ASW covers are étale torsors)**.** *Let A be an $\mathbb{F}_p$-algebra, let $r \ge 1$, and let $\vec{g} \in W_r(A)$. Let $\vec{X} = (X_0, \dots, X_{r-1})$ denote the universal Witt vector of length r over A, so that $A[\vec{X}] = A[X_0, \dots, X_{r-1}]$. Define*

$$T_{\vec{g}} := \operatorname{Spec} B, \qquad B := A[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big),$$

where the parenthesized Witt vector denotes the ideal generated by its r coordinate polynomials. Here $\vec{g} \in W_r(A) = \mathbb{W}_r(A)$ is viewed as a morphism $\operatorname{Spec} A \to \mathbb{W}_r$. Thus $T_{\vec{g}}$ is the fiber of $\wp$ over $\vec{g}$, given by the Cartesian square

$$\begin{array}{ccc} T_{\vec{g}} & \longrightarrow & \mathbb{W}_r \\ \downarrow & & \downarrow{\scriptstyle \wp} \\ \operatorname{Spec} A & \xrightarrow[\vec{g}]{} & \mathbb{W}_r \\ & \searrow & \downarrow \\ & & \operatorname{Spec} \mathbb{F}_p, \end{array}$$

where the maps to $\operatorname{Spec} \mathbb{F}_p$ are the structure maps. Then the following hold:

(i) The algebra B is finite étale and free of rank p^r over A, with basis $\{\prod_{n=0}^{r-1} X_n^{a_n} : 0 \le a_n \le p-1\}$.

(ii) Witt translation $\vec{X} \mapsto \vec{X} \oplus \vec{c}$, for $\vec{c} \in W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$, makes $T_{\vec{g}}$ a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\operatorname{Spec} A$.

Proof. For (i), the triangular shape of Witt addition ((A.8)) shows that the n-th equation in $\wp\vec{X} \ominus \vec{g} = 0$ has the form

$$X_n^p - X_n - g_n + C'_n(X_0, \dots, X_{n-1}; g_0, \dots, g_{n-1}) = 0.$$

Starting with A, first adjoin X_0 subject to the equation of index 0. Then, for each $1 \le n < r$, adjoin X_n subject to the displayed equation of index n; its coefficients lie in the algebra obtained at the preceding step. Because this equation is monic of degree p in X_n, the n-th step is free of rank p, with basis $1, X_n, \dots, X_n^{p-1}$. After all r steps we obtain B, finite free over A with the stated basis. The Jacobian matrix is lower triangular with diagonal entries -1, so B/A is étale. For (ii), the cover is surjective because B is free of positive rank. By construction, it is the pullback of $\wp : \mathbb{W}_r \to \mathbb{W}_r$ along $\vec{g}$. Recall that the kernel of the group-scheme morphism $\wp$ is the constant group scheme $W_r(\mathbb{F}_p)_{\mathbb{F}_p}$, characterized by $F\vec{u} = \vec{u}$. Translation by the base change of this kernel to $\operatorname{Spec} A$ preserves the fiber because, for $\vec{c} \in \ker(\wp)$,

$$\wp(\vec{X} \oplus \vec{c}) = \wp(\vec{X}) \oplus \wp(\vec{c}) = \wp(\vec{X}).$$

Any two points in the same fiber differ by a unique such $\vec{c}$, so this translation action makes the pullback a torsor. □

Theorem 85 (Artin–Schreier–Witt theory; [Tho05, §3–4])**.** *Let M be a field of characteristic p, and let $r \ge 1$. For $\vec{g} \in W_r(M)$, define*

$$T_{\vec{g}} := \operatorname{Spec}\Big(M[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big)\Big).$$

By Lemma 84, Witt translation makes $T_{\vec{g}}$ a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\operatorname{Spec} M$. Then the following hold:

(i) The assignment $[\vec{g}] \mapsto [T_{\vec{g}}]$ is a canonical isomorphism of abelian groups

$$W_r(M)/\wp W_r(M) \xrightarrow{\cong} H^1_{\text{ét}}\big(\operatorname{Spec} M, (\mathbb{Z}/p^r\mathbb{Z})_{\operatorname{Spec} M}\big).$$

Thus this quotient classifies $\mathbb{Z}/p^r\mathbb{Z}$-torsors over $\operatorname{Spec} M$.

(ii) *If $[\vec{g}]$ has order p^s, where $0 \le s \le r$, then $T_{\vec{g}}$ has p^{r-s} connected components, each noncanonically isomorphic to* $\operatorname{Spec} L$ *for the same cyclic extension L/M of degree p^s. Let $H \subseteq \mathbb{Z}/p^r\mathbb{Z}$ be the unique subgroup of order p^s. Each component is an H-torsor, and $T_{\vec{g}}$ is induced from any one of them. In particular, $T_{\vec{g}}$ is connected if and only if $[\vec{g}]$ has order p^r.*

(iii) *Every cyclic extension L/M of degree dividing p^r occurs in this way: for some $\vec{g} \in W_r(M)$, the connected components of $T_{\vec{g}}$ are isomorphic to* $\operatorname{Spec} L$.

Proof. For (i), put $X = \operatorname{Spec} M$, and write $\mathbb{W}_{j,M}$ for the base change of $\mathbb{W}_j$ to X. Base-changing (A.11) to X gives the short exact sequence of abelian sheaves on $X_{\text{ét}}$

$$0 \longrightarrow (\mathbb{Z}/p^r\mathbb{Z})_X \longrightarrow \mathbb{W}_{r,M} \xrightarrow{\wp} \mathbb{W}_{r,M} \longrightarrow 0.$$

Its associated long exact cohomology sequence contains

$$W_r(M) \xrightarrow{\delta} H^1_{\text{ét}}\big(X, (\mathbb{Z}/p^r\mathbb{Z})_X\big) \longrightarrow H^1_{\text{ét}}(X, \mathbb{W}_{r,M}).$$

By (A.6), $\delta(\vec{g}) = [T_{\vec{g}}]$, and δ induces the well-defined injective homomorphism $[\vec{g}] \mapsto [T_{\vec{g}}]$ on $W_r(M)/\wp W_r(M)$. Thus it remains to show that $H^1_{\text{ét}}(X, \mathbb{W}_{r,M}) = 0$. For $r = 1$, we have $\mathbb{W}_{1,M} = \mathbb{G}_{a,M}$, and this is additive Hilbert 90. For $r > 1$, Lemma 83(i)–(ii) gives the short exact sequence of abelian sheaves on $X_{\text{ét}}$

$$0 \longrightarrow \mathbb{G}_{a,M} \xrightarrow{V^{r-1}} \mathbb{W}_{r,M} \xrightarrow{t} \mathbb{W}_{r-1,M} \longrightarrow 0.$$

The associated long exact sequence contains

$$H^1_{\text{ét}}(X, \mathbb{G}_{a,M}) \longrightarrow H^1_{\text{ét}}(X, \mathbb{W}_{r,M}) \longrightarrow H^1_{\text{ét}}(X, \mathbb{W}_{r-1,M}).$$

The outer groups vanish by additive Hilbert 90 and induction, so the middle group vanishes. Hence δ is surjective, proving the claimed isomorphism.

For (ii), suppose $[\vec{g}]$ has order p^s. By (i) and Corollary 14, the character associated with $T_{\vec{g}}$ has the same order. Let H be its image. The order of the character is the smallest positive integer annihilating every element of H. Since H is cyclic, this integer is $|H|$, so $|H| = p^s$. By Theorem 13, $T_{\vec{g}}$ is induced from a connected H-torsor $\operatorname{Spec} L$, where L/M is cyclic of degree $|H|$. This induced torsor is a disjoint union of $[\mathbb{Z}/p^r\mathbb{Z} : H] = p^{r-s}$ copies of $\operatorname{Spec} L$, each an H-torsor. Translation by $\mathbb{Z}/p^r\mathbb{Z}$ permutes these components, so $T_{\vec{g}}$ is induced from any one of them. It is connected precisely when $s = r$.

For (iii), let L/M be cyclic of degree dividing p^r. Choose an injection $\operatorname{Gal}(L/M) \hookrightarrow \mathbb{Z}/p^r\mathbb{Z}$ and induce the torsor $\operatorname{Spec} L$ along it. The resulting $\mathbb{Z}/p^r\mathbb{Z}$-torsor is isomorphic to some $T_{\vec{g}}$ by the classification just proved, and its connected components are copies of $\operatorname{Spec} L$. □

Corollary 86. *Let $M \supseteq \mathbb{F}_p$ and let $\vec{g} \in W_r(M)$ have $g_0 = 0$. Then*

$$p^{r-1}[\vec{g}] = 0 \qquad \text{in } W_r(M)/\wp W_r(M).$$

Consequently, a class of order p^r has nonzero zeroth component in every representative.

Proof. By (A.9), $p^{r-1}\vec{g} = (0, \dots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. □

A.4.3 Reduced polynomial representatives

Definition 87. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced in the sense of Definition 54 and every $g_j \in x^{-1}k[x^{-1}]$.

Lemma 88 (Surjectivity on integral Witt vectors)**.** *If k is algebraically closed, then*

$$\wp : W_r(k[[x]]) \longrightarrow W_r(k[[x]])$$

is surjective.

Proof. Proceed by induction on r. For $r = 1$, given $h = \sum_{i\geq 0} b_i x^i$, choose u_0 with $u_0^p - u_0 = b_0$. For $i \geq 1$, define

$$u_i = \begin{cases} u_{i/p}^p - b_i, & p \mid i, \\ -b_i, & p \nmid i. \end{cases}$$

This is recursive because $i/p < i$. Put $u = \sum_{i\geq 0} u_i x^i \in k[[x]]$. In characteristic p,

$$u^p = \sum_{i\geq 0} u_i^p x^{pi}.$$

Thus the constant coefficient of $u^p - u$ is $u_0^p - u_0 = b_0$. For $i \geq 1$, its coefficient of x^i is $u_{i/p}^p - u_i = b_i$ if $p \mid i$, and $-u_i = b_i$ otherwise. Hence every coefficient of $u^p - u$ equals the corresponding coefficient of h, so $u^p - u = h$.

For $r > 1$, choose by induction $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp\vec{u}' = t(\vec{h})$ and lift it to $\widetilde{u} = (\vec{u}', 0)$. By Lemma 83(i), $\wp\widetilde{u} \ominus \vec{h} = V^{r-1}(g)$ for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$. Then Lemma 83(ii) gives $\wp(\widetilde{u} \oplus V^{r-1}w) = \vec{h}$. □

Proposition 89 (Reduced polynomial representatives)**.** *Let k be an algebraically closed field of characteristic p, and let $r \geq 1$. Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative.*

Proof. Proceed by induction on r. For $r = 1$, write a Laurent series as the sum of its principal part and an element of $k[[x]]$. The latter can be removed by Lemma 88. If the remaining principal part has leading term cx^{-m} with $p \mid m$, subtract

$$\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}.$$

This strictly decreases the pole order; iteration produces zero or a pole order prime to p.

For the inductive step, let $\vec{g} \in W_r(k((x)))$. By induction, there is $\vec{v} \in W_{r-1}(k((x)))$ such that

$$\vec{f}' := t(\vec{g}) \ominus \wp\vec{v}$$

is a reduced polynomial vector. Lift $\vec{v}$ to $\widetilde{\vec{v}} := (v_0, \dots, v_{r-2}, 0)$ and put

$$\vec{h} := \vec{g} \ominus \wp\widetilde{\vec{v}}.$$

Since truncation commutes with $\wp$, we have $t(\vec{h}) = \vec{f}'$. Thus $\vec{h} = (f_0', \dots, f_{r-2}', h_{r-1})$, and only its last component remains to be reduced.

By the length-one case, choose $z \in k((x))$ such that $f_{r-1} := h_{r-1} - \wp z$ is a reduced polynomial. Now set

$$\begin{aligned}\vec{f} &:= \vec{h} \ominus \wp(V^{r-1}z) \\ &= \vec{h} \ominus V^{r-1}(\wp z) = (f'_0, \dots, f'_{r-2}, f_{r-1}),\end{aligned}$$

where the last two equalities follow from Lemma 83(ii)–(iii). Every component of $\vec{f}$ is now a reduced polynomial, and

$$\vec{f} = \vec{g} \ominus \wp\big(\widetilde{\vec{v}} \oplus V^{r-1}z\big).$$

Hence $\vec{f}$ represents the same class as $\vec{g}$, as required. □

A.5 Weighted Witt addition in the cyclic calculation

The following homogeneity property of the universal addition polynomials is what makes degree estimates on Witt sums possible: even though the carry polynomials C_n of (A.8) are complicated, they are *isobaric* for the weighting in which the j-th component has weight p^j.

Lemma 90 (isobaricity of Witt addition)**.** *Assign to the variables X_j and Y_j the weight p^j. Then the n-th addition polynomial S_n is isobaric of weight p^n: every monomial of S_n has total weight exactly p^n. More generally, write $\vec{a}_i = (a_{i,0}, \dots, a_{i,r-1})$ and assign $a_{i,j}$ weight p^j. Then, for every $0 \le n < r$, the n-th component of the iterated Witt sum $\bigoplus_{i=1}^{d} \vec{a}_i$ is an isobaric polynomial of weight p^n in the variables $a_{i,j}$.*

Proof. We first prove the assertion for two summands by induction on n. For $n = 0$, we have $S_0 = X_0 + Y_0$, which is isobaric of weight 1. Suppose now that S_j is isobaric of weight p^j for every $j < n$. The ghost identities give

$$p^n S_n = \sum_{j \le n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) - \sum_{j<n} p^j S_j^{p^{n-j}}$$

Each $X_j^{p^{n-j}}$ and $Y_j^{p^{n-j}}$ has weight $p^j p^{n-j} = p^n$. Also, every monomial in the expansion of $S_j^{p^{n-j}}$ is a product of p^{n-j} monomials of weight p^j, so it has weight p^n. Thus the right-hand side is isobaric of weight p^n. This identity holds in the universal polynomial ring over $\mathbb{Z}$, where multiplication by the nonzero integer p^n does not change which monomials occur. Hence S_n is isobaric of weight p^n as well.

We next induct on the number d of summands. Write

$$\vec{A}^{(d)} := \bigoplus_{i=1}^{d} \vec{a}_i = \vec{a}_1 \oplus \cdots \oplus \vec{a}_d,$$

where each variable $a_{i,j}$ has weight p^j. For $d = 1$, the component $A_n^{(1)} = a_{1,n}$ has weight p^n. Suppose the assertion holds for $d-1$ summands. Then

$$A_n^{(d)} = S_n\big(A_0^{(d-1)}, \dots, A_n^{(d-1)}; a_{d,0}, \dots, a_{d,n}\big).$$

By induction, $A_j^{(d-1)}$ is isobaric of weight p^j. Thus each substituted polynomial has the same weight as the variable it replaces, so substitution preserves weights. Since S_n is isobaric of weight p^n, so is $A_n^{(d)}$, completing the induction. □

Appendix B

A Failure of an Artin–Schreier Induction Strategy

B.1 Motivation: can the Artin–Schreier calculation be iterated?

Part (2) of Theorem 58 proves the order-p case by an explicit Artin–Schreier calculation. Part (3) treats $G = \mathbb{Z}/p^r\mathbb{Z}$ by working directly with an Artin–Schreier–Witt equation. It is natural to ask whether the second calculation can instead be obtained by iterating the first.

Indeed, let

$$G = \mathbb{Z}/p^r\mathbb{Z}, \qquad H = p^{r-1}G \cong \mathbb{Z}/p\mathbb{Z}, \qquad G/H \cong \mathbb{Z}/p^{r-1}\mathbb{Z},$$

and factor a G-torsor $\mathcal{P} \to U$ as

$$\mathcal{P} \longrightarrow \mathcal{P}/H \longrightarrow U.$$

The tempting induction has three steps:

1. apply the induction hypothesis to the G/H-torsor $\mathcal{P}/H \to U$;
2. regard $\mathcal{P} \to \mathcal{P}/H$ as an $H \cong \mathbb{Z}/p\mathbb{Z}$-torsor and apply the Artin–Schreier calculation after compactifying its base;
3. combine the two unramifiedness statements to recover the result for the original G-torsor.

The first two steps do produce natural statements, but they live over different exceptional valuations. Passing from one to the other involves a base change induced by the comparison between the symmetric power of a torsor as a scheme and its symmetric power as a torsor. That base change can be wildly ramified. The purpose of this appendix is to formulate the proposed induction precisely, follow both inputs through the resulting valuation tower, and explain why they do not by themselves imply the required conclusion.

B.2 The attempted induction

Let G be a finite abelian group, let $H \subseteq G$ be a subgroup, and let $\mathcal{P} \to X$ be a G-torsor over an S-scheme X that is Zariski-locally quasi-projective over S. Recall the canonical factorization

$$\mathcal{P} \longrightarrow \mathcal{P}/H \longrightarrow X,$$

where $\mathcal{P}/H = (G/H) \times^G \mathcal{P}$ is a G/H-torsor over X and $\mathcal{P} \to \mathcal{P}/H$ is an H-torsor.

In addition to the kernel $K_G = \ker(G^d \to G)$ of the summation morphism, put

$$K_H := \ker(H^d \to H), \qquad K_{G/H} := \ker((G/H)^d \to G/H).$$

The snake lemma applied to the two summation maps gives an exact sequence

$$0 \longrightarrow K_H \longrightarrow K_G \longrightarrow K_{G/H} \longrightarrow 0. \tag{B.1}$$

Construction 91 (The symmetric-power comparison morphism)**.** For the G-torsor $\mathcal{P} \to X$ fixed above, put

$$\Xi_G := K_G \rtimes S_d.$$

The inclusion $S_d \hookrightarrow \Xi_G$, given by $\sigma \mapsto ((1, \dots, 1), \sigma)$, induces a canonical comparison morphism

$$q_{\mathcal{P}} : \mathcal{P}^{(d)} = S_d \backslash \mathcal{P}^{\times_S d} \longrightarrow \Xi_G \backslash \mathcal{P}^{\times_S d} = \mathcal{P}^{[d]}.$$

Indeed, the quotient map from $\mathcal{P}^{\times_S d}$ to its Ξ_G-quotient is S_d-invariant and therefore factors uniquely through the S_d-quotient.

We use the following four quotient schemes:

$$\begin{aligned}
\mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d) \backslash \mathcal{P}^{\times_S d}, \\
\mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d) \backslash \mathcal{P}^{\times_S d}, \\
(\mathcal{P}/H)^{(d)} &:= (H^d \rtimes S_d) \backslash \mathcal{P}^{\times_S d}, \\
(\mathcal{P}/H)^{[d]} &:= ((H^d + K_G) \rtimes S_d) \backslash \mathcal{P}^{\times_S d}.
\end{aligned}$$

The first is the symmetric G-power of $\mathcal{P}$. The second is the symmetric H-power relative to $\mathcal{P} \to \mathcal{P}/H$, the third is the symmetric power of the scheme $\mathcal{P}/H$, and the fourth is the symmetric G/H-power of the G/H-torsor $\mathcal{P}/H \to X$. For the last two identifications, use $(\mathcal{P}/H)^{\times_S d} \cong H^d \backslash \mathcal{P}^{\times_S d}$ and note that the preimage of $K_{G/H}$ under $G^d \twoheadrightarrow (G/H)^d$ is $H^d + K_G$.

Proposition 92. *The four quotient schemes fit into a canonical commutative diagram*

$$\begin{array}{ccc}
\mathcal{P}^{[d]_H} & \xrightarrow{\ H\ } & (\mathcal{P}/H)^{(d)} \\
\downarrow & & \downarrow{\scriptstyle q_{\mathcal{P}/H}} \\
\mathcal{P}^{[d]_G} & \xrightarrow[\ H\ ]{} & (\mathcal{P}/H)^{[d]} \\
 & {\scriptstyle G}\searrow & \downarrow{\scriptstyle G/H} \\
 & & X^{(d)},
\end{array}$$

in which every labelled arrow is a torsor under the indicated constant group. The upper square is Cartesian:

$$\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}.$$

The vertical map $q_{\mathcal{P}/H}$ is the comparison morphism of Construction 91, applied to the G/H-torsor $\mathcal{P}/H \to X$.

Proof. Apply Proposition 26 to the G-torsor $\mathcal{P} \to X$, the H-torsor $\mathcal{P} \to \mathcal{P}/H$, and the G/H-torsor $\mathcal{P}/H \to X$. This gives the three labelled torsors; the comparison map $q_{\mathcal{P}/H}$ is supplied by Construction 91, applied to $\mathcal{P}/H \to X$. The factorization

$$\mathcal{P}^{[d]_G} \longrightarrow (\mathcal{P}/H)^{[d]} \longrightarrow X^{(d)}$$

is the canonical quotient decomposition of the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ by H, because both its intermediate quotient and $(\mathcal{P}/H)^{[d]}$ are realized as

$$((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d}.$$

Let

$$T := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}.$$

Base change makes T an H-torsor over $(\mathcal{P}/H)^{(d)}$. The quotient maps give an H-equivariant morphism $\mathcal{P}^{[d]_H} \to T$. A morphism of H-torsors is an isomorphism, proving the Cartesian assertion. □

Remark. The two vertical comparison maps in the diagram need not be torsors under a constant group. For example, $K_H \rtimes S_d$ need not be normal in $K_G \rtimes S_d$, because coordinate permutations act nontrivially on K_G/K_H. Thus the map $q_{\mathcal{P}/H}$ is not another torsor layer to which the cyclic induction hypotheses automatically apply. Its ramification at the boundary is precisely the extra datum that will matter below.

We now place this quotient tower in the geometric setting relevant to the attempted induction. Let C be a smooth projective curve, let $U = C \setminus \mathfrak{m}$, and let $\mathcal{P} \to U$ be a G-torsor. Fix $p_\infty \in \mathfrak{m}$ and an integer $d \geq 1$, and assume that the ramification of $\mathcal{P}$ at p_∞ is bounded by d in the sense of Definition 41.

The G/H-torsor $\mathcal{P}/H \to U$ compactifies to a finite morphism of smooth projective curves

$$f : C' \longrightarrow C.$$

Choose $p'_\infty \in f^{-1}(p_\infty)$. The induced morphism on symmetric powers sends $d \cdot p'_\infty$ to $d \cdot p_\infty$:

$$f^{(d)} : C'^{(d)} \longrightarrow C^{(d)}.$$

Let

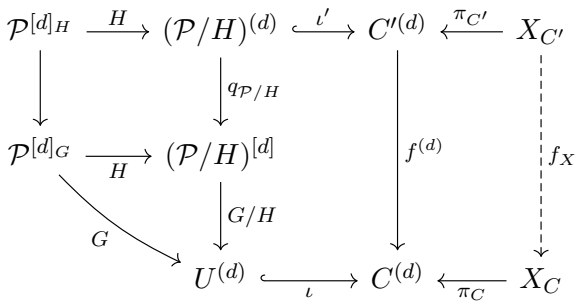
$$\pi_C : X_C \longrightarrow C^{(d)}, \qquad \pi_{C'} : X_{C'} \longrightarrow C'^{(d)}$$

be the blowups at these two diagonal points. Write $E_C, E_{C'}$ for their exceptional divisors, $\eta_C, \eta_{C'}$ for the respective generic points, and

$$V_C := \mathcal{O}_{X_C, \eta_C} \subset K(C^{(d)}), \qquad V_{C'} := \mathcal{O}_{X_{C'}, \eta_{C'}} \subset K(C'^{(d)})$$

for the corresponding DVRs.

The quotient tower, compactifications, and blowups fit into the diagram



The left block is the torsor tower of Proposition 92. The middle block records the open immersions and compactifications. The right block records the blowups. The rational lift f_X is defined at the generic point of the exceptional divisor and sends $\eta_{C'}$ to η_C, producing the extension of DVRs

$$V_C \hookrightarrow V_{C'}.$$

We can now formulate the proposed induction step. In the cyclic p-power setting of Section B.1, the two intended inputs are:

(IH$_{r-1}$) $(\mathcal{P}/H)^{[d]} \to U^{(d)}$ is unramified at η_C;

(AS) $\mathcal{P}^{[d]_H} \to (\mathcal{P}/H)^{(d)}$ is unramified at $\eta_{C'}$.

The first is the desired contribution of induction applied to the quotient G/H. The second is the desired contribution of the degree-p Artin–Schreier calculation applied to the H-torsor over the compactified curve C'. The proposed induction step asks whether these imply that

$$\mathcal{P}^{[d]_G} \longrightarrow U^{(d)}$$

is unramified at η_C. We will see that the two assertions do not, by themselves, control the ramification needed for this conclusion.

B.3 The obstruction in the valuation tower

We may reduce to the case in which $\mathcal{P} \to U$ is connected. Choose a connected component $\mathcal{P}_0 \subseteq \mathcal{P}$, and let $G_0 \subseteq G$ be the subgroup preserving it. Then $\mathcal{P}_0 \to U$ is a connected G_0-torsor. Replacing $(\mathcal{P}, G, H)$ by $(\mathcal{P}_0, G_0, H \cap G_0)$ restricts the four quotient schemes to the corresponding connected components, and both unramifiedness hypotheses remain valid. We may therefore assume that $\mathcal{P}$ is connected. We may also assume that $H \neq 0$: if $H = 0$, then $\mathcal{P}/H = \mathcal{P}$ and $G/H = G$, so (IH$_{r-1}$) is already the desired conclusion.

Under these reductions the four schemes in the Cartesian square of Proposition 92 are integral, so we may pass to function fields. Put

$$K_C := \operatorname{Frac} V_C = K(C^{(d)}), \qquad K_{C'} := \operatorname{Frac} V_{C'} = K(C'^{(d)}),$$

and

$$F := K((\mathcal{P}/H)^{[d]}), \qquad E := K(\mathcal{P}^{[d]_G}).$$

Then F/K_C is Galois with group G/H, E/F is Galois with group H, and E/K_C is Galois with group G. The Cartesian square identifies the function field of $\mathcal{P}^{[d]_H}$ with $E \cdot K_{C'}$. Its generic fibre over $(\mathcal{P}/H)^{(d)}$ is the finite domain $E \otimes_F K_{C'}$, hence a field.

We use here the following standard comparison between a Galois extension and its base change.

Lemma 93. *Let E/F be a finite Galois extension with group H, let K/F be a field extension, and form all fields inside a common overfield. Put $M := E \cap K$. Then restriction gives*

$$\operatorname{Gal}(EK/K) \cong \operatorname{Gal}(E/M),$$

and, as K-algebras,

$$E \otimes_F K \cong \prod_{i=1}^{[M:F]} EK.$$

In particular, $E \otimes_F K$ is a field if and only if E and K are linearly disjoint over F.

Proof. Write $E = F[\alpha]$ and let $g \in F[x]$ be the minimal polynomial of α. The translation theorem identifies $\mathrm{Gal}(EK/K)$ with the subgroup of $\mathrm{Gal}(E/F)$ fixing $E \cap K = M$, hence with $\mathrm{Gal}(E/M)$.

Factor $g = g_1 \cdots g_m$ into monic irreducible polynomials over K. Adjoining any root of g to K produces EK, so every g_i has degree $[E : M]$. Consequently $m = [M : F]$, and

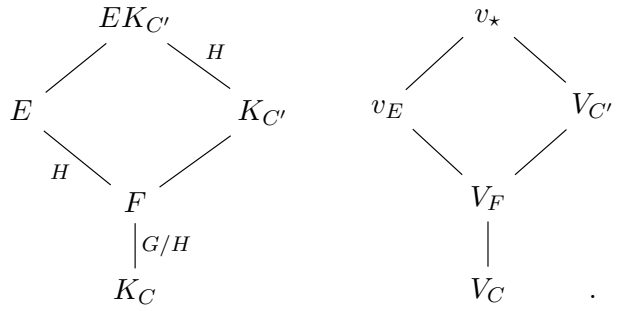
$$E \otimes_F K \cong K[x]/(g) \cong \prod_{i=1}^{[M:F]} K[x]/(g_i) \cong \prod_{i=1}^{[M:F]} EK.$$

The product is a field exactly when $M = F$. □

Applying the lemma with $K = K_{C'}$ gives

$$E \cap K_{C'} = F, \qquad \mathrm{Gal}(EK_{C'}/K_{C'}) \cong \mathrm{Gal}(E/F) = H.$$

Let $V_F := V_{C'}|_F$. Choose a place v_E of E above V_F and a place $v_\star$ of $EK_{C'}$ above both v_E and $V_{C'}$. The relevant field and valuation towers are



We can now see exactly what each part of the proposed induction contributes:

(IH_{r-1}): F/K_C is unramified at V_C. In particular, $e(V_F \mid V_C) = 1$.

(AS): $EK_{C'}/K_{C'}$ is unramified at $V_{C'}$.

Missing step: one must prove that E/F is unramified at V_F.

Goal: once the missing step holds, the tower $K_C \subset F \subset E$ shows that E/K_C is unramified at V_C.

The gap lies between the second and third items. It is measured by the following inertia comparison.

Lemma 94. *Let E/F be finite Galois and linearly disjoint from K/F. Let w be a place of EK and write w_K, w_E, w_F for its restrictions. Under the restriction isomorphism $\mathrm{Gal}(EK/K) \cong \mathrm{Gal}(E/F)$ there is an injection*

$$I(w \mid w_K) \hookrightarrow I(w_E \mid w_F).$$

After completion, the source is the inertia group of the local compositum $\widehat{E}\,\widehat{K}/\widehat{K}$. In particular, if $\widehat{E} \subseteq \widehat{K}$, then $I(w \mid w_K) = 1$, regardless of the ramification of E/F at w_F.

Proof. Let $\sigma \in I(w \mid w_K)$. It fixes K, hence F, and restriction gives $\sigma|_E \in \mathrm{Gal}(E/F)$. Because σ preserves w, its restriction preserves w_E. Moreover, σ acts trivially on the residue field at w, and therefore on the subfield given by the residue of E. Thus $\sigma|_E \in I(w_E \mid w_F)$. Injectivity follows from the restriction isomorphism. The completed statement is the usual identification of the decomposition and inertia groups with those of the corresponding extension of completed fields. □

Applied to the tower above, the lemma gives

$$I(v_\star \mid V_{C'}) \hookrightarrow I(v_E \mid V_F).$$

Statement (AS) makes the group on the left trivial, whereas the missing step requires the group on the right to be trivial. An injection with trivial source gives no such conclusion. In other words, the implication

$$EK_{C'}/K_{C'} \text{ unramified at } V_{C'} \quad \Longrightarrow \quad E/F \text{ unramified at } V_F$$

is not formal, as the following example shows.

Example 95 (Absorption of an Artin–Schreier layer)**.** Let $k = \bar{\mathbb{F}}_p$ and put $F = k((t))$. Define $K = k((\pi))$ and embed $F \hookrightarrow K$ by

$$t = \frac{\pi^p}{1 - \pi^{p-1}}.$$

Equivalently,

$$t^{-1} = \pi^{-p} - \pi^{-1} = \wp(\pi^{-1}), \qquad \wp(z) = z^p - z.$$

The element t^{-1} has reduced pole order one over F. Hence the Artin–Schreier extension

$$E = F(y), \qquad y^p - y = t^{-1},$$

is totally wildly ramified of degree p and has break one by Theorem 53. In particular,

$$I(E/F) = \mathbb{Z}/p\mathbb{Z}.$$

Over K, however, π^{-1} is already a solution of the same equation. The polynomial $X^p - X - t^{-1}$ is irreducible over F, because t^{-1} has reduced pole order one, and hence $F(\pi^{-1})/F$ has degree p. On the other hand, the displayed relation between t and π shows that K/F has degree at most p. It follows that $F(\pi^{-1}) = K$. Therefore the fields fit into the diagram

$$\begin{array}{ccc} E = F(y) & \xrightarrow[y \mapsto \pi^{-1}]{\sim} & K = F(\pi^{-1}) \\ & {\scriptstyle p} \searrow \quad \swarrow {\scriptstyle p} & \\ & F & \end{array}$$

After using the horizontal F-isomorphism to identify the two upper fields inside a common algebraic closure of F, we may regard E and K as the same subfield. Consequently

$$EK/K = K/K$$

is trivial and therefore unramified. The inertia comparison is exactly

$$\underbrace{I(EK/K)}_{1} \hookrightarrow \underbrace{I(E/F)}_{\mathbb{Z}/p\mathbb{Z}}.$$

The example shows directly why the missing implication is not a formal property of base change: E/F is wildly ramified, whereas after the wild base change K/F, the extension EK/K is trivial. We do not claim that this particular extension K/F arises from the comparison map $q_{\mathcal{P}/H}$, so this is not a counterexample to part (3) of Theorem 58. It shows only that the desired conclusion requires more than a general base-change argument. The direct Witt-vector calculation supplies that additional input by treating the full $\mathbb{Z}/p^r\mathbb{Z}$-torsor at once.

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